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EPB41L4A-AS1 long noncoding RNA acts in both *cis*- and *trans*-acting transcriptional regulation and controls nucleolar biology

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eLife Assessment

This paper provides **important** findings towards understanding the role of the lncRNA EPB41L4A-AS1 in a human cell line. The data is generally **convincing**, supported by extensive and clever integrative analysis. The work provides insights into how this lncRNA regulates gene expression via complex mechanisms; however the biological relevance awaits validation in other models.

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Abstract

Mammalian genomes are pervasively transcribed into long noncoding RNAs (lncRNAs), whose functions and modes of action remain poorly understood. *EPB41L4A-AS1* is an evolutionary conserved, broadly and highly expressed lncRNA that produces the H/ACA snoRNA *SNORA13* from one of its introns. We studied the consequences of *EPB41L4A-AS1* perturbation in breast cancer cells and found that it acts both *in cis*, to enhance transcription of the proximal *EPB41L4A* gene and additional genes in its two flanking topologically associated domains, and *in trans* by broadly regulating gene expression, including expression of snoRNAs, transcription of genes involved in nucleolar biology and the distribution of nucleolar proteins. These effects are phenocopied by the loss of SUB1, an interactor of *EPB41L4A-AS1*, and are observed following transient perturbations of *EPB41L4A-AS1* that do not affect steady-state *SNORA13* levels or the rRNA modification it helps install. Exogenous expression of the full-length *EPB41L4A-AS1* locus but not *SNORA13* expression can rescue the *trans*-acting transcriptional effects of its perturbation. The *EPB41L4A-AS1* gene is thus a versatile locus producing RNA molecules acting on multiple levels for key cellular functions.

Introduction

Long noncoding RNAs (lncRNAs) are a highly heterogeneous group of RNA molecules, defined as being >500 nt long and not displaying a significant protein-coding potential^{1–4}. Given this broad definition and the pervasive transcription of the human genome^{5–8}, lncRNAs are now the RNA class with the largest number of annotated genes⁹. However, only a limited number of lncRNAs have well-defined mechanisms of action. Long noncoding RNAs exhibit a wide range of expression levels, driven by varying transcription rates and half-lives. Similarly to protein-coding mRNAs, lncRNAs are transcribed by RNA polymerase II (Pol II), 5' capped with 7-methyl guanosine (m⁷G), polyadenylated at their 3' ends, and have a canonical exon–intron structure. Interestingly, small peptides arising from short and poorly conserved translated open reading frames (ORFs) have been reported in several lncRNAs^{10–13}, and studies using ultra-deep sequencing of ribosome profiling data and deep proteomic analysis found evidence of translation of ORFs in hundreds of

lncRNAs¹⁴. Yet it remains unclear how many of these are encoding functional peptides, mainly because these ORFs are typically poorly conserved and the protein products do not appear to be abundant¹⁵.

One way to functionally classify lncRNAs is by the nature of the functional elements harbored within their loci. One class comprises genes in which the DNA element has a function unrelated to the transcription of the locus¹⁶. The other includes cases in which the act of transcription is functionally relevant, while the resulting RNAs are “byproducts” of Pol II activity, which, for example, can tune the expression of another locus in 3D proximity^{1,17–19}. This class is potentially broad because it can explain the prevalence of lncRNA loci in which the lncRNAs are expressed at very few copies and/or have sequences that are evolving without any apparent selection. For the cases where the RNA product is functional, a further distinction can be made between *cis*-acting lncRNAs, the function of which is restricted to the locus from which they are transcribed, and *trans*-acting lncRNAs, which are transcribed, processed, and then move away by diffusion or active transport to exert their function elsewhere. The latter RNAs require a relatively stable RNA molecule because they have to interact with one or more other RNA species and/or proteins, whereas, for *cis*-acting lncRNAs, such an RNA molecule may or may not be necessary, with the act of transcription and concomitant chromatin remodeling reported to be sufficient in several cases²⁰.

Small nucleolar RNAs (snoRNAs) are intermediate-sized RNAs whose main function is to guide the installation of RNA modifications on other RNAs, mainly ribosomal RNAs (rRNAs). In mammalian cells, snoRNAs are produced from introns of polyadenylated transcripts, some of which code for proteins, and others are instead lncRNAs. Noncoding snoRNA host genes (or ncSHNG) are characterized by several characteristic features, including high and broad expression levels, relatively inefficient splicing, and sensitivity to nonsense-mediated decay²¹. While lncRNA functions have been shown for some ncSHNGs, most remain poorly understood. Here, we describe a bioinformatic screen that converged on *EPB41L4A-AS1*, a very abundant and conserved lncRNA/snoRNA-host gene acting as a regulator in *cis*, and our observations that this lncRNA also works in *trans* to regulate nucleolar biology.

Results

A computational screen for putative *cis*-acting lncRNAs

We hypothesized that lncRNAs likely to act in *cis* will be occasionally co-regulated with their target genes and also found in spatial proximity to the target promoters. We sought such lncRNA-target pairs in the GeneHancer database²², that connects regulatory elements and their targets by combining tissue co-expression data, enhancer-enriched histone modifications, transcription factor binding, Hi-C data, and eQTLs. We complemented this resource with RNA-seq datasets from the breast cancer cell line MCF-7 exposed to different treatments^{23–25} (Fig. 1A [↗](#) and Table S1). This analysis (see Methods) led us to focus on four lncRNA-target pairs connected in GeneHancer and in which both genes were significantly either upregulated or downregulated in the treatment conditions, compared to controls (Fig. S1A–D [↗](#)). We knocked down (KD) each candidate for 72 hours with two distinct GapmeRs—antisense oligonucleotides with a DNA core that induces the degradation of target nascent RNA via RNase H activity. We then tested the expression of the target genes using RT-qPCR. Among the four lncRNAs, KD of *EPB41L4A-AS1* with both GapmeRs significantly repressed *EPB41L4A* (Fig. 1B [↗](#)), whereas the other lncRNAs had no substantial effects on the putative target genes.

EPB41L4A-AS1 is a complex lncRNA gene

The *EPB41L4A-AS1* locus is ~2 kb (kilobases) long, and the main transcribed isoform terminates ~300 nt from the 3' end of the main isoform of the convergently transcribed *EPB41L4A* gene, and >250 kb away from the *EPB41L4A* transcription start site (TSS, Fig. 1C [↗](#)). An H/ACA snoRNA, *SNORA13*, a homolog of yeast *snR35*, is embedded within the first intron of *EPB41L4A-AS1*. This snoRNA guides pseudouridylation on position 1248 of the small subunit 18S rRNA²⁶. This

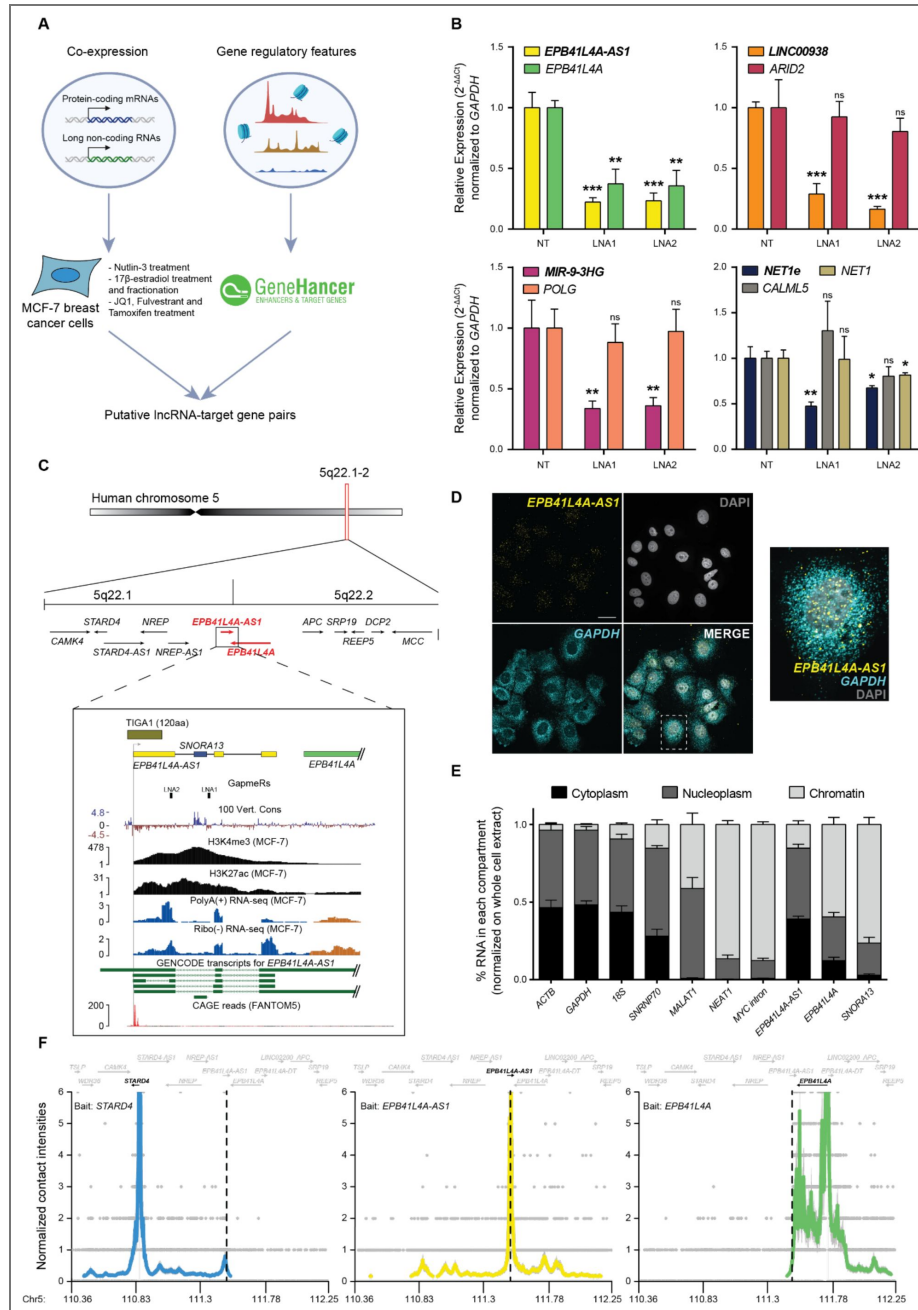


Figure 1. EPB41L4A-AS1 is a highly expressed lncRNA with a widespread cellular distribution.

(A) Graphical overview of the screen to identify candidate *cis*-acting lncRNAs. (B) RT-qPCR to assess the expression of each candidate *cis*-acting lncRNA and their predicted *cis*-regulated target genes, following KD with two unique GapmeRs. (C) (Top) Overview of the 5q22.1-2 locus, with the bars highlighting the two TADs. (Bottom) Zoom in on the EPB41L4A-AS1 locus, with tracks for the PhyloP conservation score (UCSC genome browser), H3K4me3 and H3K27ac histone modifications (ENCODE), PolyA+ and Ribo(-) RNA-seq coverage (this study), GENCODE transcripts and CAGE reads (FANTOM5). The location of the GapmeRs used in this study is also reported. (D) A representative single-molecule RNA FISH (smFISH) image shows EPB41L4A-AS1 intra-cellular localization. GAPDH and DAPI were used to label the cytoplasm and the nucleus, respectively (scale bar = 20 μ m, 100X magnification). (E) RT-qPCR analysis of subcellular fractionation experiments. The percentages of RNA in each compartment were obtained by normalizing the expression in the different fractions to that in whole cells. (F) UMI-4C contact profiles using baits targeting the TSSes of STARD4 (left), EPB41L4A-AS1 (center), or EPB41L4A (right). The dotted line represents the center of the EPB41L4A-AS1 locus. All experiments were performed in n=3 biological replicates, except UMI-4C with n=2, with the error bars in the barplots representing the standard deviation. ns = P>0.05; * = P<0.05, ** = P<0.01, *** = P<0.001 (two-sided Student's t-test).

modification is the first step in the installation of one of the most complex RNA modifications known to date—N1-methyl-N3-aminocarboxypropyl-pseudouridine (m(1)acp(3)Ψ)—that is located next to the tRNA P-site and is often lost in human cancers²⁶. Recently, *SNORA13* has been reported to negatively regulate ribosome biogenesis by decreasing the incorporation of RPL23 into maturing 60S subunits, eventually triggering p53-mediated cellular senescence²⁷.

Several groups reported different functions of *EPB41L4A-AS1*. It was found to be differentially expressed across the cell cycle, and its KD affected cell cycle progression and reduced proliferation in multiple cancer cell lines²⁸. Another study implicated *EPB41L4A-AS1* in glucose metabolism through interactions with HDAC2 and NPM1²⁹. Silencing of *EPB41L4A-AS1* led to HDAC2 translocation into the nucleoplasm, where it binds the *VDAC1* and *VHL* promoter regions, among others²⁹, and effects on the expression of additional genes were reported in other systems as well^{30–33}. In human cancers, *EPB41L4A-AS1* is also downregulated and correlates with a poor prognosis in a variety of cancers, including breast cancer (BRCA), which we also observed by analyzing the TCGA data (Fig. S2C), aligning well with the widespread loss of the m(1)acp(3)Ψ modification. *EPB41L4A* instead is a poorly studied gene, reported to be involved in predisposition to papillary thyroid carcinoma by modulating the WNT/β-catenin pathway^{34,35}, possibly playing a role in connecting the cytoskeleton to the plasma membrane, as well as playing unspecified roles throughout development³⁶.

***EPB41L4A-AS1* is highly expressed in both the nucleus and cytoplasm and is regulated in multiple conditions**

Using single-molecule fluorescence *in-situ* hybridization (smFISH) and subcellular fractionation we found that *EPB41L4A-AS1* is expressed at an average of 33.37 ± 3.95 molecule per cell, and displays both nuclear and cytoplasmic localization in MCF-7 cells (Fig. 1D), with a minor fraction associated with chromatin as well (Fig. 1E). In contrast, both *EPB41L4A* and *SNORA13* were mostly found in the chromatin fraction (Fig. 1E), the former possibly due to the length of its pre-mRNA (>250 kb), which would require substantial time to transcribe^{37–40}. Coding-Potential Assessment (CPAT)⁴¹ analysis showed the sequence of *EPB41L4A-AS1* has little coding potential. A translated ORF in the first exon of *EPB41L4A-AS1* has been reported to produce a small peptide (TIGA1), which localizes to the mitochondria and induces growth arrest⁴² (Fig. 1C). Applying CPAT to each exon separately, found evidence for some coding potential only in the first exon that harbors the TIGA1 peptide (Fig. S3A). Importantly, analysis of the RNA-seq signal coverage shows that the common *EPB41L4A-AS1* isoform in MCF-7 cells begins downstream of the AUG codon at the start of TIGA1, and TIGA1 doesn't have another in-frame AUG in its sequence. Consistently, the vast majority of CAGE tags at the *EPB41L4A-AS1* are also downstream of the AUG of TIGA1, consistent with the start sites of most isoforms of *EPB41L4A-AS1* annotated in GENCODE (Fig. 1E). We therefore conclude that while some fraction of *EPB41L4A-AS1* might produce TIGA1 peptides, most RNA products of this gene in MCF-7 do not produce TIGA1 or other peptides predicted to be functional.

Consistently with a previous study⁴², we found a strong upregulation of *EPB41L4A-AS1*, *EPB41L4A*, and *SNORA13* in MCF-7 cells upon serum starvation (Fig. S3B). In contrast with a previous report²⁸, we did not detect *EPB41L4A-AS1* to be differentially expressed across the cell cycle in MCF-7 cells (Fig. S3C). We further explored the effect of different stresses and environmental conditions on *EPB41L4A-AS1* by analyzing its expression in a published dataset comprising five different cell types exposed to a panel of 50 different substances⁴³ (Fig. S3D), as well as exposing MCF-7 cells to LPS (bacterial infection), H₂O₂ (oxidative stress), thapsigargin (ER stress and unfolded protein response) and etoposide (DNA damage) (Fig. S3E). Strikingly, *EPB41L4A-AS1* expression was induced in nearly all conditions, often together with *EPB41L4A* but not *SNORA13*, suggesting the levels of the snoRNA are typically detached from that of the lncRNA⁴⁴, likely because the latter is substantially more stable (see below).

EPB41L4A-AS1 regulates EPB41L4A transcription

We assessed the 3D chromatin architecture around the *EPB41L4A-AS1* locus by performing UMI-4C-seq⁴⁵ with baits targeting the lncRNA, *EPB41L4A* and the promoter of *STARD4*—a flanking gene found ~650 Kb upstream of the *EPB41L4A-AS1* promoter. *EPB41L4A-AS1* formed contacts with both genes (Fig. 1F⁴⁵, middle), with prominent peaks corresponding to the promoter of *EPB41L4A*. Interestingly, the promoter of *EPB41L4A* displays several contacts with its own gene body (Fig. 1F⁴⁵, right panel), consistent with a “stripe” in HiC data where a single anchor interacts with the whole topologically associated domain (TAD) at high frequency⁴⁶, with those frequencies reduced to below background levels upstream of the *EPB41L4A-AS1* locus. Conversely, contacts from the *STARD4* promoter also show a peak near *EPB41L4A-AS1* and then reduced below the background (Fig. 1F⁴⁵, left panel). These data indicate that *EPB41L4A-AS1* is located at a TAD boundary region, forming strong spatial contacts with sites upstream and downstream, including the promoter of *EPB41L4A*. In agreement with this, *EPB41L4A-AS1* is located between the 5q22.1 and 5q22.2 chromosomal bands, delimiting two TADs as evident in both our 4C-seq and Micro-C data from H1 cells (Fig. 1F⁴⁵ and 2A⁴⁵). Both *EPB41L4A* and *EPB41L4A-AS1* are characterized by CTCF binding near their promoters and their chromatin is extensively demarcated with H3K27ac (Fig. 2A⁴⁵). A recent study identified this boundary as one of the most insulated in human embryonic stem cells (hESCs)⁴⁷.

We then focused on a possible reciprocal regulation between *EPB41L4A-AS1* and *EPB41L4A*. *EPB41L4A-AS1* KD with antisense GapmerRs reduced the expression of *EPB41L4A* mRNA and its promoter-associated divergent lncRNA *EPB41L4A-DT* (Fig. 2B⁴⁵), as well as the *EPB41L4A* protein levels (Fig. 2C⁴⁵). CUT&RUN-qPCR of the H3K27ac mark—which is associated with active promoters and enhancers—showed a reduction of H3K27ac at the *EPB41L4A* promoter (Fig. 2D⁴⁵). In contrast to the effects of GapmerRs, which mostly act in the nucleus and on the nascent RNA^{50,51}, KD of *EPB41L4A-AS1* with siRNAs, which instead mainly act on the mature RNA in the cytoplasm, did not affect *EPB41L4A* levels (Fig. S4A⁴⁵). Levels of *EPB41L4A* were not affected by increased expression of *EPB41L4A-AS1* from the endogenous locus by CRISPR activation (CRISPRa), nor by its exogenous expression from a plasmid (Fig. S4B and S4C⁴⁵). The former suggests that endogenous levels of *EPB41L4A-AS1*—that are far greater than those of *EPB41L4A*—are sufficient to sustain the maximal expression of this target gene in MCF-7 cells. Moreover, downregulation (Fig. S4D and S4E⁴⁵) or upregulation (Fig. S4F and S4G⁴⁵) of *EPB41L4A* did not have any consistent effects on *EPB41L4A-AS1* expression, suggesting a unidirectional effect of *EPB41L4A-AS1* transcription of its nascent RNA product on *EPB41L4A*. Consistently with a *cis*-acting regulation, the reduced *EPB41L4A* mRNA levels after *EPB41L4A-AS1* knockdown were rescued only after CRISPRa of the endogenous *EPB41L4A-AS1* locus (Fig. S4H and S4I⁴⁵). To study the potential mode of action of this regulation, we inspected UMI-4C data anchored at the *EPB41L4A-AS1* promoter and found a reduction in the contact frequency with the *EPB41L4A* promoter upon treatment with *EPB41L4A-AS1*-targeting GapmerRs, that was statistically significant for LNA1 (Fig. 2E⁴⁵ and S4J⁴⁵). We conclude that the transcription of the RNA product of *EPB41L4A-AS1* acts unidirectionally and in *cis* to promote the expression of *EPB41L4A*, possibly by ensuring spatial proximity between the TAD boundary and the *EPB41L4A* promoter.

Genome-wide transcriptional alterations upon depletion of EPB41L4A-AS1

To evaluate the global transcriptional response to *EPB41L4A-AS1* levels, we sequenced polyadenylated RNA from cells transfected with *EPB41L4A-AS1*-, *EPB41L4A*-targeting, or control GapmerRs by RNA-seq (Table S2). Because LNA1 induced the expression of the stress responsive gene *CDKN1A* (p21)—a known offtarget effect of some GapmerR^{52–54} (Fig. S5A–C⁴⁵)—we have decided to initially sequence only the cells transfected with LNA2. Consistently with the RT-qPCR data, KD of *EPB41L4A-AS1* reduced *EPB41L4A* expression, and also reduced expression of several, but not all other genes in the TADs flanking the lncRNA (Fig. 2F⁴⁵). Based on these data, *EPB41L4A-AS1* is a significant *cis*-acting activator according to TransCistor⁵⁵ (P=0.005 using the

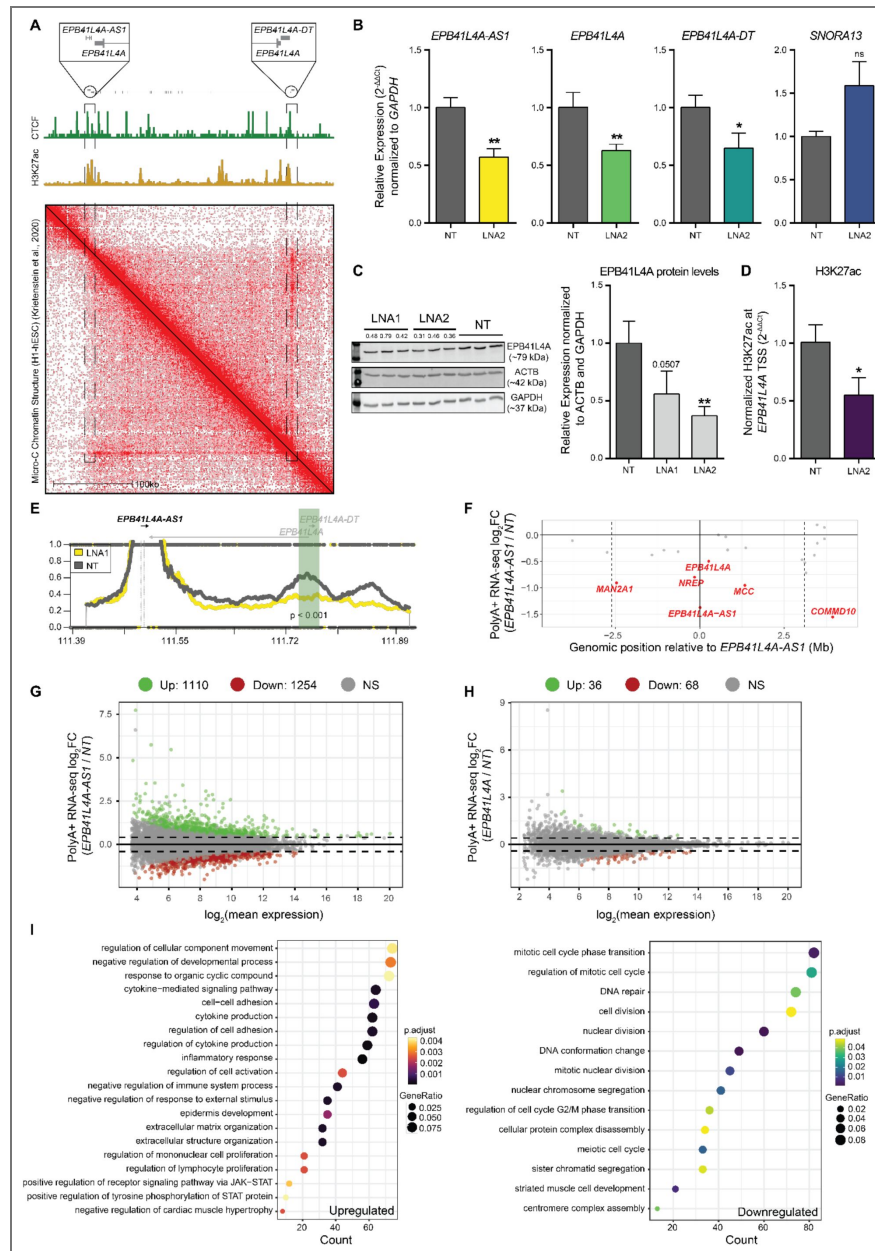


Figure 2. *EPB41L4A-AS1* is a cis-acting lncRNA affecting genome-wide gene expression.

(A) (Top) In scale view of the *EPB41L4A-AS1-EPB41L4A* locus, with zoomed areas corresponding to the two TSS. (Middle) CTCF and H3K27ac coverage across this region. (Bottom) Micro-C data in H1-hESCs⁴⁸ show continuous contacts throughout the *EPB41L4A* gene body. **(B)** RT-qPCR and **(C)** Western blot for the indicated genes and proteins following *EPB41L4A-AS1* KD. **(D)** H3K27ac CUT&RUN-qPCR following *EPB41L4A-AS1* KD with GapmeRs using primers targeting the promoter of *EPB41L4A*. **(E)** UMI-4C contact profiles in control and LNA1-transfected cells using baits targeting the TSS of *EPB41L4A-AS1*. The green area represents the quantified genomic interval, and the p-value was calculated using a Chi-squared test. **(F)** Changes in gene expression for the genes in the flanking TADs of the lncRNA in cells transfected with GapmeRs targeting *EPB41L4A-AS1*. The vertical dotted lines represent the TAD boundaries (as assessed by TADmap⁴⁹), the continuous vertical line the lncRNA locus and inter-TAD boundary, and the horizontal continuous line a log2Fold-change equal to 0. The dots represent individual genes, with the significant ones highlighted. **(G)** MA plot showing the changes in genome-wide gene expression in cells transfected with GapmeRs targeting *EPB41L4A-AS1*. **(H)** Same as (G), but with GapmeRs targeting *EPB41L4A*. **(I)** GO enrichment analysis for the upregulated (left) and downregulated (right) genes after *EPB41L4A-AS1* KD. All experiments were performed in n=3 biological replicates, except UMI-4C with n=2, with the error bars in the barplots representing the standard deviation. ns - P>0.05; * - P<0.05; ** - P<0.01; *** - P<0.001 (two-sided Student's t-test). A gene was considered to be differentially expressed if both adjusted P<0.05 and |log2Fold-change| >0.41 (corresponding to a change of 33%).

digital mode). The *cis*-regulated genes reduced by *EPB41L4A-AS1* KD included *NREP*, a gene important for brain development, whose homolog was downregulated by genetic manipulations of regions homologous to the lncRNA locus in mice⁴⁷. Depletion of *EPB41L4A-AS1* thus affects several genes in its vicinity.

Downregulation of *EPB41L4A-AS1* led to broad changes in gene expression beyond the proximal TADs (Fig. S5D [↗](#)). A total of 2,364 genes were differentially expressed (1,110 up, 1,254 down, adjusted $P < 0.05$ and $|\log_2\text{Fold-change}| > 0.41$) (Fig. 2G [↗](#)), in contrast to only 104 (36 up, 68 down) when *EPB41L4A* was targeted (Fig. 2H [↗](#)). A total of 25 (69%) of the upregulated, and 22 (32%) of the downregulated genes after *EPB41L4A* KD were also regulated in the same direction by the KD of *EPB41L4A-AS1* (Fig. S5E [↗](#), $P < 10^{-15}$ for both). In agreement with studies on *EPB41L4A* function^{34,35}, these genes were enriched with those functioning in cell-cell adhesion and cell surface receptor signaling, although with a borderline significance (Fig. S5F [↗](#)). Most of the consequences of the depletion of *EPB41L4A-AS1* are thus not directly explained by changes in *EPB41L4A* levels. An additional *trans*-acting function for *EPB41L4A-AS1* would therefore be consistent with its high expression levels compared to most lncRNAs detected in MCF-7 (Fig. S5G [↗](#)). To strengthen these findings, we have transfected MCF-7 cells with LNA1 and a second control GapmeR (NT2), as well as the previous one (NT1) and LNA2, and sequenced the polyadenylated RNA fraction as before. Notably, the expression levels (in FPKMs) of the replicates of both control samples are highly correlated with each other (Fig. S6A [↗](#)), and the global transcriptomic changes triggered by the two *EPB41L4A-AS1*-targeting LNAs are largely concordant (Fig. S6B and S6C [↗](#)). Because of this concordance and the cleaner (i.e., no CDKN1A upregulation) readout in LNA2-transfected cells, we focused mainly on these cells for subsequent analyses.

Several GO terms were found to be enriched among both up- and down-regulated genes following *EPB41L4A-AS1* KD. Upregulated genes were linked to cell-cell adhesion, inflammatory response and JAK/STAT/ERK pathway, and the downregulated ones to cell cycle, division and DNA repair, among others (Fig. 2I [↗](#)). When considering cellular components, the downregulated genes were associated with ribonucleoprotein and spliceosomal complexes, as well as to ribosome subunit and, interestingly, the nucleolus (Fig. S7A and S7B [↗](#)). Thus, since *EPB41L4A-AS1* is a snoRNA-host gene (SNHG) and snoRNAs typically localize to and act in the nucleolus, we aimed to investigate whether this lncRNA might interact with some factors that can explain the observed enrichment in nucleolar-associated genes.

EPB41L4A-AS1 binds SUB1 and affects the expression of its target genes

In order to identify potential factors that might be associated with *EPB41L4A-AS1*, we inspected protein-RNA binding data from the ENCODE eCLIP dataset⁵⁶. The exons of the *EPB41L4A-AS1* lncRNA were densely and strongly bound by SUB1 (also known as PC4) in both HepG2 and K562 cells (Fig. 3A [↗](#)). SUB1 interacts with all three RNA polymerases and was reported to be involved in transcription initiation and elongation, response to DNA damage, chromatin condensation^{57–60}, telomere maintenance^{61,62} and rDNA transcription⁶³. SUB1 normally localizes throughout the nucleus in various cell lines, yet staining experiments show a moderate enrichment for the nucleolus (source: Human Protein Atlas; <https://www.proteinatlas.org/ENSG00000113387-SUB1/subcellular> [↗](#))⁶³. Another nucleolus-related protein, NPM1, which binds C/D-box snoRNAs and directs 2'-O-methylation of rRNAs⁶⁴, had a single moderately confident binding site within *EPB41L4A-AS1* intron overlapping *SNORA13* (Fig. 3A [↗](#)). Interestingly, both *SUB1* and *NPM1* mRNAs were significantly downregulated following *EPB41L4A-AS1* KD (Fig. 3B [↗](#)).

We hypothesized that loss of *EPB41L4A-AS1* might affect SUB1, either via the reduction in its expression or by affecting its functions. We stratified SUB1 eCLIP targets into confidence intervals, based on the number, strength and confidence of the reported binding sites. Indeed, eCLIP targets of SUB1 (from HepG2 cells profiled by ENCODE) were significantly downregulated following *EPB41L4A-AS1* KD in MCF-7, with more confident targets experiencing stronger downregulation (Fig. 3C [↗](#)). Importantly, this still holds true when controlling for gene expression levels (Fig. S7C [↗](#)), suggesting that this negative trend is not due to differences in their baseline expression.

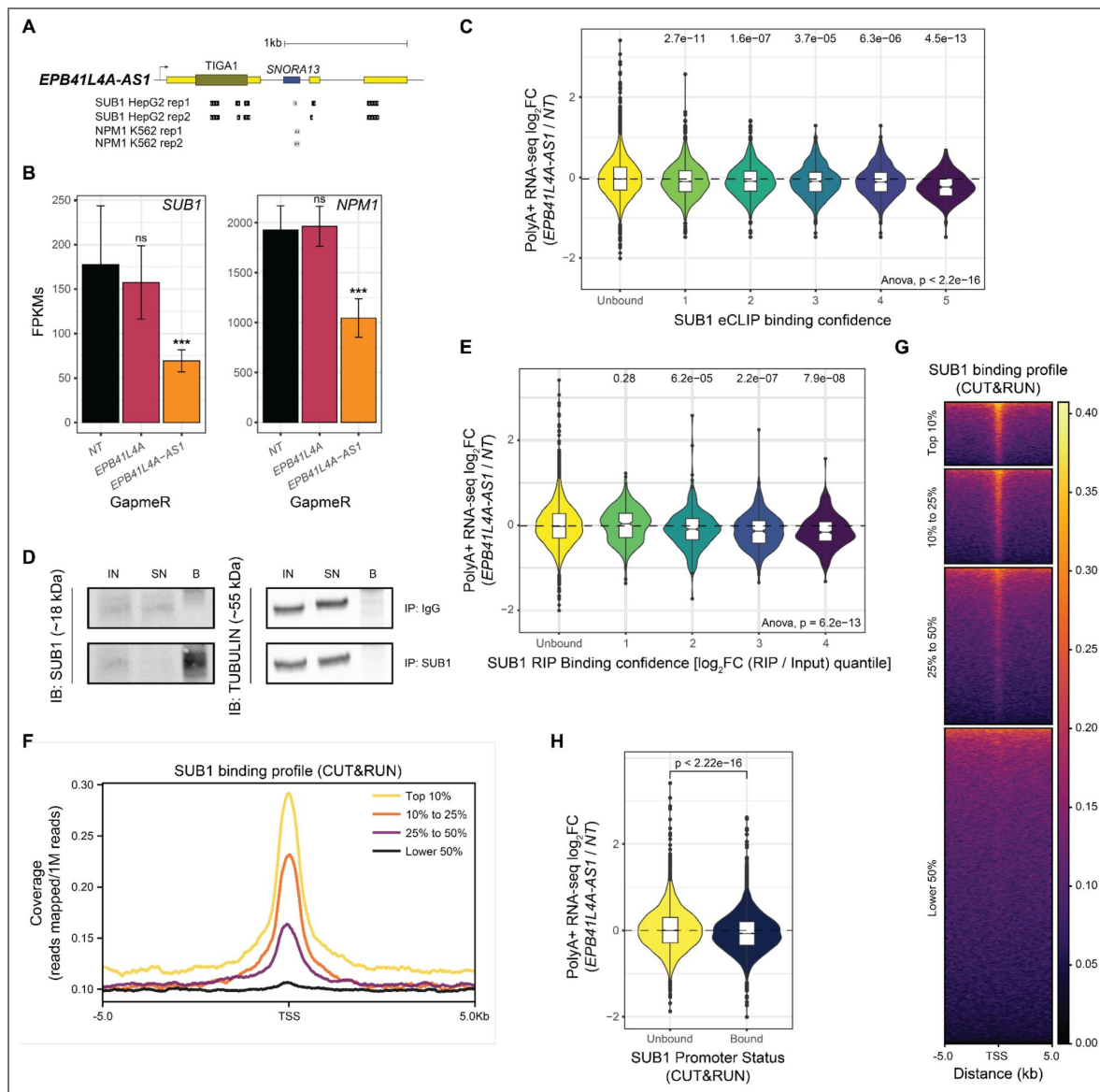


Figure 3. SUB1 interacts with *EPB41L4A-AS1* and affects gene expression at the DNA and RNA levels.

(A) Schematics of the *EPB41L4A-AS1* locus with tracks depicting the eCLIP peaks for both SUB1 and NPM1 (source: ENCODE). (B) Average expression levels (in FPKMs) of *SUB1* and *NPM1* in cells transfected with GapmeRs targeting either *EPB41L4A-AS1* or *EPB41L4A*, with the error bars representing the standard deviation across the $n = 3$ replicates. DESeq2 adjusted P-values compared to control GapmeR are also reported. (C) Changes in gene expression upon *EPB41L4A-AS1* KD of the genes ranked by the SUB1 eCLIP binding confidence. (D) Western blot following RIP using either a SUB1 or IgG antibody. TUBULIN was used as a negative control, IN - input, SN - supernatant/unbound, B - bound. (E) Same as (C), but with genes ranked by their enrichment in the RIP data (log₂FC RIP / Input). (F) Metagene profile around TSS of the normalized SUB1 CUT&RUN signal, stratified by gene expression levels in MCF-7 cells. (G) Heatmap of SUB1 reads coverage around the TSSs of all human genes, stratified by gene expression levels in MCF-7 cells. (H) Changes in gene expression upon *EPB41L4A-AS1* KD with GapmeRs for genes with and without a high-confidence SUB1 peak in their TSS. All experiments were performed in $n = 3$ biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA p-value is also reported. In all cases, ns - $P > 0.05$; * - $P < 0.05$; ** - $P < 0.01$; *** - $P < 0.001$.

To obtain SUB1-associated transcripts in MCF-7 cells, we performed a native RNA immunoprecipitation followed by sequencing of polyA⁺ RNAs (RIP-seq) (Fig. 3D [↗](#), S7D and S7E [↗](#)). As expected, *EPB41L4A-AS1* precipitated with SUB1, as did other top hits from the eCLIP data (Fig. S7D [↗](#)). A total of 1,470 genes were enriched in the IP fraction (adjusted $P < 0.05$ and $|\log_2\text{Fold-change}| > 0.41$) (Fig. S7E [↗](#)), and our MCF-7 RIP data was in agreement with that of the eCLIP in HepG2 cells (Fig. S7F [↗](#)). Similarly to what was done with the eCLIP data, we divided the RIP-defined targets into confidence intervals based on the significance and overall enrichment in the IP fraction. Strikingly, SUB1-associated transcripts were downregulated upon *EPB41L4A-AS1* KD (Fig. 3E [↗](#)), similar to what was observed for the eCLIP-based targets. Since SUB1 is also a transcriptional regulator associated with chromatin, we performed CUT&RUN to uncover the genome-wide chromatin occupancy of SUB1 in MCF-7 cells (Table S3). SUB1 binding was found to be enriched around TSSs of active genes (Fig. 3F and 3G [↗](#)), and among its 835 high-confidence genomic peaks, 418 (~50%) overlapped with the promoters of 411 genes. Of these genes, transcripts from only 56 were also bound by SUB1 at the RNA level, suggesting that SUB1 largely targets distinct gene sets at the DNA and RNA levels. Genes with promoter occupancy of SUB1 were enriched for biological processes related to genome maintenance, stress response, and nuclear transport (Fig. S7G [↗](#)), which is in line with previous studies^{57–60}. These genes were also slightly, yet highly significantly downregulated following *EPB41L4A-AS1* KD (Fig. 3H [↗](#)). Taken together, *EPB41L4A-AS1* interacts with SUB1 and its KD led to both a decrease in SUB1 mRNA level and the concomitant downregulation of SUB1 targets on chromatin and at the RNA level, positioning this lncRNA as a modulator of SUB1 activity.

Global de-regulation of snoRNAs and histone mRNA expression upon loss of *EPB41L4A-AS1*

Since many classes of noncoding RNAs—including those localizing to the nucleolus, such as snoRNAs—are not polyadenylated, we next sequenced total rRNA-depleted RNAs from *EPB41L4A-AS1* KD and control MCF-7 cells. As expected, using this approach we could quantify various RNA species lacking polyA tails, such as small nuclear RNAs (snRNAs), snoRNAs and histone mRNAs (Fig. 4A [↗](#)), and could also better quantify intronic expression. Consistent with a *cis*-acting regulation at the transcriptional level, both the exonic and intronic reads of *EPB41L4A* were significantly reduced after *EPB41L4A-AS1* downregulation (Fig. S8A [↗](#)). When considering different transcript classes, we observed a marked upregulation of most snRNAs and snoRNAs and a significant downregulation of histone mRNAs (Fig. 4A [↗](#)). To quantify SNHG de-regulation at higher resolution, we created a custom annotation in which we divided each SNHG into exonic regions, the snoRNA-encoding region, regions in the snoRNA-containing introns that are found upstream ('pre-') and downstream ('post-') of the snoRNA, and other snoRNA-less introns ('Other') of the SNHG transcripts, while considering the various isoforms annotated for each SNHG (see Methods) (Fig. 4B [↗](#)). We then quantified the levels of each region in control and KD conditions. Intriguingly, while SNHG exons and unrelated introns were not affected by *EPB41L4A-AS1* KD, read coverage was increased over both the snoRNAs and the flanking intronic regions (Fig. 4C [↗](#)). These changes were shared among all the snoRNA classes (Fig. S8B [↗](#)) and were concordant within individual introns when both (pre- and post-) had enough coverage to be quantified (Fig. S8C [↗](#)). Notably, the effect sizes of these changes were small, and they were not evident when plotting metagenes of these regions—that show the median coverage level—which is close to zero even in rRNA-depleted data (Fig. S8D [↗](#)), therefore suggesting that snoRNA processing remains an efficient process even in cells depleted of *EPB41L4A-AS1*. Therefore, most of the effect we observe is likely in the increased accumulation of the mature and stable snoRNAs, eventually hinting that *EPB41L4A-AS1* perturbations specifically affect the snoRNA production and/or stability rather than SNHG transcription.

To test if the effects on snoRNA biogenesis are related to the changes in SUB1 expression and/or activity, we sequenced the rRNA-depleted RNA from MCF-7 cells in which *SUB1* was depleted by siRNAs (Fig. S8E and S8F [↗](#)). We identified a total of 243 upregulated and 154 downregulated genes (Fig. S8G [↗](#)), of which 41 (~17%) and 74 (~48%) were concordantly affected by *EPB41L4A-AS1*

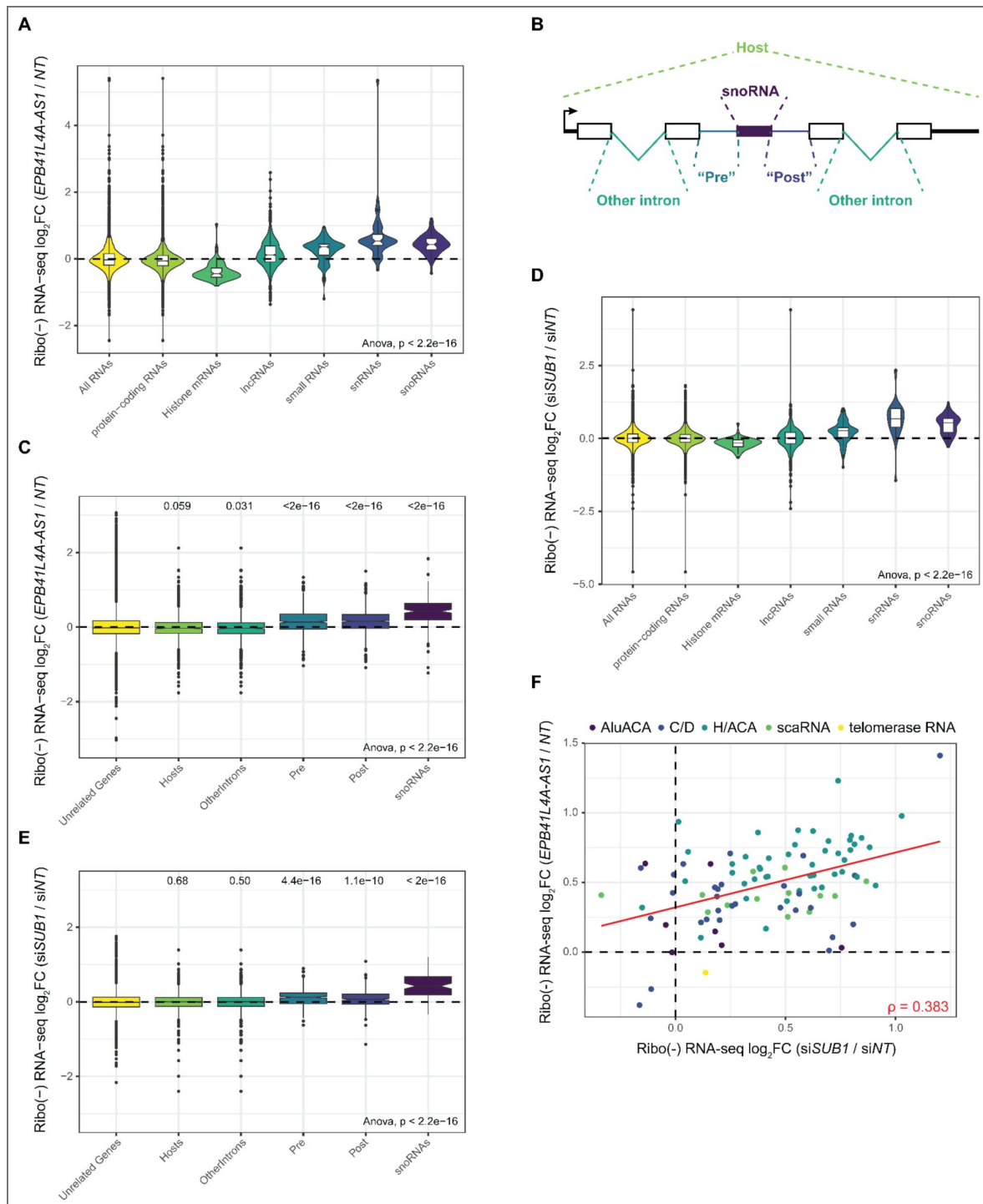


Figure 4. EPB41L4A-AS1 and SUB1 depletion results in a widespread accumulation of mature snoRNAs.

(A) Changes in gene expression upon EPB41L4A-AS1 KD with GapmeRs of the indicated RNA classes. (B) Schematics depicting the different regions which were separately quantified in each SNHG. (C) Changes in RNA-seq read coverage in different regions upon EPB41L4A-AS1 KD with GapmeRs. (D) As in (B) for SUB1 KD with siRNAs of the indicated RNA classes. (E) as in (C) for SUB1 KD. (F) Correspondence between changes in snoRNAs expression after EPB41L4A-AS1 and SUB1 KD. The color indicates the different snoRNA classes, and Spearman's correlation coefficient is shown. All experiments were performed in $n = 3$ biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA P-value is also reported.

depletion, and the magnitudes of the genome-wide responses were moderately, but significantly correlated (Spearman's $R = 0.138$, $P < 10^{-15}$). Strikingly, *SUB1* KD led to a similar increase in snoRNA levels and a decrease in histone mRNAs (Fig. 4D, 4E and 5H). Interestingly, the changes in snoRNAs expression were highly concordant with those driven by the KD of *EPB41L4A-AS1* (Fig. 4F, $R = 0.383$) and intronic regions flanking the snoRNAs were also increased, suggesting that *EPB41L4A-AS1* and *SUB1* affect the abundance of the same pool of snoRNAs in MCF-7 cells through likely a shared mechanism.

Lastly, we tested whether the effects of *EPB41L4A-AS1* perturbation could be a generic outcome of a noncoding SNHG (ncSNHG) perturbation. To this end, we used GapmeRs to deplete *GAS5*, another abundant SNHG which encodes 11 snoRNAs within its introns (Fig. S9A and S9B). As expected from the various functions associated with this ncSNHG, *GAS5* KD led to substantial changes in gene expression (Fig. S9C), including a marked suppression of histone mRNA transcription, in line with its anti-proliferative function (Fig. S9D). However, snoRNAs or their host introns were largely unaffected (Fig. S9D-F), suggesting that the effect of *EPB41L4A-AS1* on snoRNA expression is not a common feature of ncSNHGs.

Depletion of *EPB41L4A-AS1* affects *SUB1* distribution in the nucleus

The results described so far point to *EPB41L4A-AS1* being important for *SUB1* function. To test whether *EPB41L4A-AS1* is important for its proper intracellular distribution, we immunostained for *SUB1* and *NPM1* in MCF-7 cells transfected with either control or *EPB41L4A-AS1*-targeting GapmeRs. A substantial change in subcellular localization could be observed for both proteins (Fig. 5A and 5B). Specifically, *SUB1* normally displays a pan-nuclear localization, sometimes slightly enriching into DAPI-low areas corresponding to the nucleoli (Fig. 5A and 5C). Upon *EPB41L4A-AS1* depletion, *SUB1* shifts towards a stronger nucleolar localization while retaining a presence throughout the nucleus. *NPM1* instead regularly shows enrichment inside the nucleoli; however, in GapmeR-transfected cells, this is accompanied by the appearance of globular aggregates within the nucleus (Fig. 5B and 5D), reminiscent of the previously reported nucleolar damage/stress response.

To test if the observed patterns of changes in *SUB1* and *NPM1* distribution are triggered by nucleolar stress, we exposed MCF-7 cells to CX-5461, an RNA polymerase I inhibitor that is currently being evaluated as a therapeutic agent for solid tumors with DNA repair deficiencies. This led to a similar *NPM1* pattern as that triggered by *EPB41L4A-AS1* depletion as early as 1h after treatment with the higher dose, and the change was more pronounced in later time points (Fig. S10A and S10B). *SUB1* localization also changed, but only at the last time point (24hrs) (Fig. S10C and S10D). Lastly, we tested whether *SUB1* depletion could affect *NPM1* distribution, and indeed *SUB1* KD via siRNAs made *NPM1* assume a similar pattern to that caused by *EPB41L4A-AS1* loss (Fig. 5F and 5G), suggesting that *SUB1* acts upstream of *NPM1* and this lncRNA might modulate this regulation.

EPB41L4A-AS1 perturbations do not affect steady-state levels of *SNORA13*

A recent study reported an independent function for *SNORA13*, the snoRNA hosted by *EPB41L4A-AS1* in ribosomal assembly and prevention of nucleolar stress in BJ fibroblasts and mice. We therefore wondered if the effects we observe in our experimental system result from changes in expression levels of this snoRNA. *SNORA13* levels did not significantly change in RT-qPCR analysis of perturbed cells (Fig. 2B), or in rRNA-depleted RNA-seq data (Fig. 6A). We also performed a Northern blot for *SNORA13*, which also did not indicate changes in expression level or changes in the RNA size of *SNORA13* upon GapmeR-mediated depletion of *EPB41L4A-AS1* (Fig. 6B). To complement these data, we assessed the levels of the 18S:1248m¹acp³Ψ modification, which includes a pseudouridylation guided by *SNORA13*. We used an aminocarboxyl propyl reverse transcription (aRT)-PCR assay as previously described, and found no difference in HinfI cleavage between control cells and those transfected with either *EPB41L4A-AS1*- or *EPB41L4A*-targeting GapmeRs (Fig. 6C), suggesting that the MACP modification is not significantly affected as well. In

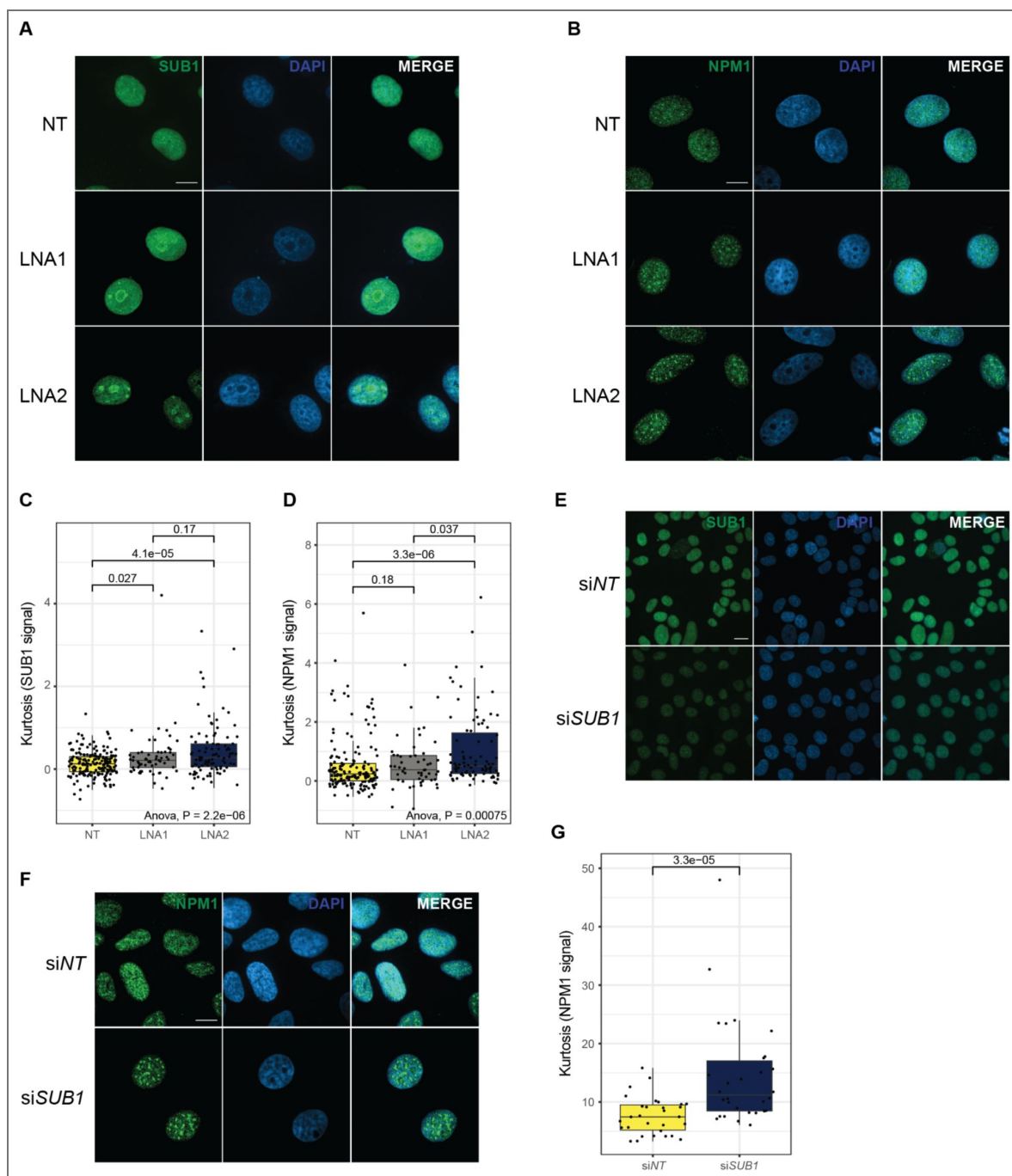


Figure 5. Loss of *EPB41L4-AS1* results in an altered nuclear patterning of SUB1 and NPM1.

(A) Representative immunofluorescence images for SUB1 after *EPB41L4 A-AS1* depletion with two distinct GapmeRs (scale bar = 20 μ m, 100X magnification). (B) Same as in (A), but for NPM1. (C) Quantification of the kurtosis of SUB1 nuclear signal in the indicated conditions. (D) Same as in (C), but for NPM1. (E) Representative immunofluorescence images for SUB1 after *SUB1* KD with siRNAs (scale bar = 20 μ m, 60X magnification). (F) Same as in (E), but for NPM1. (G) Same as in (D), but after *SUB1* depletion. All experiments were performed in $n = 3$ biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA P-value is also reported. In each boxplot, points represent individual measurements (cell nuclei).

addition, variant calling at 18S:1248 exploiting the residual rRNA reads in our polyA⁺ RNA-seq dataset revealed no distinction in allele frequency between control and transfected cells (Fig. 6D), concordant with the aRT-PCR results. In conclusion, we show that in our experimental system—MCF-7 cells transiently transfected with GapmeRs targeting its host—we do not observe changes in steady state *SNORA13* levels nor abundance of the only known RNA modification driven by this snoRNA. Therefore, we postulate that the observed endophenotypes are likely due to changes in the *EPB41L4A-AS1* lncRNA levels and its cooperation with SUB1.

A full-length, intron- and *SNORA13*-containing *EPB41L4A-AS1* is required to partially rescue snoRNA expression

We reasoned that if *EPB41L4A-AS1* indeed regulates snoRNA expression, this function likely occurs *in trans*, because none of the genes in the vicinity of the *EPB41L4A-AS1* locus are known to be involved in snoRNA biogenesis or stability. To test this, we performed rescue experiments in which we co-transfected lncRNA-targeting GapmeRs with different GapmeR-resistant *EPB41L4A-AS1* variants expressed from a plasmid driven by a constitutive CMV promoter. We confirmed that the transfected vector is well spliced (Fig. 6E), and results in an increase in both *EPB41L4A-AS1* and *SNORA13* expression (Fig. 6F, left).

We then assessed both KD dynamics and the abundance of snoRNAs selected based on their sensitivity to *EPB41L4A-AS1* KD in the RNA-seq data at 24, 48 and 72 hours after transfection. *EPB41L4A* expression decreased concomitantly with that of *EPB41L4A-AS1*, in line with the transcription of the lncRNA regulating this gene in *cis* (Fig. 6F, right). The expression of most *EPB41L4A-AS1*-sensitive snoRNAs increased as early as 24 hours after transfection, and they generally remained upregulated throughout the next two days (Fig. 6G). We then designed four different *EPB41L4A-AS1* expression vectors to attempt a rescue: 1) full length, including introns; 2) full length, including introns but without *SNORA13* (Δ *SNORA13*); 3) full length, including introns but with either the H or ACA *SNORA13* boxes deleted (Δ H and Δ ACA); and 4) *SNORA13* alone (Fig. 6H). We then compared gene expression between cells transfected with the overexpressing vectors (OE) or an empty control (EV). We were able to obtain a partial rescue of snoRNA expression only when co-transfecting the full length *EPB41L4A-AS1*, and not any other vector not expressing any or just a mutated *SNORA13* (Fig. 6I), indicating that the *SNORA13* sequence is involved in the process. However, the expression of a stand-alone *SNORA13*, using the expression vector used and validated previously²⁷, was incapable to drive any rescue of the snoRNAs abundance (Fig. 6I). This suggests that the entire *EPB41L4A-AS1* transcript is required to counteract *EPB41L4A-AS1* knockdown, most likely because the intact *SNORA13* sequence is necessary to undergo the characteristic processing steps of a SNHG.

EPB41L4A-AS1 affects RNA metabolism of both the genes found in *cis* and of snoRNAs

To distinguish between the effects of *EPB41L4A-AS1* depletion on RNA synthesis and RNA decay, we performed SLAM-seq⁸⁰ using ribo-depleted total RNA in *EPB41L4A-AS1* KD cells (Table S4). Following 48 hours after control or *EPB41L4A-AS1*-targeting GapmeRs transfection, we collected RNA at 0, 4, 8 and 24 hours post-addition of 4sU to the media, while replenishing fresh 4sU at each time point (see Methods) (Fig. 7A). *EPB41L4A* showed decreased synthesis rate and unchanged half-life (Fig. 7B), as expected from a gene regulated at the level of transcription. As a group, genes upregulated after *EPB41L4A-AS1* KD in the regular steady-state RNA-seq data had substantially higher synthesis rates, and those downregulated had lower synthesis rates, when compared to unchanged ones ($P=3.25\times 10^{-14}$ for the up-regulated genes and $P=0.002$ for the down-regulated genes, Mann-Whitney test, Fig. 7C), whereas only a minor difference was observed in terms of half-lives ($P=0.011$ for the up-regulated genes and $P=1.82\times 10^{-8}$ for the down-regulated genes, Mann-Whitney test, Fig. 7D). This suggests that the *trans*-regulated genes are also mostly regulated at the level of their transcription. Additionally, when comparing the effects across genes in the two neighboring TADs (“Cis”), other TADs on the same chromosome (“Chr5”), or other

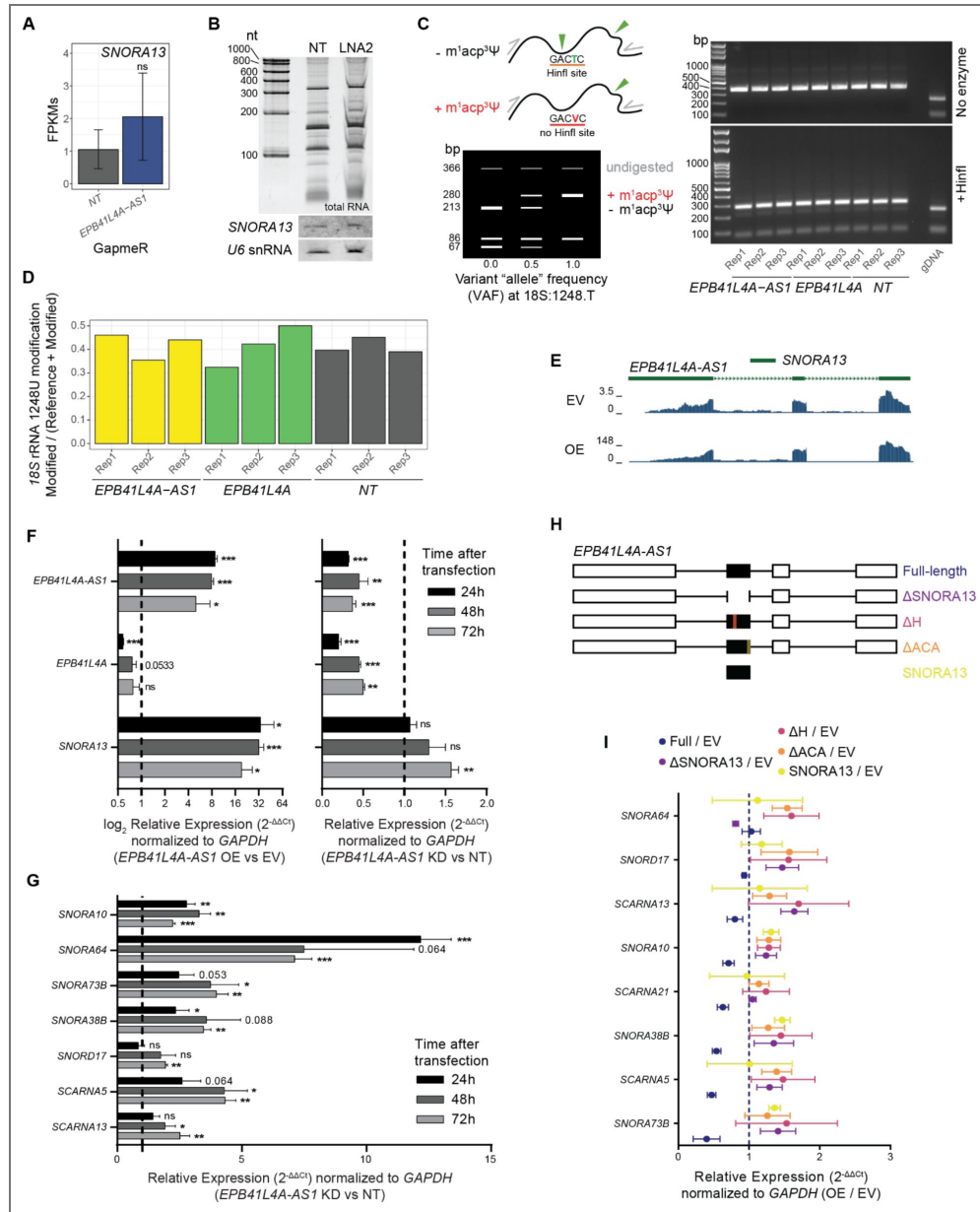


Figure 6. *EPB41L4A-AS1* but not the expression of *SNORA13* is required for proper snoRNAs expression.

(A) Average expression levels (in FPKM) of *SNORA13* in cells transfected with GapmeRs targeting *EPB41L4A-AS1* (rRNA-depleted RNA-seq), with the error bars representing the standard deviation across the three replicates. DESeq2 adjusted P-values compared to control GapmeR are also reported. (B) Northern blot for *SNORA13* in cells transfected with GapmeRs targeting *EPB41L4A-AS1*. *U6* was used as a loading control, and total RNA stain is also shown on top. (C) Schematics of the expected fragment sizes following HinfI digestion in the aRT-PCR assay (left), and agarose gel following HinfI digestion of the PCR-amplified 18S rRNA in cells transfected with GapmeRs targeting either *EPB41L4A-AS1* or *EPB41L4A* (right). No enzyme and genomic DNA (gDNA) were used as negative controls. (D) Allele frequency at 18S:1248 in the polyA+ RNA-seq dataset, using rRNA reads. (E) UCSC Genome Browser view of the *EPB41L4A-AS1* locus, with tracks showing the rRNA-depleted RNA-seq coverage of *EPB41L4A-AS1* OE and control cells. (F) RT-qPCR for the indicated genes upon *EPB41L4A-AS1* KD with GapmeRs (right) or overexpression with a full-length unspliced vector (left) over the course of three days post transfection. (G) RT-qPCR for the indicated snoRNAs upon *EPB41L4A-AS1* KD with GapmeRs over the course of three days post transfection. (H) Schematics of the different *EPB41L4A-AS1* and *SNORA13* overexpressing vectors used in the rescue experiments. (I) The ratio between changes in gene expression detected by RT-qPCR in cells transfected with an *EPB41L4A-AS1* or *SNORA13* overexpressing vector vs control cells. All experiments were performed in n = 3 biological replicates, with the error bars in the barplots and forest plots representing the standard deviation. In all cases, ns - P>0.05; * - P<0.05; ** - P<0.01; *** - P<0.001 (two-sided Student's t-test).

chromosomes (“Trans”), the genes proximal to *EPB41L4A-AS1* specifically exhibit a reduced synthesis rate compared to the others (Fig. S11A), but no changes in their half-lives (Fig. S11B).

Compared to their hosts and other genes, snoRNAs were, as expected, relatively stable in both KD and control conditions (Fig. 7E and 7F), while the intronic regions exhibited high turnover. When comparing *EPB41L4A-AS1* KD to the control GapmeRs, a slight increase in the synthesis rates of different parts of SNHG was observed. The snoRNA half-life is difficult to measure accurately due to their relative stability and their short length, which means that the number of incorporated 4sU that can be detected is much lower than that in longer transcripts. With these caveats in mind, there appeared to be a relative increase in snoRNA stability compared to a reduction observed in other genes and other parts of the SNHG (Fig. 7G and 7H). Altogether, this indicates that the increase of snoRNAs level driven by *EPB41L4A-AS1* knockdown is primarily due to the increased transcription of their hosts, with a possible additional component of relative stabilization.

EPB41L4A-AS1 loss affects cell proliferation and invasion

Lastly, we aimed to assess the phenotypic consequences of *EPB41L4A-AS1* loss. As mentioned above, the genes affected by the KD were found to be enriched for GO terms related to cell adhesion, movement and division (Fig. 2I). Thus, we measured cell migration and proliferation by performing wound healing and cell counting assays in cells with reduced *EPB41L4A-AS1* levels. Cells transfected with *EPB41L4A-AS1* targeting GapmeRs exhibited significantly higher migration rates, compared to controls (Fig. 8A and 8B), with cells bordering the wounded region displaying a flattened and translucent morphology (Fig. 8A). This higher migration is not due to higher cellular proliferation/division, as cells with reduced lncRNA levels proliferate less (Fig. 8C), and this deficiency could be rescued by co-transfecting a vector encoding the GapmeR-resistant, full-length *EPB41L4A-AS1* (Fig. 8C), suggesting that this modulation takes place in *trans*.

These findings suggest that proper *EPB41L4A-AS1* expression is required for cellular proliferation, whereas its deficiency results in the onset of more aggressive and migratory behavior, likely linked to the increase of the gene signature of epithelial to mesenchymal transition (EMT) (Fig. 8D). Because EMT is not characterized by a unique gene expression program and rather involves distinct and partially overlapping gene signatures⁸¹, we checked the expression level of marker genes linked to different types of EMTs (Fig. 8E). The most upregulated gene in Fig. 8D is *TIMP3*, a matrix metalloproteinase inhibitor associated with a particular EMT signature that is less invasive and more profibrotic (EMT-T2)⁸¹. Interestingly, we observed a stark upregulation of other genes linked to EMT-T2, such as *TIMP1*, *FOSB*, *SOX9*, *JUNB*, *JUN* and *KLF4*, whereas *MPP* genes (linked to EMT-T1, which is highly proteolytic and invasive) are generally downregulated or not expressed. This suggests that the downregulation of *EPB41L4A-AS1* is primarily linked to a specific EMT program (EMT-T2), and future studies aimed at uncovering the exact mechanisms and relevance will shed light upon a possible therapeutic potential of this lncRNA.

Discussion

We show here that the *EPB41L4A-AS1* gene locus harbors a pleiotropic gene that, within a single biological system—MCF-7 breast cancer cells—performs diverse functions (Fig. 9). At the site of its own transcription, which overlaps a strong TAD boundary, *EPB41L4A-AS1* is required to maintain expression of several adjacent genes, which are regulated at the level of transcription. Strikingly, the promoter of *EPB41L4A-AS1* ranks in the 99.8th percentile of the strongest TAD boundaries in human H1 embryonic stem cells^{47,82}. It features several CTCF binding sites (Fig. 2A), and in MCF-7 cells, we demonstrate that it blocks the propagation of the 4C signal between the two flanking TADs (Fig. 1F). Future studies will help elucidate how *EPB41L4A-AS1* transcription and/or the RNA product regulate this boundary. So far, we found that *EPB41L4A-AS1* did not affect CTCF binding to the boundary, and while some peaks in the vicinity of *EPB41L4A-AS1* were significantly affected by its loss, they did not appear to be found near genes that were dysregulated by its KD (Fig. S11C). We also found that the KD of *EPB41L4A-AS1* for 72 hours—

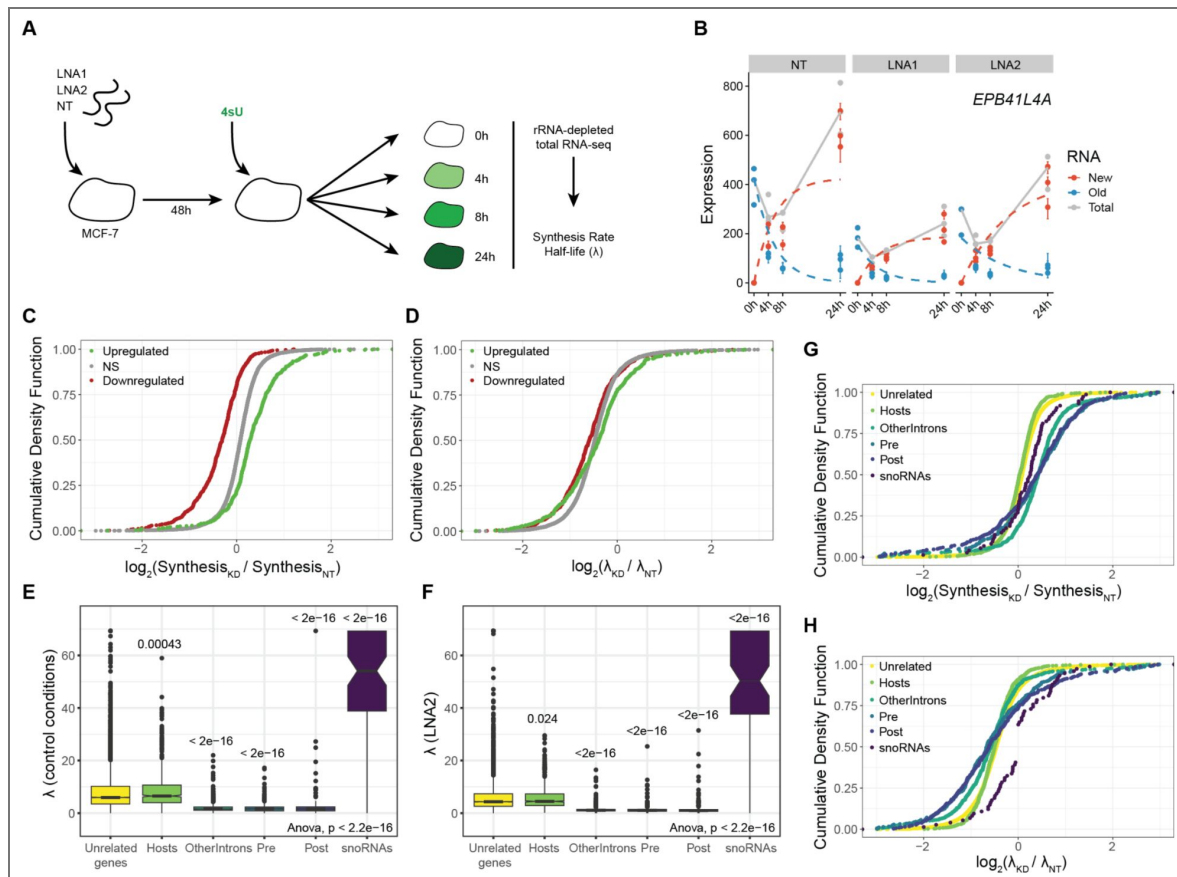


Figure 7. The increased abundance of snoRNAs is primarily due to their hosts' increased transcription and stability.

(A) Workflow of the SLAM-seq experiment. MCF-7 cells were transfected with the indicated GapmeRs for 48 hours, after which the media was replaced with media containing 4sU and the cells were harvested at different time points. (B) Fitted model depicting the synthesis and decay rates of the *EPB41L4A* mRNA upon *EPB41L4A-AS1* KD with GapmeRs. (C) Changes in synthesis rate upon *EPB41L4A-AS1* KD with GapmeRs for the indicated group of genes. (D) Same as in (C), but for changes in half lives. (E) Half lives in control conditions of the indicated SNHG regions as described in Fig. 4B. P-values refer to the comparison with the unrelated genes. (F) Same as in (E), but after *EPB41L4A-AS1* KD with LNA2. (G) Changes in synthesis rate upon *EPB41L4A-AS1* KD with GapmeRs for the indicated SNHG regions. (H) Same as in (G), but for changes in half lives. All experiments were performed in $n = 3$ biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA P-value is also reported. In all cases, ns - $P > 0.05$; * - $P < 0.05$; ** - $P < 0.01$; *** - $P < 0.001$.

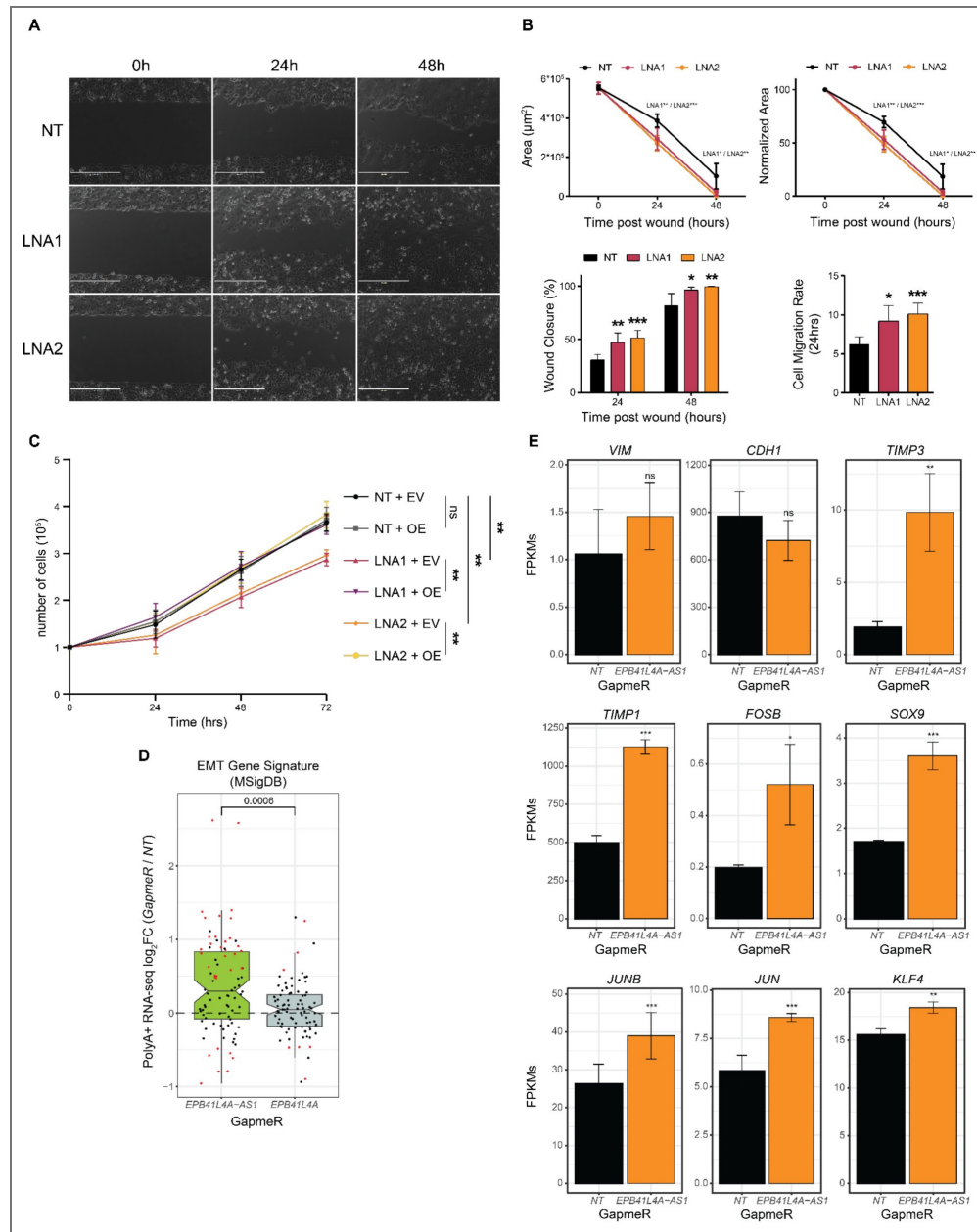


Figure 8. Cells with reduced *EPB41L4A-AS1* expression display reduced proliferation and increased invasion capacity.

(A) Representative brightfield images of the wound at the indicated time points and conditions. (B) Wound area (top-left), normalized wound area (top-right), closure percentage (bottom-left) and migration rate (bottom-right) at the indicated time points and conditions. The significance was calculated by a two-sided Student's t-test. (C) Growth curve of the cells in the indicated conditions over the course of three days. The significance was calculated by a two-sided Student's t-test. (D) Changes in gene expression of the EMT signature genes (from MSigDB) after KD with GapmeRs targeting either *EPB41L4A-AS1* or *EPB41L4A* (polyA+ RNA-seq data). The significance of the comparison was computed by a Mann-Whitney test. (E) Average expression levels (in FPKMs) following *EPB41L4A-AS1* KD with GapmeRs of selected genes that have been previously linked to EMT, with the error bars representing the standard deviation across the $n = 3$ replicates. DESeq2 adjusted P-values compared to control GapmeR are also reported. All experiments were performed in $n = 3$ biological replicates. The error bars in the barplots represent the standard deviation. In the boxplot, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points. The points in the boxplot represent individual genes, and their color whether they were found to be significantly (adjusted $P < 0.05$ and $|\log_2\text{Fold-change}| > 0.41$) dysregulated (red) or not (black). In all cases, ns - $P > 0.05 = \text{ns}$; * - $P < 0.05 = *$; ** - $P < 0.01 = **$; *** - $P < 0.001 = ***$ (two-sided Student's t-test).

which depletes the RNA product, but may also affect the nascent RNA transcription^{53,54}—reduces the spatial contacts between the TAD boundary and the *EPB41L4A* promoter (Fig. 2E [↗](#) and Fig. 5J [↗](#)). Further elucidation of the exact functional entity needed for the *cis*-acting regulation will require detailed genetic perturbations of the locus, that are difficult to carry out in the polyploid MCF-7 cells, without affecting other functional elements of this locus or cell survival as we were unable to generate deletion clones despite several attempts.

A recent study found that *SNORA13* negatively regulates ribosome biogenesis in TERT-immortalized human fibroblasts (BJ-HRAS^{G12V}), by decreasing the incorporation of RPL23 into the maturing 60S ribosomal subunits, eventually triggering p53-mediated cellular senescence²⁷. In their experimental setting, *SNORA13* was constitutively reduced by either CRISPRi of the *EPB41L4A-AS1* locus for 7 days or its KO leaving the lncRNA sequence and expression unaltered, whereas our approach relies solely on transient knockdown for 48–72hrs that does not appreciably affect *SNORA13* (Fig. 1B [↗](#), 2B [↗](#), 6A and 6B [↗](#)), or the levels of the RNA modification that it guides (Fig. 6C and 6D [↗](#)). Therefore, while both studies implicate *EPB41L4A-AS1* in nucleolar biology and its perturbation in nucleolar stress, there is an apparent discrepancy between our results on the role that *SNORA13* plays. In our hands, the differential expression of snoRNAs and cell proliferation can be rescued by the expression of *EPB41L4A-AS1* with the snoRNA, and not by separate expression of *SNORA13*, suggesting the snoRNA alone is not playing a central role in the observed phenotypes. Also, the study by Cheng et al. showed that activity of the p53 pathway was less active in *SNORA13* knockout cells, with or without induction of oncogenic stress, whereas we see an opposite trend of some up-regulation of the p53 pathway upon transient *EPB41L4A-AS1* KD (Fig. S11D [↗](#)). We cannot however completely rule out potential contributions of this snoRNA to the phenotypes observed in this study. For instance, the phenotype of snoRNAs expression is partially rescued by *EPB41L4A-AS1* overexpression in *trans* only when a full-length, unspliced and *SNORA13*-containing vector is used, and not when the snoRNA is deleted, mutated or expressed alone (Fig. 6I [↗](#)). This would suggest that the processing of *EPB41L4A-AS1* is involved in its mode of action, although further experiments will be needed to address whether *SNORA13* or its DNA sequence are indeed required to regulate the expression of the snoRNAs. We note that compensatory effects are possible in cells that grow for extended periods of time without *SNORA13*.

Beyond its site of transcription, *EPB41L4A-AS1* associates with SUB1, an abundant protein linked to various functions, and these two players are required for the proper distribution of at least two nuclear proteins. Their dysregulation results in large-scale changes in gene expression, including up-regulation of snoRNA expression, mostly through increased transcription of their hosts, and possibly through a somewhat impaired snoRNA processing and/or stability. To further hinder our efforts in discerning between these two possibilities, the exact molecular pathways involved in snoRNAs biogenesis, maturation and decay are still not completely understood. One of the genes that most consistently was affected by *EPB41L4A-AS1* KD with GapmeRs is *MTREX* (also known as *MTR4*), that becomes downregulated at both the RNA and protein levels (Fig. S12A–C [↗](#)). Interestingly, *MTREX* is part of the NEXT and PAXT complexes⁸³, that target several short-lived RNAs for degradation, and the depletion of either *MTREX* or other complex members leads to the upregulation of such RNAs, that include PROMPTs, uaRNAs and eRNAs, among others. It is therefore tempting to hypothesize a role for *MTREX*-containing complexes in trimming and degrading those introns, and releasing the mature snoRNAs. Future studies specifically aimed at uncovering novel players in mammalian snoRNA biology will both conclusively elucidate whether *MTREX* is indeed involved in these processes.

Several features of the response to *EPB41L4A-AS1* resemble nucleolar stress, including altered distribution of NPM1^{74,75}. SUB1 was shown to be involved in many nuclear processes, including transcription⁵⁸, DNA damage response^{84,85}, telomere maintenance⁶¹, and nucleolar processes including rRNA biogenesis^{63,86}. Our results suggest a complex and multi-faceted relationship between *EPB41L4A-AS1* and SUB1, as *SUB1* mRNA levels are reduced by the transient (72 hours) KD of the lncRNA (Fig. 3B [↗](#)), the distribution of the protein in the nucleus is altered (Fig. 5A and 5C [↗](#)), while the protein itself is the most prominent binder of the mature *EPB41L4A-AS1* in

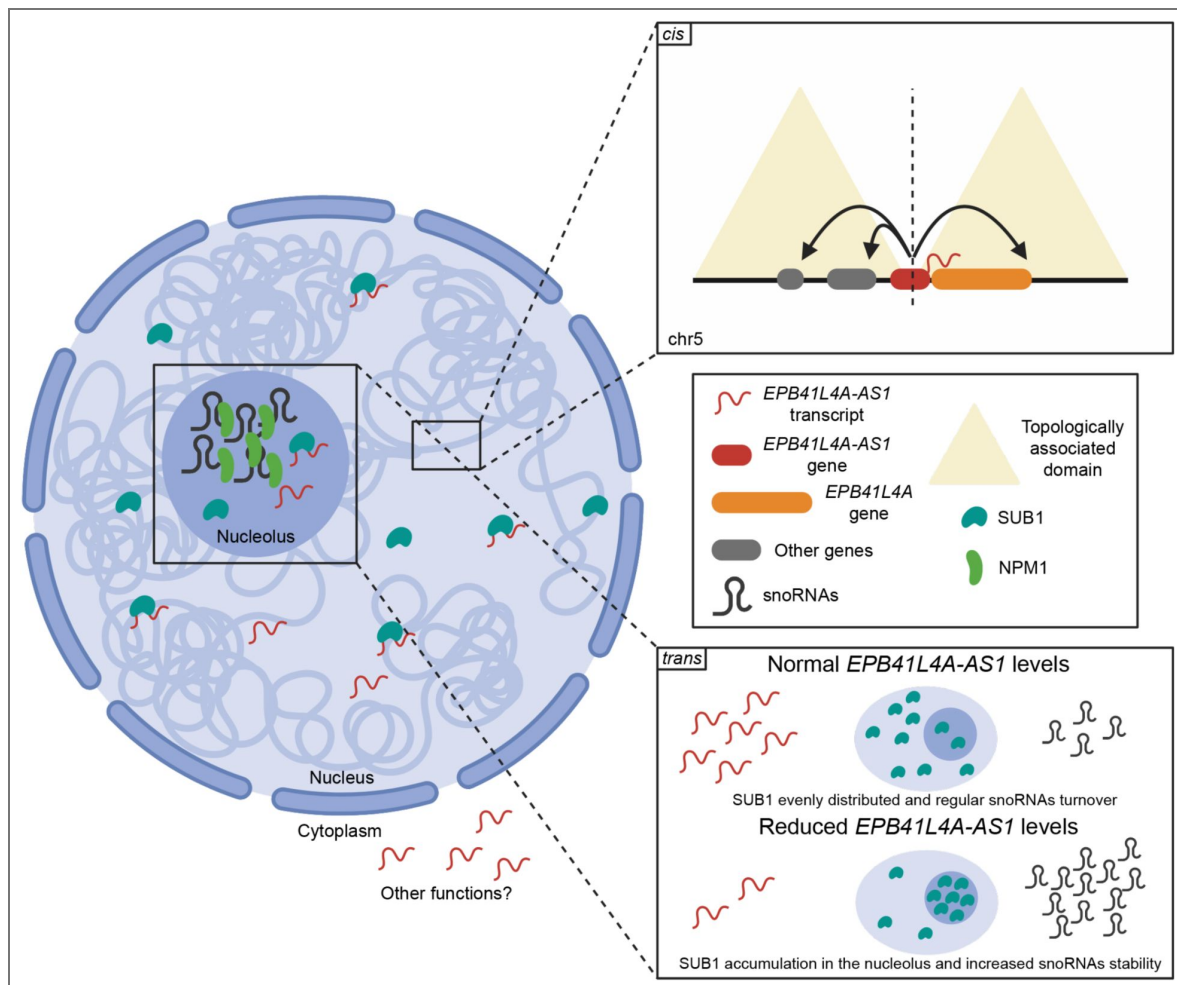


Figure 9. Proposed model of the *cis* and *trans* functions of the *EPB41L4A-AS1* lncRNA.

Under normal conditions, the *EPB41L4A-AS1* locus is transcribed and ensures the proper expression of several genes located in *cis*. In *trans*, it interacts with SUB1 which eventually is evenly distributed throughout the nucleus. When there are reduced levels of *EPB41L4A-AS1*, the expression of the genes in *cis* is reduced, and SUB1 accumulates in the nucleolus. As a consequence, snoRNAs abundance increases, likely due to their increased stability, and MCF-7 cells acquire a more invasive *in vitro* phenotype. Created with BioRender.com.

ENCODE eCLIP data (Fig. 3A). The most striking connection between *EPB41L4A-AS1* and SUB1 is the similar phenotype triggered by their loss (Fig. 4). We note that a recent study has shown that SUB1 is required for Pol I-mediated rDNA transcription in the nucleolus⁶³. In the presence of nucleolar SUB1, rDNA transcription proceeds as expected, but when SUB1 is depleted or its nucleolar localization is affected—by either sodium butyrate treatment or inhibition of KAT5-mediated phosphorylation at its lysine 35 (K35)—the levels of the 47S pre-rRNA are significantly reduced. In our settings, SUB1 enriches into the nucleolus following *EPB41L4A-AS1* KD; thus, we might expect to see a slightly increased rDNA transcription or no effect at all, given that SUB1 localizes in the nucleolus in baseline conditions as well. It is however difficult to determine which of the connections between these two genes is the most functionally relevant and which may be indirect and/or feedback interactions. For example, it is possible that *EPB41L4A-AS1* primarily acts as a transcriptional regulator of *SUB1* mRNA, or that its RNA product is required for proper stability and/or localization of the SUB1 protein, or that *EPB41L4A-AS1* acts as a scaffold for the formation of protein-protein interactions in which SUB1 is involved.

In the context of cancer, a systematic study of TCGA data, which is consistent with our analysis (Fig. S2) found that *EPB41L4A-AS1* is downregulated in tumors compared to tumor-adjacent normal tissues²⁹, in accordance with both the downregulation of the TIGA1 protein²⁹ and the reduction in the *SNORA13*-guided rRNA modification²⁶. The association with poor prognosis²⁹ is consistent with the observed increased invasiveness that we find in MCF-7 cells. Effects of *EPB41L4A-AS1* on migration were also reported in colorectal cancer cells³². The notion that cells with reduced *EPB41L4A-AS1* levels exhibit lower proliferation (Fig. 8C), yet increased invasion (Fig. 8A and 8B) is compatible with a function as an oncogene by promoting EMT (Fig. 8D and 8E). Cells undergoing this process may reduce or even completely halt proliferation/cell division, until they revert back to an epithelial state^{87,88}. Notably, downregulated genes following *EPB41L4A-AS1* KD are enriched in GO terms related to cell proliferation and cell cycle progression (Fig. 2I), whereas those upregulated for terms linked to EMT processes. Thus, while we cannot rule out a potential function as tumor suppressor gene, our data better fits the idea that this lncRNA promotes invasion, and thus, primarily functions as an oncogene.

We note that this is not the first instance in which this lncRNA has been investigated. Several recent studies have described the functions of *EPB41L4A-AS1* in various biological systems, highlighting its dysregulation in cancer^{29,32,89}, diabetes³¹, aging^{90,91}, and Alzheimer's disease (AD)⁹¹. Both p53 and PGC-1 α were shown to act upstream of *EPB41L4A-AS1* and regulate its expression²⁹. In these different systems, different pathways were described as regulated by this lncRNA, including glycolysis, glutamine metabolism²⁹, and autophagy⁹¹. Mechanistically, it was reported that this lncRNA regulates gene expression via its association with and upregulation of KAT2A (GCN5/GNC5L2)^{31,91}. These prior studies generally relied on candidate-gene or co-expression approaches to identify the genes regulated by *EPB41L4A-AS1*, and did not systematically examine its *cis*- and *trans*-acting effects, with the exception of a *cis*-acting regulation, that was briefly hinted in a preprint posted four years ago⁹², but not published so far to the best of our knowledge. These studies also did not systematically investigate the *trans*-regulated targets of *EPB41L4A-AS1*, did not separately deplete the nascent and the mature forms of the lncRNA, and did not attempt to decouple its functions from those of *SNORA13*.

Methods

Cell culture, transfection and treatments

MCF-7 cells were cultured in DMEM supplemented with 4.5 g/L of glucose, 4 mM L-Glutamine, 10% FBS, 1% PenStrep solution, grown in incubators maintained at 37°C with 5% CO₂, and were detached for passaging using a Trypsin (0.05%) – EDTA (0.02%) solution. DNA plasmid, siRNAs (final concentration 25 nM) and GapmeRs (50 nM) transfections were performed by using Lipofectamine 3000 (ThermoFisher Scientific), according to the manufacturer's protocol protocol. In all cases, cells were plated the day before the transfection in an antibiotic-free medium which

was subsequently changed prior to adding the transfection mix, and again after 24 hr to maximize cell viability. All transfections were carried out for 72 hours, unless specified otherwise. The complete list with the sequences of both siRNAs and LNA/GapmeRs is reported in Table S5.

Cell synchronization was performed using a standard double-thymidine block. Briefly, thymidine (Sigma) was added to the culture medium at a final concentration of 2 mM, and the cells were incubated for 18 hours. The medium was then replaced with standard complete medium for 9 hours, followed by the addition of fresh thymidine-containing medium. After an additional 15 hours, the medium was replaced with standard complete medium to release the cells from the G1/S boundary into the S phase.

Lipopolysaccharide (LPS) was purchased from Sigma, diluted in DNase/RNase-free H₂O and added to the cell culture media at the indicated final concentrations. Hydrogen peroxide (H₂O₂) was purchased from Sigma and added directly to the cell culture media at the indicated final concentrations. Thapsigargin and Etoposide (Sigma) were diluted in tissue culture-grade DMSO and added to the cell culture media at the indicated final concentrations. Cells were cultured with these stressors for a total of three days, and cells supplemented with an equal volume of the respective solvent were used as controls. CX-5461 (Sigma) was diluted in sterile-filtered 50 mM NaH₂PO₄ at a stock concentration of 10mM and added to the cell culture media at a final concentration of 0.5, 1 or 5 μM. As a control, cells were supplemented with an equal volume of 50 mM NaH₂PO₄.

Plasmids construction

The sequence corresponding to the cDNA of human *EPB41L4A-AS1* (ENST00000688370.2_3) was synthesized and cloned by Hylabs inside a pcDNA3.1(+) vector. The vector to express the *EPB41L4A* ORF (NM_022140.5) was purchased from Genecopoeia (EX-A8889-Lv201). The vector expressing a full length and unspliced *EPB41L4A-AS1* (pLV[ncRNA]-Puro-CMV) was purchased from Vectorbuilder, and the target site of LNA2 was mutated by introducing four mismatches (AACTTAAAGCAGCGT → ACCTTAGAAGTAGAGT) via restriction-free cloning.

To generate the Δ*SNORA13*, Δ*H*, and Δ*ACA* vectors, we ordered *EPB41L4A-AS1* gene fragments with the desired mutations from TWIST. These fragments corresponded to the sequence between the unique SalI and MluI restriction sites and were inserted via restriction digestion. The pLKO.1 *SNORA13* plasmid²⁷ was a gift from Prof. Mendell (UT Southwestern). All gRNAs were ordered as single-stranded oligos (Sigma) and cloned inside pKLV-Puro vectors as previously described⁹³. The complete list of gRNAs and their sequences is available in Table S5. All plasmids were sent for Sanger and whole plasmid (Plasmidsaurus) sequencing prior to use.

Wound healing assay and analysis

Wound healing experiments were carried out as previously described⁹⁴. To perform the wound healing assay, we employed the μ-Plate 24 Well system (Ibidi). Specifically, we transfected the targeting and control GapmeRs via reverse transfection, diluting the transfection mixes directly into the cell mixtures while seeding the cells. After reaching confluence overnight (O/N), the culture inserts were gently removed with sterile forceps and the wells were washed with 1X PBS. Brightfield images were taken using an EVOS Cell Imaging System (Thermo Fisher Scientific) at different time points. Data analysis was then performed using the Wound_healing_size_tool⁹⁵ plugin for ImageJ, which calculates the total (μm²) and percentage (%) of the wounded area. From these data, we calculated the rate of cell migration (RM) as:

$$R_M = \frac{W_i - W_t}{t}$$

Where *W_i* represents the initial average wound width, and *W_t* is the average wound width at a given time *t*. Additionally, we derived the percentage of wound closure defined as:

$$\text{Wound Closure (\%)} = \frac{A_{t=0} - A_{t=\Delta t}}{A_{t=0}}$$

With $A_{t=0}$ and $A_{t=\Delta t}$ being the wounded area (in μm^2) at the beginning and after time t , respectively.

Gene expression analysis

RNA was extracted by using TRI-Reagent as previously described⁹⁴, according to the manufacturer's instructions. The isolated RNA was then resuspended in DNase/RNase-free H₂O and DNase-treated using the Baseline-ZERO™ DNase (Biosearch Technologies), following the manufacturer's recommendations and incubating the RNAs for 1 hour at 37°C. RNA was then retrotranscribed using the qScript™ Flex cDNA Synthesis Kit (Quanta Bio), following standard conditions as suggested by the manufacturer. The resulting cDNA was finally diluted 1:5-1:10 with DNase/RNase-free H₂O and used as input for Quantitative Real-Time PCR using the Fast SYBR™ Green Master Mix (Applied Biosystems). Each reaction was amplified using the following PCR program: 20'' at 95°C, 40 cycles by denaturing for 1'' at 95°C followed by annealing/extension for 20'' at 60°C. Results were analyzed using the $\Delta\Delta\text{Ct}$ method. The complete list of primers and their sequences is available in Table S5.

Aminocarboxyl propyl reverse transcription (aRT)-PCR assay to estimate 18S:1248m¹acp³Ψ

The aminocarboxylpropyl reverse transcription (aRT)-PCR assay to detect and quantify the 1248.m1acp³Ψ modification was performed as previously described²⁶, with minor modifications. Briefly, total RNA was extracted as described above, and 1 μg per sample was DNase-treated using TURBO DNase (Invitrogen), according to the manufacturer's protocol. Retrotranscription was then carried out using SuperScript III (Invitrogen), following the manual's recommendations and by using only random hexamers. The cDNA was then diluted 1:5 with DNase/RNase-free H₂O and used for end-point PCR using the Q5 High-Fidelity DNA Polymerase (NEB), according to the standard protocol and using the primers listed in the original protocol²⁶ and available in Table S5. From each PCR reaction, 5 μL were digested in duplicate with HinFI (NEB) in rCutSmart buffer (NEB) and an additional 5 μL were incubated without the enzyme as a negative control. Lastly, the resulting fragments were then run on a 2% agarose gel.

Northern Blot

Northern Blot to detect *SNORA13* was performed as previously described²⁷, with minor modifications. A total of 20 μg of RNA per sample was loaded onto an 8% TBE-UREA polyacrylamide gel, which had been pre-equilibrated by running at low voltage in 1X TBE. After electrophoresis, the gel was stained for 15 minutes in 0.5X TBE with a few drops of ethidium bromide while rocking, to assess the quality of the run. The RNA was then transferred to an HyBond-NX membrane (GE Healthcare) in 0.5X TBE at room temperature (RT) using a constant current of 0.25 A for 2 hours. Following the transfer, the membrane was rinsed in 2X SSC, crosslinked at 1200 J/cm² in a UV oven, and pre-hybridized for 30 minutes at 42°C in 10 mL of pre-warmed ULTRAhyb™ Ultrasensitive Hybridization Buffer (ThermoFisher Scientific) while rotating. IR fluorescent probes targeting *U6* and *SNORA13* were then added to a final concentration of 10 nM, and the membrane was incubated O/N at 42°C while rotating. The next day, the membrane was washed twice in 2X SSC, 0.1% SDS at 42°C for 5 minutes while rotating, followed by two additional washes in 0.1X SSC, 0.1% SDS, and then finally imaged. The complete list of probes is available in Table S5.

Protein extraction and Western Blot

Proteins were extracted using RIPA as previously described⁹⁴. Briefly, cells were detached using a 0.05% Trypsin – 0.02% EDTA solution, pelleted by centrifugation, resuspended in ice-cold RIPA buffer, and incubated on ice for 20 minutes with intermittent vortexing. The lysates were then centrifuged at 15,000g for 15 minutes at 4°C, after which the supernatants were collected and quantified. For Western blots, equal amounts of protein were loaded onto a polyacrylamide gel and run in an SDS running buffer. Proteins were then transferred onto a pre-activated PVDF membrane in a cold Tris-Glycine buffer supplemented with 20% methanol, with a constant current of 0.30 A for 2 hours. Membranes were then blocked in 5% milk in 1X PBS-0.1% Tween 20 for 1 hour, followed by O/N incubation at 4°C with primary antibodies diluted in the same blocking solution. The next day, membranes were washed three times with 1X PBS-0.1% Tween 20, incubated for 2 hours at room temperature with the appropriate secondary antibodies, and washed three more times before image acquisition. The complete list of primary and secondary antibodies and relative dilutions is available in Table S5.

Immunostaining

Immunofluorescence was performed as previously described⁹⁴. Cells were plated and transfected on either sterile glass coverslips or 8-well chambers (Ibidi). They were first washed once with 1X PBS, then fixed with cold 4% paraformaldehyde (PFA) in 1X PBS for 15 minutes at room temperature. After two 5-minute washes with 1X PBS, cells were permeabilized by incubating them in a permeabilization/blocking solution containing 5% horse serum, 1 mg/mL BSA, and 0.1% Triton X-100 in 1X PBS for 30 minutes at room temperature. Following three washes with 1X PBS, cells were incubated O/N at 4°C with primary antibodies diluted in blocking solution (5% horse serum, 1 mg/mL BSA in 1X PBS). The next day, cells were washed three times with 1X PBS and incubated for 2 hours at room temperature with secondary antibodies in blocking solution, protected from light. After three additional washes with 1X PBS, cells were stained with DAPI and mounted using ProLong™ Glass Antifade Mountant (Invitrogen). The slides were left to cure O/N in the dark at room temperature before imaging with a Zeiss Spinning Disk confocal microscope. The complete list of primary and secondary antibodies and relative dilutions is available in Table S5.

Subcellular fractionation

To fractionate cells and isolate the cytoplasmic, nucleoplasmic and chromatin RNA fractions we employed a published protocol⁹⁶ with minor modifications. Confluent 10 cm plates were harvested with a Trypsin (0.05%) – EDTA (0.02%) solution, pelleted by centrifugation, washed once with ice-cold 1X PBS and resuspended in 380 µL of ice-cold HLB buffer (10 mM Tris-HCl, pH 7.5, 10 mM NaCl, 3 mM MgCl₂, 0.3% NP-40, 10% Glycerol) freshly supplemented with RNase Inhibitors (EURX). The samples were then incubated on ice for 20' with occasional vortexing, spun down for 3' at 4°C / 1000 g, the supernatant was then transferred to a new tube, centrifuged again for 10' at 4°C / 1000 g, after which the supernatant (cytoplasmic fraction) was moved to a new tube, supplemented with 1 mL of RNA precipitation solution (RPS, 150mM Sodium Acetate in 100% ethanol) and stored at -20°C for at least 1 h. The semi-pure nuclei were then washed three times with ice-cold HLB and centrifugations for 2' at 4°C / 500 g, after which the pellets were resuspended in 380 µL of ice-cold MWS (10 mM Tris-HCl, pH 7.0, 4 mM EDTA, pH 8.0, 0.3 M NaCl, 1 M Urea, 1% NP-40) freshly supplemented with RNase Inhibitors, quickly vortexed and incubated on ice for 5 minutes, vortexed again and incubated for an additional 10 minutes. The tubes were then centrifuged for 3' at 4°C / 1000 g, the supernatant (nucleoplasmic / nuclear soluble fraction) was saved to a new tube, supplemented with 1 mL of RPS and stored at -20°C for at least 1 h. The chromatin pellets were finally washed three times with ice-cold MWS and centrifugations for 2' at 4°C / 500 g, after which 1 mL of TRI-Reagent was added and samples were stored at -20°C. After 1+ hours at -20°C, the tubes with the cytoplasmic and nucleoplasmic fractions in RPS were briefly vortexed, centrifuged for 15' at 4°C / 15000 g, the pellets were washed with 70% ethanol by vortexing and centrifuged again for 15' at 4°C / 15000 g, after which they were air-dried and resuspended in 1 mL of TRI-Reagent. To each sample in 1 mL of TRI-Reagent, 10 µL of 0.5 M EDTA,

pH 8.0 were added followed by an incubation with vortexing for ~10 minutes at 65°C, or until the RNA pellets fully dissolved. The RNA was then extracted following the standard TRI-Reagent protocol.

Native RNA Immunoprecipitation (RIP)

To perform Native RNA Immunoprecipitation, we have employed a slightly modified version of the ENCODE IP protocol⁹⁷. Briefly, 5×10^6 MCF-7 cells were harvested by trypsinization, washed once with 1X PBS, and flash frozen as pellets in 50 μ L of 1X PBS. The pellets were then thawed on ice, resuspended in 450 μ L of ice-cold lysis buffer (50 mM Tris-HCl, pH 7.4; 100 mM NaCl; 1% NP-40; 0.1% SDS; 0.5% sodium deoxycholate) supplemented with EDTA-free Protease Inhibitor Cocktail (APEX-BIO) and RNase Inhibitor (EURX), and incubated on ice for 20' with occasional vortexing to allow complete lysis. The lysates were then centrifuged at max speed for 15' at 4°C and the supernatants were collected into a new tube. A total of 20 μ L/sample of Protein A/G beads were then washed twice with ice-cold lysis buffer and resuspended in 250 μ L of lysis buffer, of which 50 μ L were used to pre-clear the lysate by incubation at RT for 30' with gentle rotation, and the remaining 200 μ L were instead incubated for 1 h/RT with 3 μ g of antibody. After 30' of pre-clearing, the clear lysate was moved to a new tube in ice, and 25 μ L (5%) of it was saved as INPUT sample. After 1 h, the antibody/beads were washed three times with ice-cold lysis buffer and incubated with the pre-cleared lysate O/N at 4°C while gently rotating. The following day, 25 μ L (5%) of the supernatant was saved as UNBOUND sample, and the beads (BOUND sample) were gently washed twice with ice-cold wash buffer (20 mM Tris-HCl pH 7.4, 10 mM MgCl₂, 0.2% Tween-20) freshly supplemented with RNase Inhibitors (EURX). Finally, 500 μ L of TRI-Reagent was added to all three fractions and RNA isolation was carried out as previously described. The complete list of antibodies and relative dilutions is available in Table S5.

Single-molecule RNA FISH with HCR technology (smFISH HCR)

To visualize the subcellular localization of EPB41L4A-AS1 in vivo, we performed single-molecule fluorescence in situ hybridization (smFISH) using HCRTM amplifiers. Probe sets ($n = 30$ unique probes) targeting *EPB41L4A-AS1* and *GAPDH* (positive control) were designed and ordered from Molecular Instruments. We followed the Multiplexed HCR v3.0 protocol with minor modifications. MCF-7 cells were plated in 8-well chambers (Ibidi) and cultured O/N as described above. The next day, cells were fixed with cold 4% PFA in 1X PBS for 10 minutes at RT and then permeabilized O/N in 70% ethanol at -20°C. Following permeabilization, cells were washed twice with 2X SSC buffer and incubated at 37°C for 30 minutes in hybridization buffer (HB). The HB was then replaced with a probe solution containing 1.2 pmol of *EPB41L4A-AS1* probes and 0.6 pmol of *GAPDH* probes in HB. The slides were incubated O/N at 37°C. To remove excess probes, the slides were washed four times with probe wash buffer at 37°C for 5 minutes each, followed by two washes with 5X SSCT at RT for 5 minutes. The samples were then pre-amplified in amplification buffer for 30 minutes at RT and subsequently incubated O/N in the dark at RT in amplification buffer supplemented with 18 pmol of the appropriate hairpins. Finally, excess hairpins were removed by washing the slides five times in 5X SSCT at RT. The slides were mounted with ProLongTM Glass Antifade Mountant (Invitrogen), cured O/N in the dark at RT, and imaged using a Nikon CSU-W1 spinning disk confocal microscope. In order to estimate the RNA copy number, we imaged multiple distinct fields, extracted the number of *EPB41L4A-AS1*/*GAPDH* molecules in each field using the “Find Maxima” tool in ImageJ/Fiji, and divided them by the number of cells (as assessed by DAPI staining).

Cleavage Under Targets and Release Using Nuclease (CUT&RUN)

CUT&RUN reactions were performed following the V3 of the protocol⁹⁸, with minor modifications which were described previously⁹⁴. An equal number of cells was harvested and washed once with 1X PBS at room temperature. After three washes with Wash Buffer (20 mM HEPES-NaOH pH 7.5, 150 mM NaCl, 0.5 mM Spermidine supplemented with EDTA-free Protease Inhibitor Cocktail (APEX-BIO)) at RT, cells were resuspended in 1 mL Wash Buffer and incubated for 10 minutes while gently rotating with 20 μ L of pre-activated Concanavalin A-coated magnetic beads (EpiCypher).

The buffer was then removed, and the beads were resuspended in 150 μ L of Antibody Buffer (Wash Buffer with 0.1% Digitonin (Sigma) and 2 mM EDTA, pH 8) containing the antibody of interest, and left gently rotating O/N with a 180° angle at 4°C. On the next day, cells were washed two times with ice-cold Dig-Wash Buffer (Wash Buffer with 0.1% Digitonin), resuspended in 150 μ L of Dig-Wash Buffer supplemented with 1 μ L of custom-made pA/G-MNase every mL and left rotating for 1h at 4°C. After that, cells were washed again two times with Dig-Wash Buffer, resuspended in 100 μ L of the same buffer, and chilled into a thermoblock sitting on ice. To initiate the cleavage reaction, 2 μ L of 100 mM CaCl₂ was added to each tube, and the tubes were left at 0°C for 30 minutes. To halt the reaction, 100 μ L of a 2X STOP Buffer (340 mM NaCl, 20 mM EDTA pH 8, 4 mM EGTA pH 8, 0.05% Digitonin, 100 μ g/mL RNase A, 50 μ g/mL Glycogen) were added to each tube, and cleaved DNA fragments were released by incubating the samples for 30' at 37°C. The tubes were then centrifuged for 5 minutes at 4°C / 16,000g and the supernatant was collected. Finally, the DNA was purified by standard Phenol/Chloroform extraction using the SPRINT Phase Lock Gel Heavy tubes (QuantaBio), and the success of the CUT&RUN reaction was assessed by running the positive control (H3K27me3 or H3K4me3) on a TapeStation (Agilent Technologies) using a High Sensitivity D1000 ScreenTape (Agilent Technologies). The complete list of primary and secondary antibodies and relative dilutions is available in Table S5.

RNA-seq and CUT&RUN Library Preparation

RNA-seq libraries were prepared using the CORALL mRNA-Seq Kit V2 (Lexogen), following the manufacturer's instructions. Prior to library generation, 1 μ g of total RNA was either PolyA-enriched using the Poly(A) RNA Selection Kit V1.5 (Lexogen), or depleted of rRNAs using the RiboCop rRNA depletion kit for HMR V2 (Lexogen). To prepare the CUT&RUN libraries, we followed the original protocol⁹⁸, with slight modifications which were described previously⁹⁴. The libraries were then quality-checked by both dsDNA Qubit (Thermo Fisher Scientific) and TapeStation (Agilent Technologies). All libraries were sequenced on an Illumina NovaSeq 6000 or Novaseq X instrument, aiming for either 10M (CUT&RUN) or 15M (RNA-seq) reads per sample.

RNA-seq data analysis

Analysis of the RNA-seq libraries was performed as previously described⁹⁴. Raw FASTQ files were processed by a customized Lexogen CORALL analysis pipeline script (https://github.com/Lexogen-Tools/corall_analysis⁹⁴). Briefly, adaptors were trimmed by using Cutadapt⁹⁹, and the resulting trimmed FASTQ files were used as input for mapping to the hg19 human genome with STAR aligner¹⁰⁰, using the ‘--quantMode GeneCounts’ option in order to count the number of reads mapping to each feature provided by an appropriate GTF file. To call for differential expression, we used DESeq2¹⁰¹, considering a gene to be differentially expressed when adjusted $P < 0.05$ and absolute $\log_2FC > 0.41$ (~33% increase or decrease). Prior to any subsequent analysis, we filtered out poorly expressed genes, pseudogenes, and genes with an exonic length <200nt. To perform GO enrichment analysis and GSEA, we used the ClusterProfiler R package¹⁰² by considering a term to be enriched when the adjusted $P < 0.05$. Whenever several redundant terms were found, we employed the *simplify* function with a cutoff=0.7 to reduce the redundancy.

CUT&RUN data analysis

Raw FASTQ files were aligned to the hg19 reference genome with Bowtie2¹⁰³, using the -p 8, -X 2000 and --no-unal options. Peaks were called using the MACS2 callpeak function¹⁰⁴. Peak quantification and differential expression were performed using the HOMER tool suite¹⁰⁵, using the *annotatePeaks.pl* and *getDiffExpression.pl* functions, respectively. Lastly, to plot the metagene profiles and coverage heatmaps, we employed deepTools¹⁰⁶.

Computational screen to identify putative *cis*-acting lncRNAs

We reasoned that lncRNAs that are likely to act in *cis* will be occasionally co-regulated with their target genes and also found in spatial proximity to the target promoters. We therefore sought such lncRNA-target pairs using the GeneHancer database²² that connects between regulatory elements and their targets by combining tissue co-expression data, ChIP-seq of enhancer-enriched histone

modifications and transcription factor binding, Hi-C data and eQTLs within putative enhancer regions. We complemented this by analyzing several RNA-seq datasets of MCF-7 breast cancer cells exposed to different conditions^{23–25}, to infer genome-wide transcriptional alterations across a variety of different experimental settings. We then focused on those cases in which a lncRNA 1) overlaps a high-confidence GeneHancer (GH) element, 2) is differentially expressed in at least one of the conditions analyzed, and 3) the targets of the aforementioned GH element are also concomitantly differentially expressed. Because this approach identified over 600 candidate lncRNA-target pairs (Table S1), we first discarded lncRNAs which have been already reported to act in *cis*, and prioritized those cases in which the lncRNA is well expressed and/or the target genes are biologically interesting. This led us to focus on four lncRNA-target pairs connected in GeneHancer and in which both genes were significantly either upregulated or downregulated in the treatment condition, compared to controls (Fig. S1A–D⁴⁵).

Chromosome Conformation Capture (3C)

Between 5 and 10×10⁶ cells for each condition/replicate were collected and washed once with 1X PBS at RT. Cells were then resuspended in 10 mL of ice-cold 1X PBS with 10% FBS, supplemented with another 10 mL of 1X PBS with 10% FBS and 4% Formaldehyde, and left gently rotating for 5' at RT. Quenching was achieved by adding 1020 µL 2.5 M Glycine to each tube and placing them on ice to cool down, after that the cells were centrifuged for 8' at 4°C / 300 g and washed once with ice-cold 1X PBS. The pellet was then resuspended in 10 mL of Nuclear Permeabilization Buffer (10 mM Tris-HCl, pH 8.0, 10 mM NaCl, 0.5% NP-40) supplemented with EDTA-free Protease Inhibitor Cocktail (APEX-BIO), gently rotated for 1 h in a cold room and centrifuged for 5' at 4°C / 600 g. After discarding the supernatant, the nuclei were resuspended in ~1 mL of ice-cold 1X PBS, moved to a 1.5 mL centrifuge tube, centrifuged and washed again with 1 mL ice-cold 1X PBS, and eventually frozen as a pellet at -80 °C.

After 1+ days at -80 °C, each pellet was thawed on ice and gently resuspended in 175.5 µL of nuclease-free H₂O, after which 24.5 µL of 10X DpnII Reaction Buffer was added and the tubes were moved to a thermomixer set at 37°C. Each reaction was then supplemented with 3 µL of pre-warmed 20% SDS, incubated for 1 h at 37°C / 900g, supplemented again with 20 µL of pre-warmed 20% Triton-X100 and incubated for an additional 1 h at 37°C / 900g, after which 15 µL of DpnII (NEB, 50.000 U/mL) were added and the reaction was finally incubated O/N at 37°C / 900g. The following day, the restriction reaction was removed by centrifugation for 5' at 4°C / 600 g, and the pellets were washed once with ice-cold 1X PBS, eventually leaving a small volume (~30 µL) on top of the pellet. The tubes were then incubated for 20' at 65°C, to inactivate DpnII activity, followed by thorough resuspension in 445 µL of nuclease-free H₂O and were transferred to ice. After chilling, 50 µL of 10X Ligation Buffer (NEB), 0.5 µg BSA and 5 µL of Quick Ligase (NEB) were added to each tube, and left incubating O/N at 16°C. After the O/N incubation, the ligation reaction was removed by centrifugation for 5' at 4°C / 600 g, the pellets were washed once with ice-cold 1X PBS and eventually resuspended in 250 µL 10 mM Tris-HCl, pH 7.5. A total of 30 µg of RNase A were then added to each sample, which were subsequently incubated for 45' at 37°C / 750g, after which 15 µL of PCR-grade Proteinase K were added and incubated O/N at 65°C / 750g. The following day, the de-crosslinked reactions (3C template) were cleaned with 0.5X Sera-Mag Size Selection Beads (Cytiva Life Sciences), according to manufacturer's instructions, eluted in 200 µL Elution Buffer (Qiagen) and quantified by dsDNA Qubit.

UMI Circular Chromosome Conformation Capture (UMI-4C)

Quantitative UMI-4C was performed as previously described⁴⁵, with minor modifications. An equal amount of 3C template (5–10 µg) for each sample/replicate was diluted to a final volume of 200 µL with Elution Buffer and sonicated to an extent the fragments size ranges between 250 and 1000bp. To the sonicated fragments, 20 µL of 10X End-Repair Buffer (NEB) and 10 µL of End Repair Mix (NEB) were added and the samples were incubated for 30' at 20°C, after which they were cleaned with 2.2X Sera-Mag Selection Beads and eluted in 78 µL 10 mM Tris-HCl, pH 8.0. To 76 µL of the cleaned DNA, 10 µL 10X NEBuffer 2, 10 µL 10 mM dATP, and 4 µL Klenow (3'-5' exo, NEB)

were added, and the reaction was incubated for 30' at 37°C. After this, the enzymes were inactivated by incubating the samples for 20' at 75°C, then 2 µL of FastCIP (NEB) were added and the reactions were further incubated for 1 h at 50°C, followed by purification with 2X Sera-Mag Size Selection Beads and elution in 67 µL of 10mM Tris-HCl, pH 8.0. Illumina-compatible adapters were ligated by adding 80 µL Quick Ligase Buffer (NEB), 10 µL Quick Ligase (NEB) and 4 µL of 15 µM Y-Shaped Adaptors (IDT) to 66 µL of the eluted DNA from the previous step, followed by an incubation at 25°C for 15'. After that, the samples were denatured for 2' at 95°C, placed on ice, brought to volume by adding 160 µL of nuclease-free H₂O, cleaned with 1X Sera-Mag Size Selection Beads and eluted in 40 µL 10 mM Tris-HCl, pH 8.0. To quantify the resulting 4C template, 1 µL was diluted in 4 µL of 10mM Tris-HCl, pH 8.0, denatured for 5' at 95°C, placed on ice and analyzed by ssDNA Qubit.

Libraries were then prepared using a nested PCR approach, preparing five reactions for each sample and employing three bait primers for each target (EPB41L4A-AS1 TSS, EPB41L4A TSS, STARD4 TSS, see next for design criteria) to increase complexity of the 4C profiles. For the first PCR (PCR1), upstream bait primers were designed close (<80-85 bp) to a DpnII restriction site (hg19 genome build); for the second PCR (PCR2), the downstream bait primers were designed to follow – but not overlap – the upstream ones, ending ideally <20bp from the DpnII site (“pad” sequence), and containing a sequence consisting of the Universal Illumina P5 on their 5' end. Both primers were designed to have a standard annealing temperature of ~60°C, and their sequences are available in Table S5. PCR1 consisted of 200 ng of 4C template, 10 µL of 10X Phusion Buffer (NEB), 1 µL of 10 mM dNTPs, 2.5 µL of pooled upstream bait primers with a final concentration of 10 µM, 2.5 µL of 10 µM Universal Illumina P7 primer, 0.5 µL Phusion Polymerase (NEB), and nuclease-free H₂O up to a final volume of 50 µL. The reactions were incubated for 30'' at 98°C, amplified with 20 cycles consisting of 10'' at 98°C, 30'' at 56°C and 60'' at 72°C, with a final extension of 10' at 72°C, individually purified with 0.8X Sera-Mag Size Selection Beads and eluted in 20 µL of 10 mM Tris-HCl, pH 8.0. PCR2 consisted of 15 µL of purified template from PCR1, 10 µL of 10X Phusion Buffer (NEB), 1 µL of 10 mM dNTPs, 2.5 µL of pooled downstream bait primers with a final concentration of 10 µM, 2.5 µL of 10µM Universal Illumina P7 primer, 0.5 µL Phusion Polymerase (NEB), and nuclease-free H₂O up to a final volume of 50 µL. The reactions were incubated with the same program of PCR1 but with 18 cycles of amplification instead of 20, all five replicates of each sample were then pooled together, purified with 0.8X Sera-Mag Size Selection Beads and eluted in 40 µL of Elution Buffer. The libraries were then quality-checked by both dsDNA Qubit and TapeStation, and eventually pooled according to the relative number of reads needed. All libraries were sequenced on an Illumina NovaSeq 6000 instrument, aiming at ~5M reads per sample.

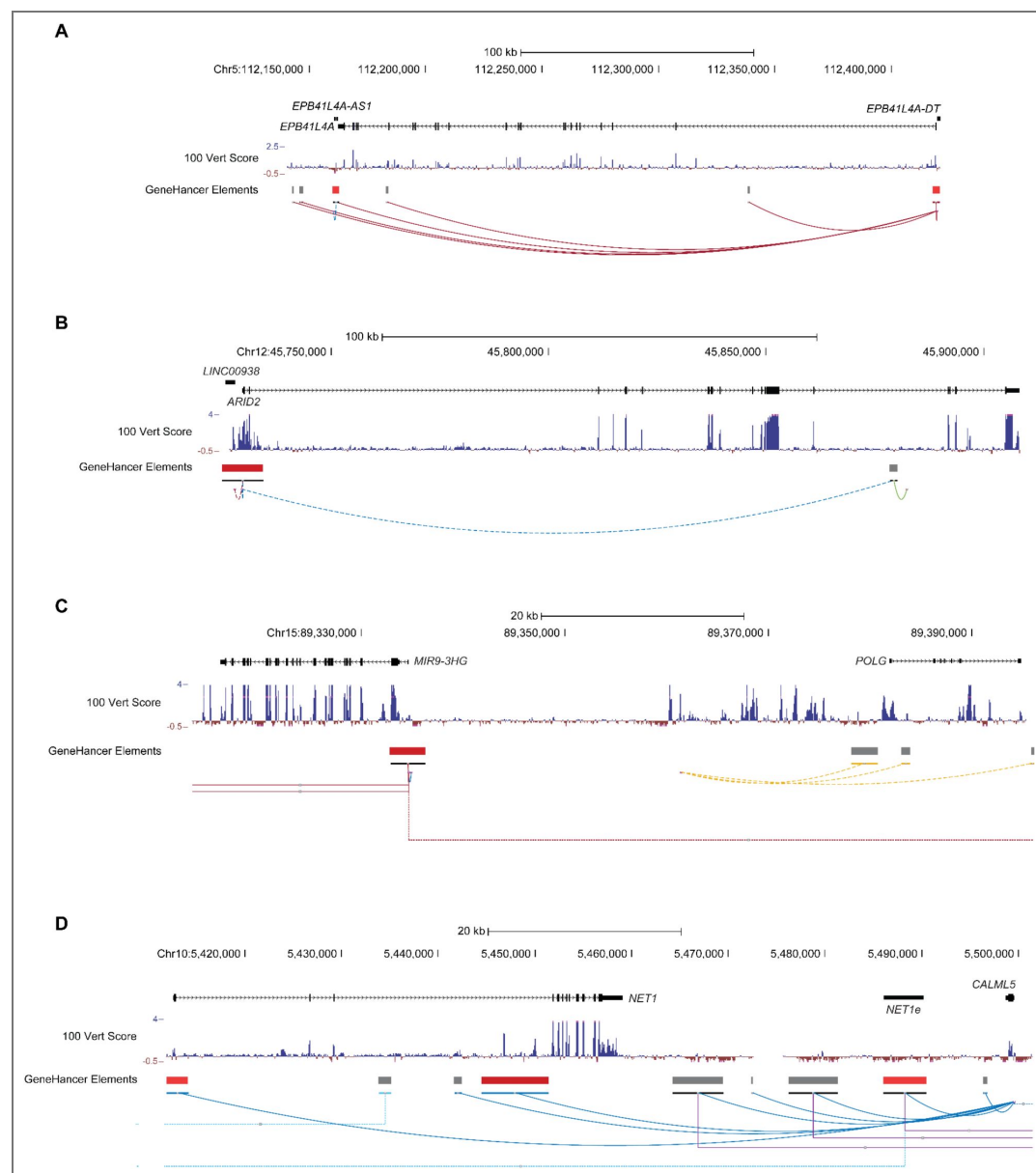
UMI-4C Data Analysis

To analyze the UMI-4C data we have employed the `umi4cpackage` R package (<https://github.com/tanaylab/umi4cpackage>). After setting up the configuration file as described, they were loaded with the `p4cLoadConfFiles` function, and the raw fastq files were analyzed using the `p4cCreate4CseqTrack` function. The 4C profiles were then built and plotted using the `p4cNewProfile` and `plot` functions, respectively, specifying the *cis* window and overlaying two profiles when needed. The p-values and log₂FC between conditions and across a genomic region of interest were finally calculated using the `p4cIntervalsMean` function.

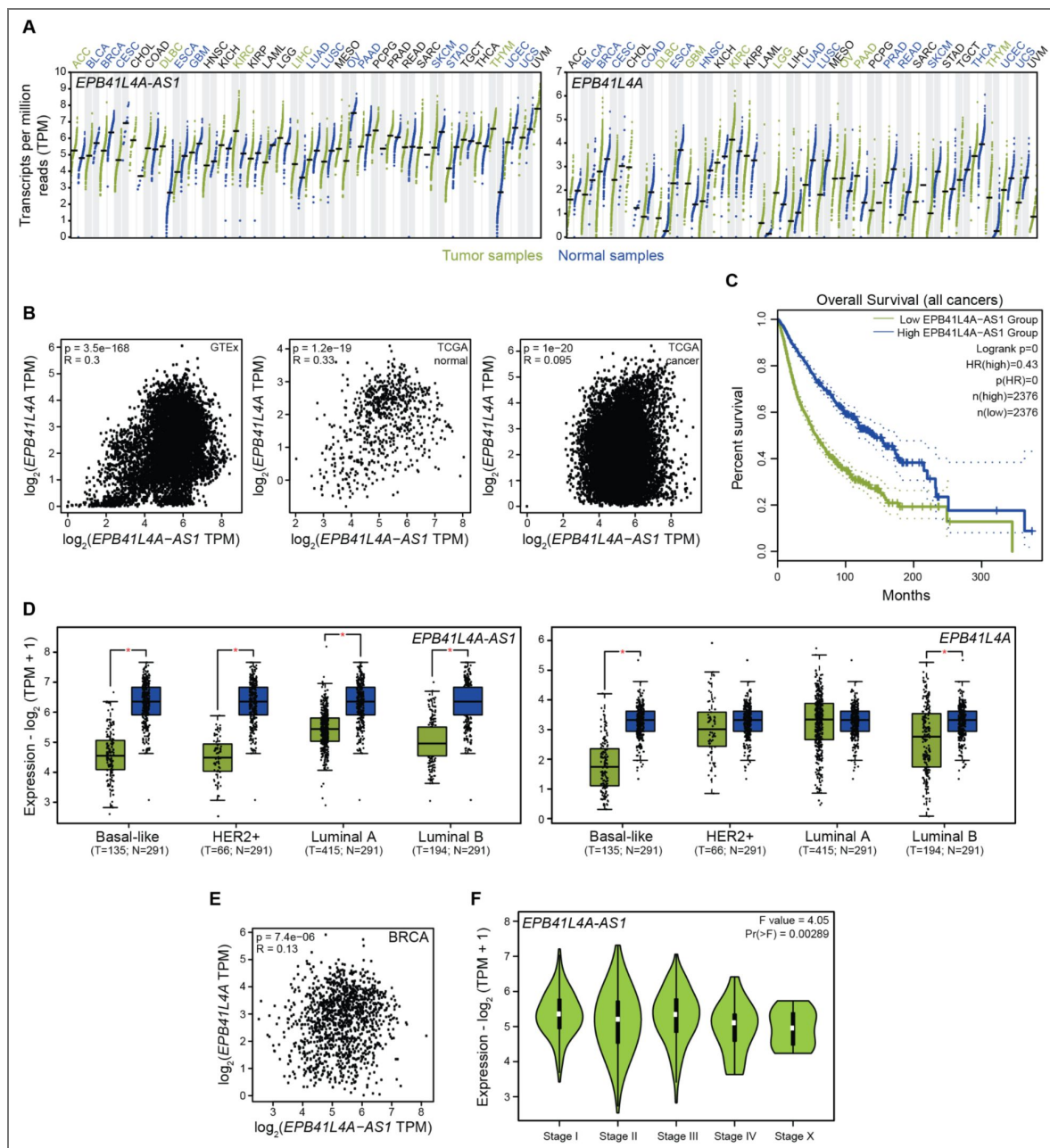
Metabolic labeling, SLAM-seq and data analysis

For half-life and synthesis rate estimation upon the KD of *EPB41L4A-AS1*, cells were transfected with control and targeting GapmeRs as described above. Metabolic labeling and SLAM-seq were then performed as previously described¹⁰⁷, with minor modifications. All steps were performed in the dark until the chemical conversion with iodoacetamide (IAA). After 48h hours, the growth medium was replaced with fresh media supplemented with 500 µM 4-Thiouridine (4sU, Sigma), and replaced again with fresh 4sU after every time point. In total, we collected RNA from cells at time 0 (unlabeled), and after 4, 8 and 24h hours post-4sU addition. RNA extraction was carried out as described above with the addition of DTT (0.1 mM final concentration) to keep the samples

under reducing conditions, and IAA conversion was performed as described previously^{80,107}. Ribosomal RNAs were then depleted using the NEBNext rRNA Depletion Kit V2 (NEB) and libraries were generated with the NEBNext Ultra II Directional Library Kit (NEB), both according the manufacturer's instructions. Estimation of half-lives and synthesis rates was performed using GRAND-SLAM/grandR¹⁰⁸, and are listed in Table S4. Notably, the estimation assumes steady state expression in control samples, but not in the KD ones.

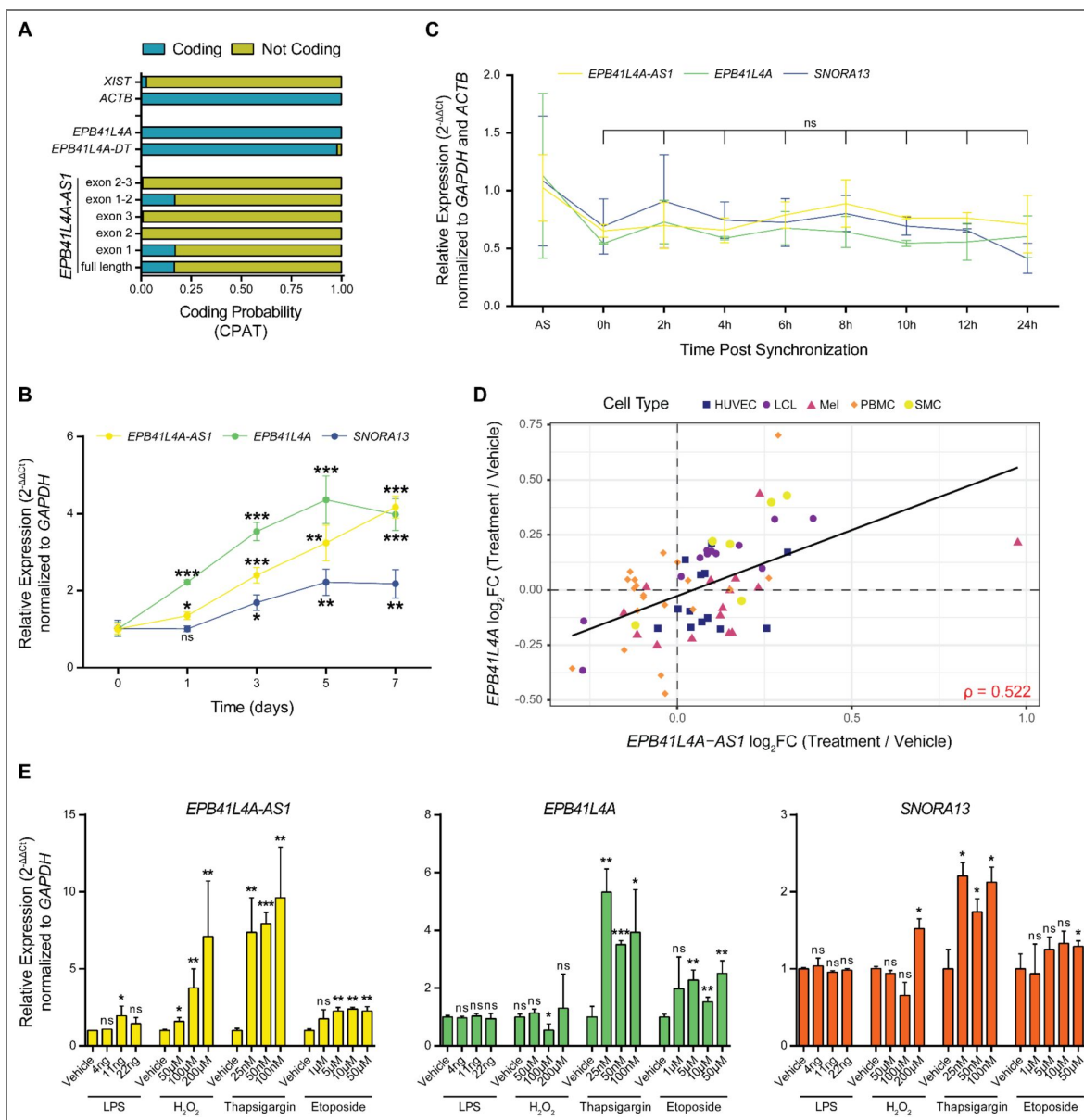


Supplementary Figure 1. UCSC Genome Browser view of the loci containing the selected lncRNAs and target genes for validation. View of the *EPB41L4A-AS1 - EPB41L4A* (A), *LINC00938 - ARID2* (B), *MIR9-3HG - POLG* (C) and *NET1e - NET1 - CALML5* (D) loci. In each view, the PhyloP conservation score and GeneHancer (GH) elements are also reported.



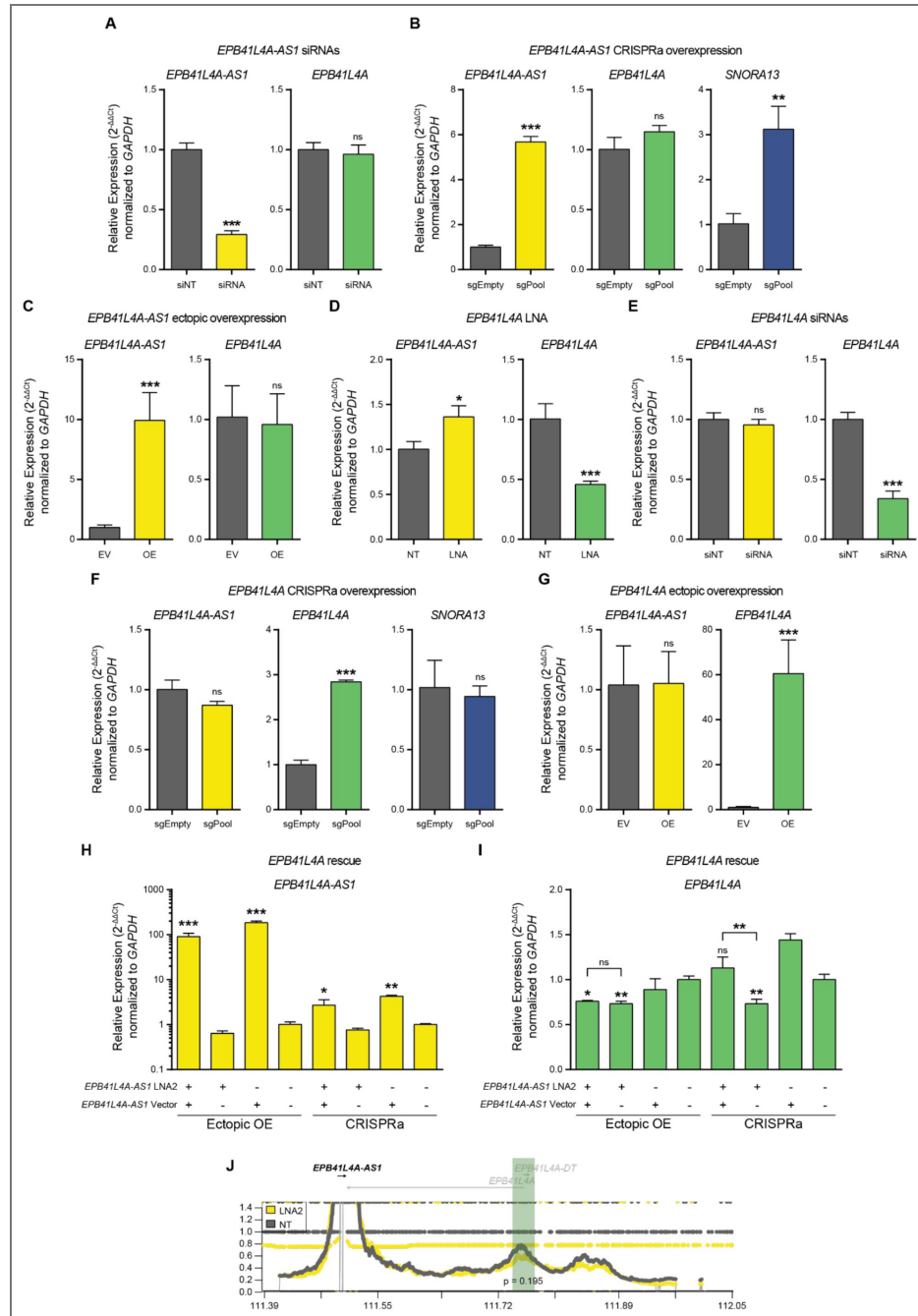
Supplementary Figure 2. *EPB41L4A-AS1* is dysregulated in several cancer types and correlates with survival.

(A) Expression level of *EPB41L4A-AS1* (left) and *EPB41L4A* (right) across the TCGA cohort. The thick lines and each point represent the median and an individual sample, respectively, and the color on top reflects whether the indicated genes are significantly more expressed in control or tumor conditions. (B) Correspondence between the expression of *EPB41L4A-AS1* and *EPB41L4A* in GTEx v8 (left), TCGA control (center) and tumor (right) samples. The Spearman's correlation coefficient and p-value is also reported. (C) Kaplan-Meier survival curve stratified into individuals with the lower and higher 50% expression level of *EPB41L4A-AS1*. (D) Expression level of *EPB41L4A-AS1* (left) and *EPB41L4A* (right) in patients with the different BRCA subtypes and matched controls. The thick lines, edges of the box, whiskers and each point represent the median, first and third quartiles, the upper and lower 1.5 interquartile ranges (IQRs), and an individual sample, respectively. (E) Correspondence between the expression of *EPB41L4A-AS1* and *EPB41L4A* in BRCA samples. (F) Expression level of *EPB41L4A-AS1* in patients at different BRCA stages. The white dot and edges of the box represent the median and the first and third quartiles, respectively. The one-way ANOVA p-value for differential expression is also reported. In all cases, $P < 0.05 = *$. Data was accessed using the GEPIA2 portal.¹⁰⁹



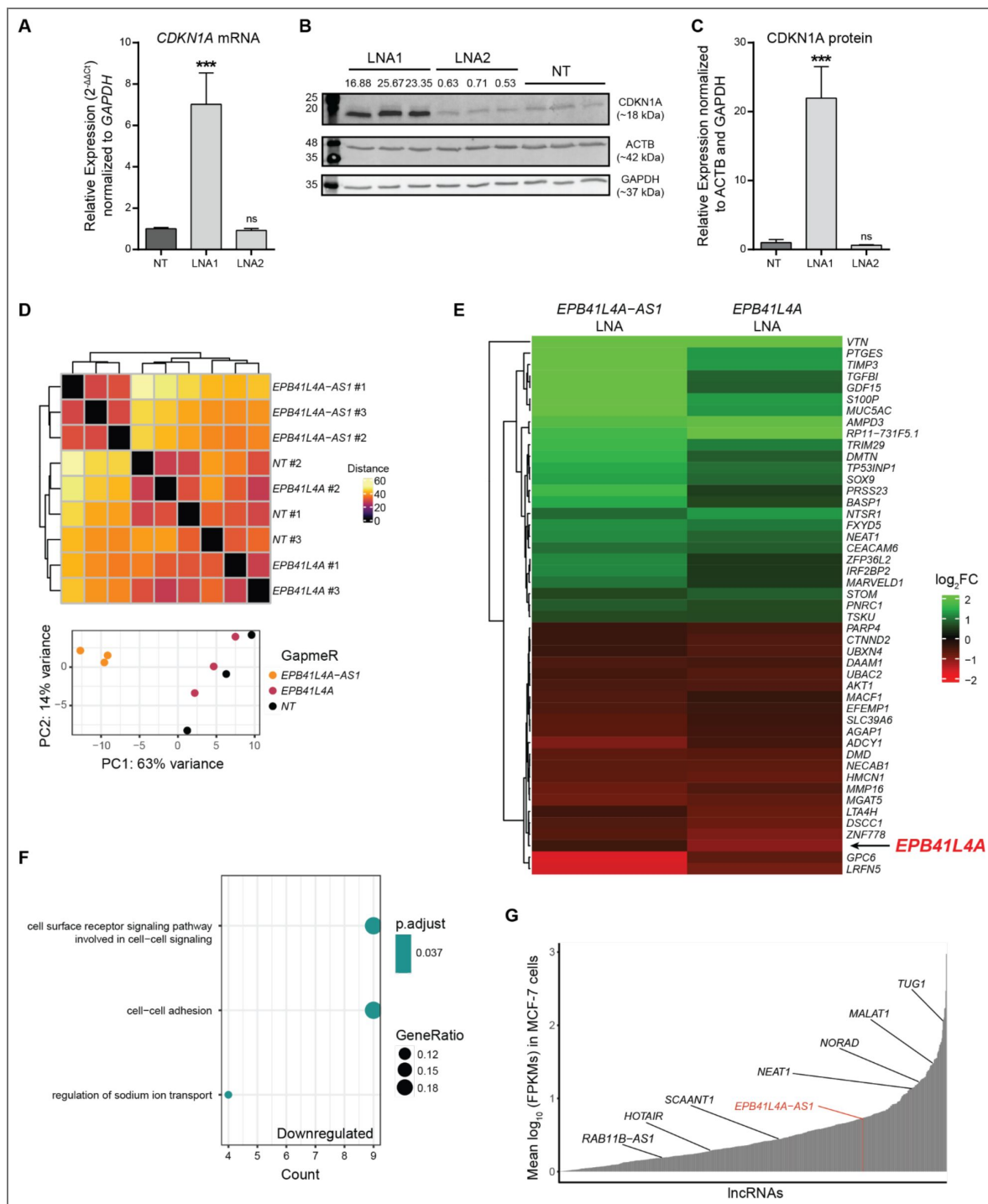
Supplementary Figure 3. *EPB41L4A-AS1* expression is altered upon multiple stimuli.

(A) CPAT⁴¹ analysis of the indicated transcripts. As controls for a noncoding and a coding transcript, *XIST* and *ACTB* are reported. Changes in gene expression for the indicated genes (B) during a time course of 7 days in serum starvation conditions, (C) following the release from a double thymidine cell cycle block, (D) in a panel of 250 unique cell type-treatment combination (the Spearman's correlation coefficient and trendline are also reported), (E) and after exposing MCF-7 cells to LPS, H₂O₂, thapsigargin and etoposide. When applicable, all experiments were performed in n = 3 biological replicates, with the error bars in the dotplots and boxplots representing the standard deviation. In all cases, ns - P>0.05 = ns; * - P<0.05 = *; ** - P<0.01 = **; *** - P<0.001 = *** (two-sided Student's t-test).



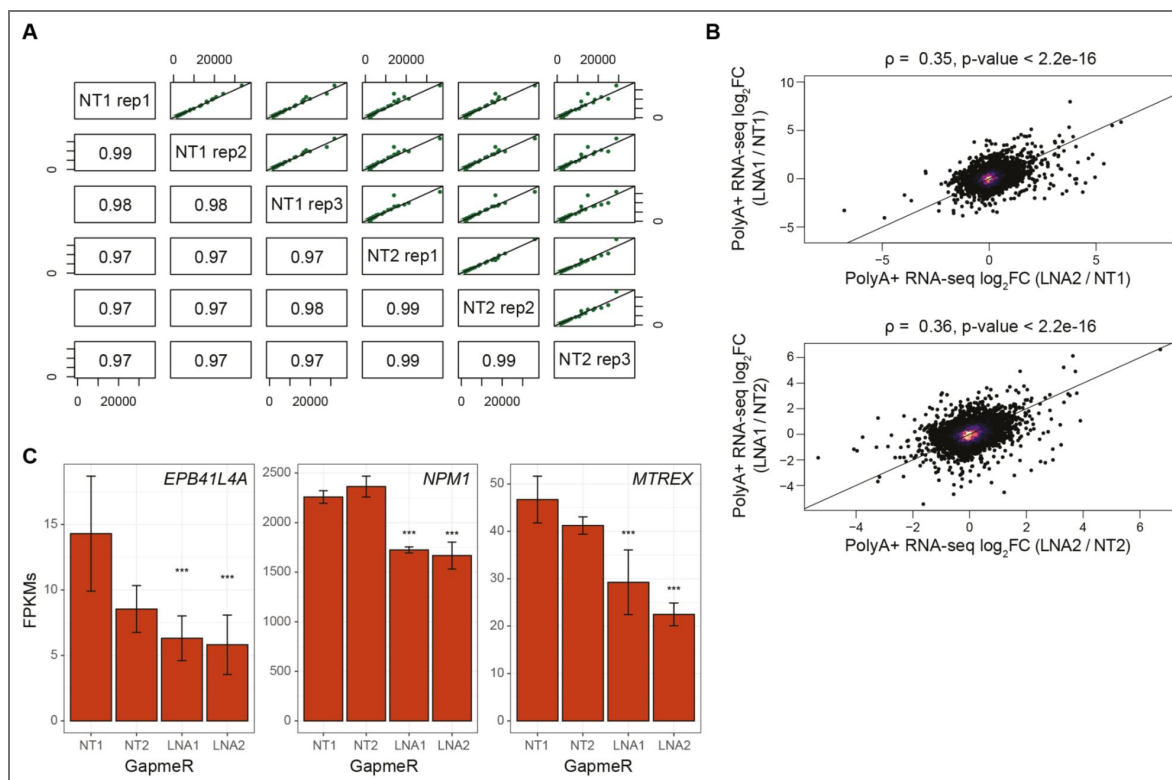
Supplementary Figure 4. *EPB41L4A-AS1* unidirectionally facilitates *EPB41L4A* expression in *cis*.

RT-qPCR to assess the expression of the reported genes after (A) transfection with siRNAs against *EPB41L4A-AS1*, (B) CRISPRa with guides targeting the *EPB41L4A-AS1* promoter, (C) transfection with plasmid encoding the *EPB41L4A-AS1* cDNA, transfection with GapmeRs (D) and siRNAs (E) targeting *EPB41L4A*, (F) CRISPRa with guides targeting the *EPB41L4A* promoter and (G) transfection with plasmid encoding the *EPB41L4A* cDNA. (H) Changes in *EPB41L4A-AS1* expression after rescuing *EPB41L4A-AS1* with an ectopic plasmid or CRISPRa following its KD with GapmeRs. In both panels (Ectopic OE and CRISPRa) the “-” samples represent those transfected with the Empty Vector or sgControl. Asterisks indicate significance relative to the -/- control (transfected with both the control GapmeR and vector). (I) Same as in (H), but for changes in *EPB41L4A* expression. (J) UMI-4C contact profiles in control and LNA2-transfected cells using baits targeting the TSS of *EPB41L4A-AS1*. The green area represents the quantified genomic interval, and the p-value was calculated using a Chi-squared test. All experiments were performed in n = 3 biological replicates, except UMI-4C with n=2, with the error bars in the boxplots representing the standard deviation. In all cases, ns - P>0.05 = ns; * - P<0.05 = *; ** - P<0.01 = **; *** - P<0.001 = *** (two-sided Student’s t-test).



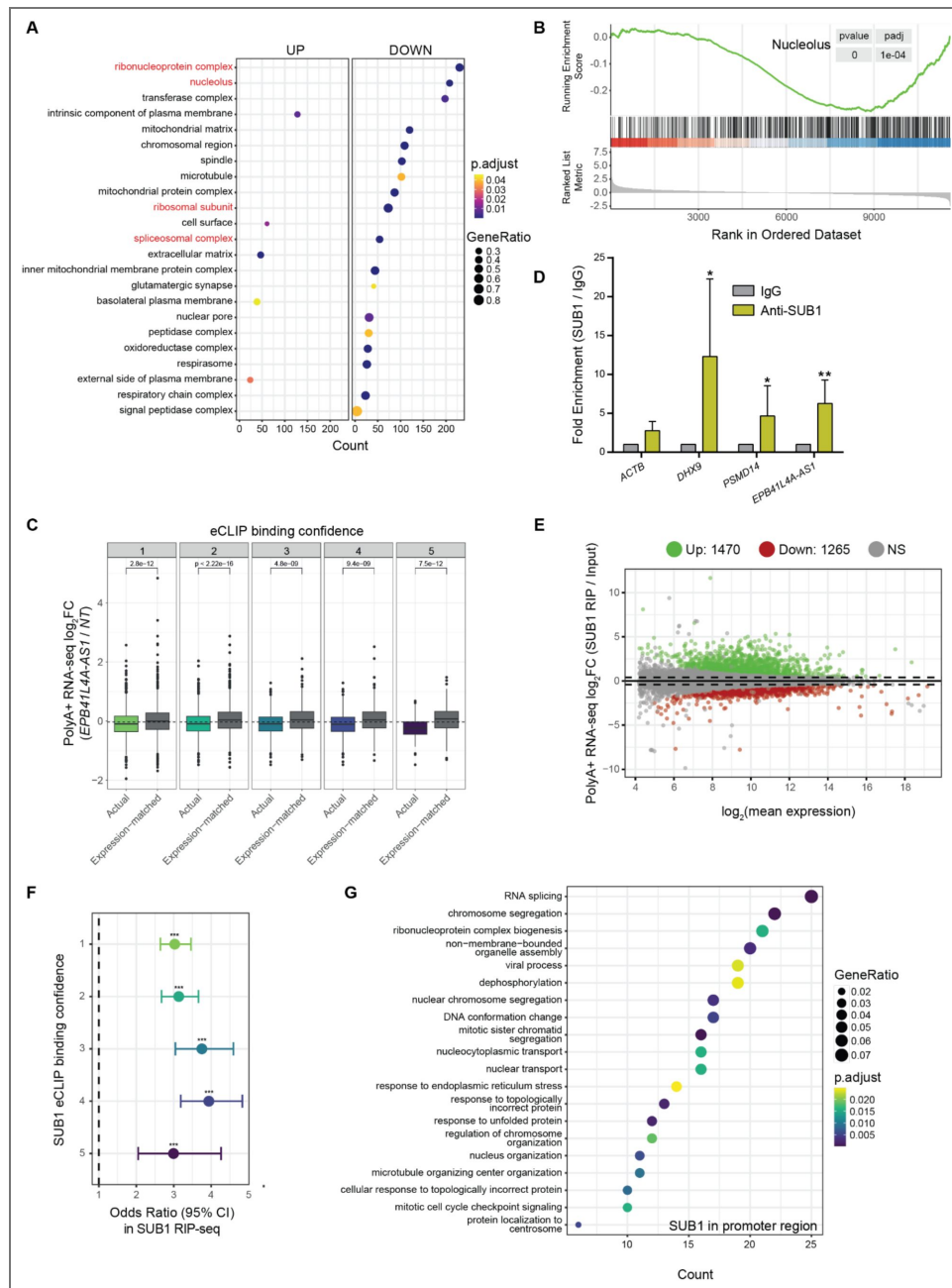
Supplementary Figure 5. Most of the transcriptomic changes following *EPB41L4A-AS1* downregulation are not explained by *EPB41L4A*.

(A) RT-qPCR and (B) Western blot for *CDKN1A* mRNA and protein following *EPB41L4A-AS1* KD with LNA1 and LNA2. (C) Quantification of *CDKN1A* protein levels from (B). (D) Hierarchical clustering (top) and Principal Component Analysis (PCA, bottom) of the polyA⁺ RNA-seq libraries. The color in the heatmap represents the Euclidean distance between the libraries. (E) Heatmap of the differentially expressed genes shared between *EPB41L4A-AS1* and *EPB41L4A* depletions. Color intensity reflects the changes in gene expression ($\log_2 FC$). (F) GO enrichment analysis of the genes downregulated following *EPB41L4A* depletion. (G) Barplot of the lncRNAs (annotated in GENCODE) detected in MCF-7 cells, ordered by their expression. *EPB41L4A-AS1* and other representative *cis*- or *trans*-acting lncRNAs are highlighted. All experiments were performed in $n=3$ biological replicates, with the error bars in the barplots representing the standard deviation. ns - $P>0.05$; * - $P<0.05$; ** - $P<0.01$; *** - $P<0.001$ (two-sided Student's t-test). A gene was considered to be differentially expressed if both adjusted $P<0.05$ and $|\log_2 \text{Fold-change}| > 0.41$ (corresponding to a change of 33%).



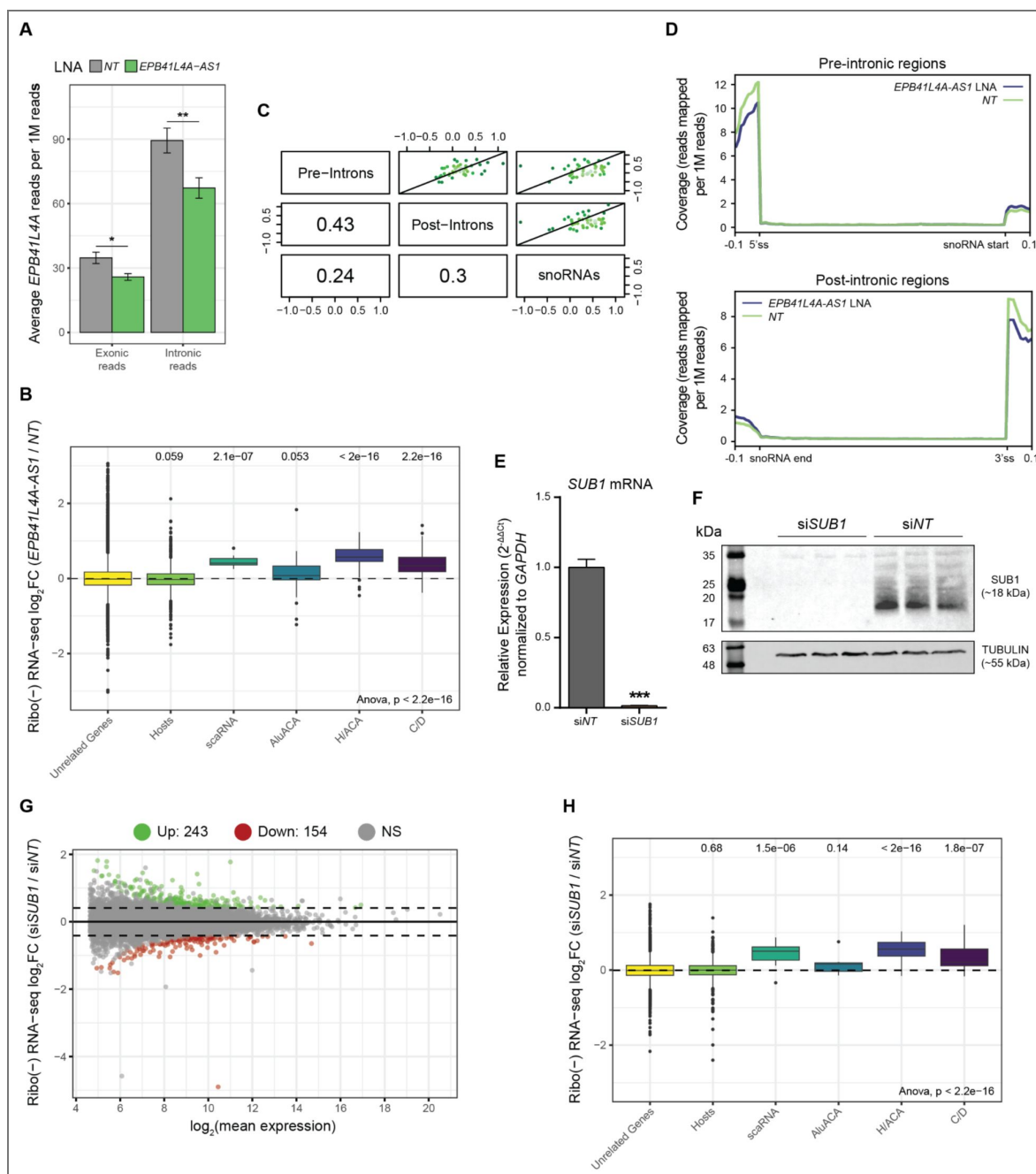
Supplementary Figure 6. LNA1 and LNA2 induce a similar transcriptional response.

(A) Correspondence of the gene expression levels (in FPKMs) between each individual replicate of the two control GapmeRs. The values depict the Spearman's correlation coefficients of each pairwise correlation. **(B)** Correspondence between changes in gene expression after *EPB41L4A-AS1* KD using either LNA1 or LNA2, normalized on either NT1 (top) or NT2 (bottom). **(C)** Average expression levels (in FPKMs) of *EPB41L4A*, *NPM1* and *MTREX* in cells transfected with the two GapmeRs targeting *EPB41L4A-AS1*, with the error bars representing the standard deviation across the $n = 3$ replicates. DESeq2 adjusted P-values compared to control GapmeR #1 (NT1) are also reported. All experiments were performed in $n=3$ biological replicates, with the error bars in the barplots representing the standard deviation. ns - $P>0.05$; * - $P<0.05$; ** - $P<0.01$; *** - $P<0.001$.



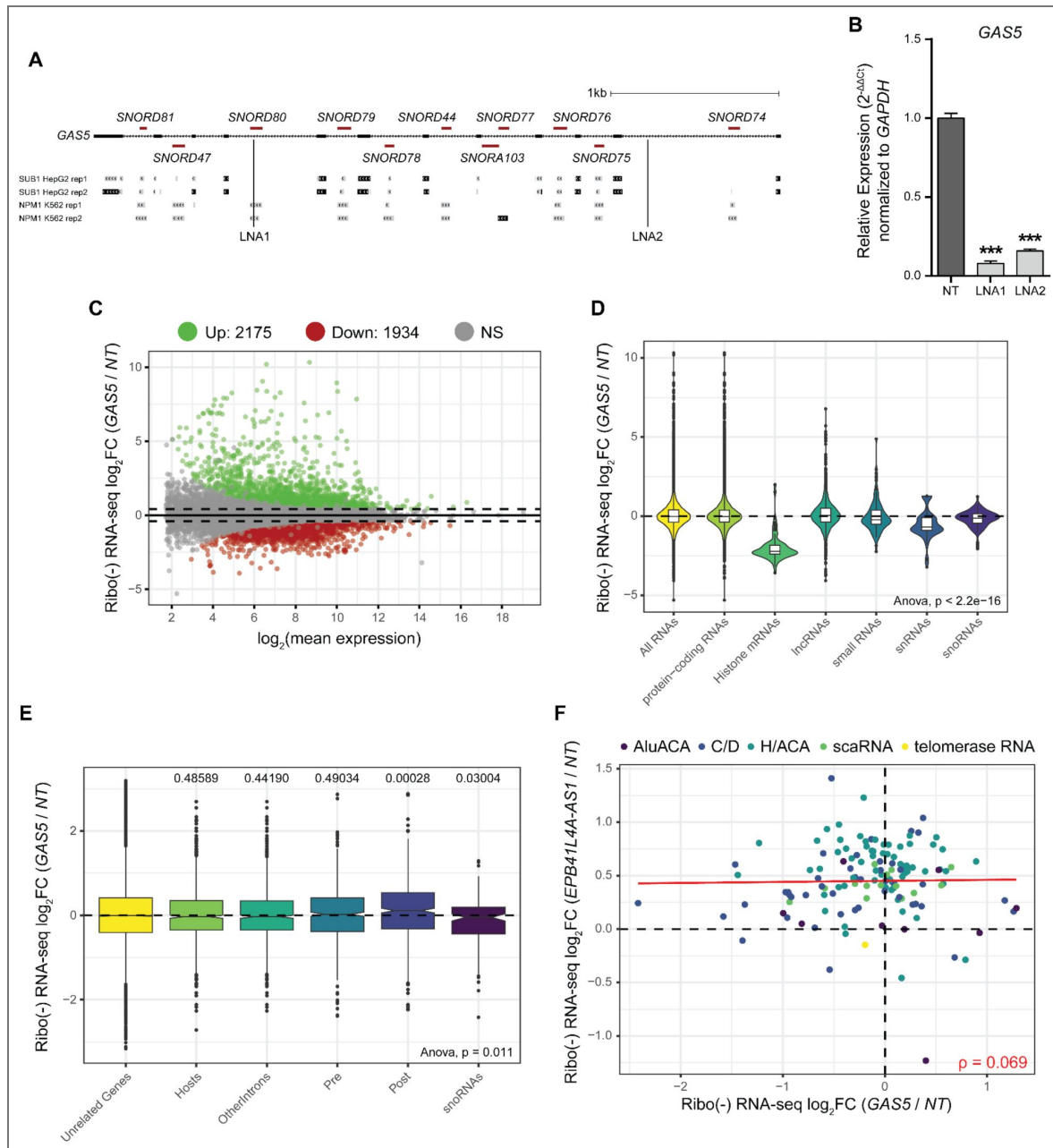
Supplementary Figure 7. SUB1 is both a chromatin-associated and an RNA-binding protein.

(A) Gene set enrichment analysis (GSEA) using *cellular component* as ontology. (B) Running score and preranked list of the nucleolus-associated genes in the GSEA analysis. (C) Changes in gene expression upon *EPB41L4A-AS1* KD of the genes ranked by the SUB1 eCLIP binding confidence, for both the actual targets (as in Fig. 3C) and expression-matched controls (in grey). (D) RT-qPCR for the indicated genes after RIP with either an anti-SUB1 or control IgG antibody. (E) MA plot showing the relative enrichment in RIP-seq after using either an anti-SUB1 or control IgG antibody. (F) Forest plot for the enrichment of the genes in the ENCODE SUB1 eCLIP in our RIP-seq data. The genes in the eCLIP dataset were ranked according to their enrichment score. (G) GO enrichment analysis of the genes with a SUB1 peak as determined from our CUT&RUN data. All experiments were performed in $n = 3$ biological replicates, with the error bars representing the standard deviation (C) or the 95% confidence interval (E). In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test. A gene was considered to be differentially expressed if both adjusted $P < 0.05$ and $|\log_2 \text{Fold-change}| > 0.41$ (corresponding to a change of 33%). In all cases, ns - $P > 0.05 = \text{ns}$; * - $P < 0.05 = *$; ** - $P < 0.01 = **$; *** - $P < 0.001 = ***$ (two-sided Student's t-test in (C), Fisher's exact test in (E)).



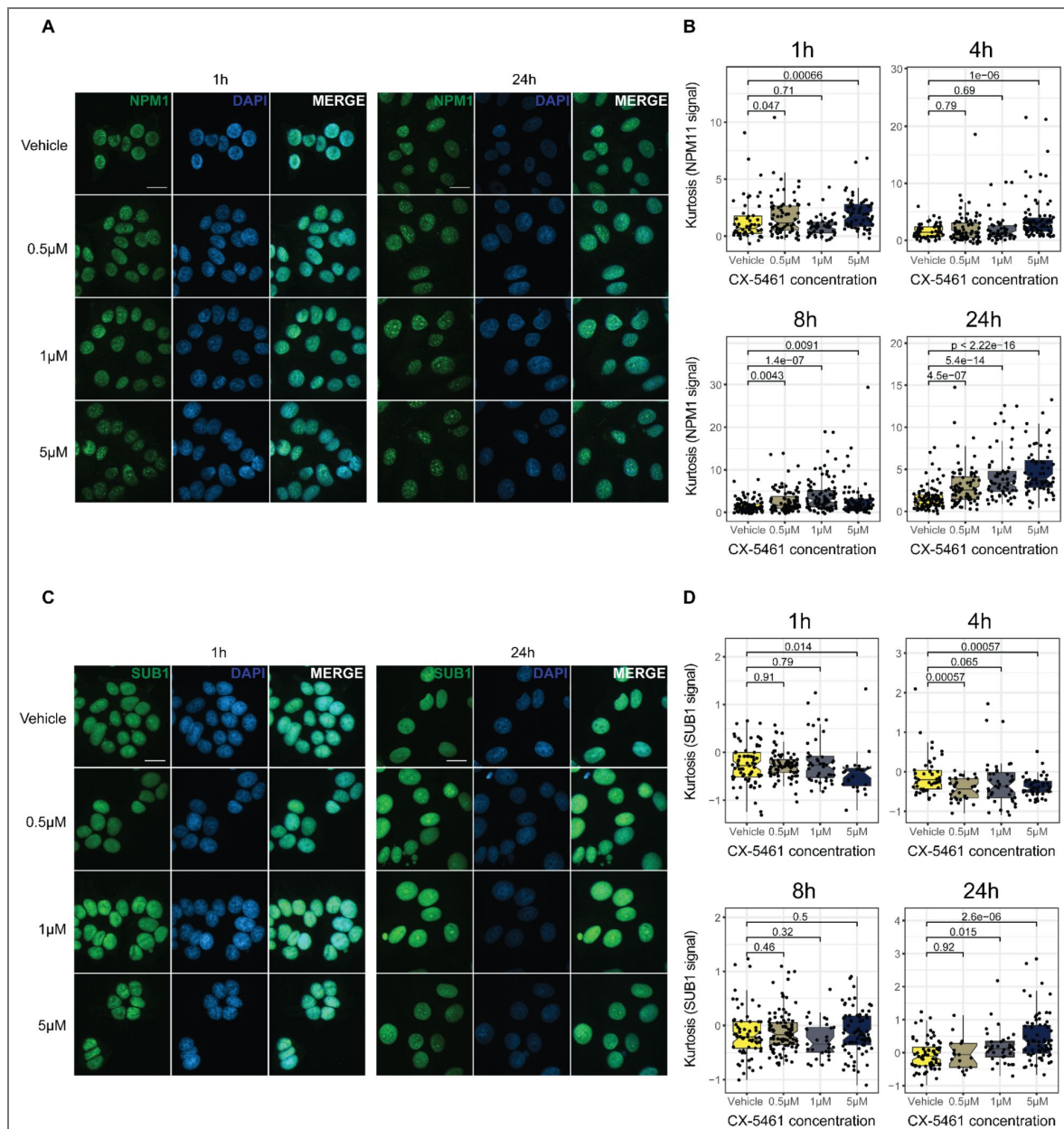
Supplementary Figure 8. EPB41L4A-AS1 and SUB1 depletion affect the expression of different classes of snoRNAs.

(A) Barplot depicting the normalized read density in the Ribo(-) RNA-seq data in EPB41L4A exons and introns. (B) Boxplot of gene expression changes (log₂FC) after EPB41L4A-AS1 depletion for the indicated RNA categories. (C) Correspondence of the changes in gene expression after EPB41L4A-AS1 KD between the pre-intronic, post-intronic and snoRNA regions. Spearman's correlation coefficients are also reported. (D) Normalized Ribo(-) RNA-seq signal over the intronic regions before (top) and after (bottom) the expressed snoRNAs. RT-qPCR (E) and Western blot (F) upon SUB1 depletion with siRNAs to assess KD efficiency. (G) MA plot showing the genome-wide gene expression changes after SUB1 KD with siRNAs. (H) Same as in (B), but after SUB1 KD with siRNAs. All experiments were performed in n = 3 biological replicates, with the error bars in the barplots representing the standard deviation. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA p-value is also reported. A gene was considered to be differentially expressed if both adjusted P < 0.05 and |log₂Fold-change| > 0.41 (corresponding to a change of 33%). In all cases, ns - P > 0.05 = ns; * - P < 0.05 = *; ** - P < 0.01 = **; *** - P < 0.001 = *** (two-sided Student's t-test).



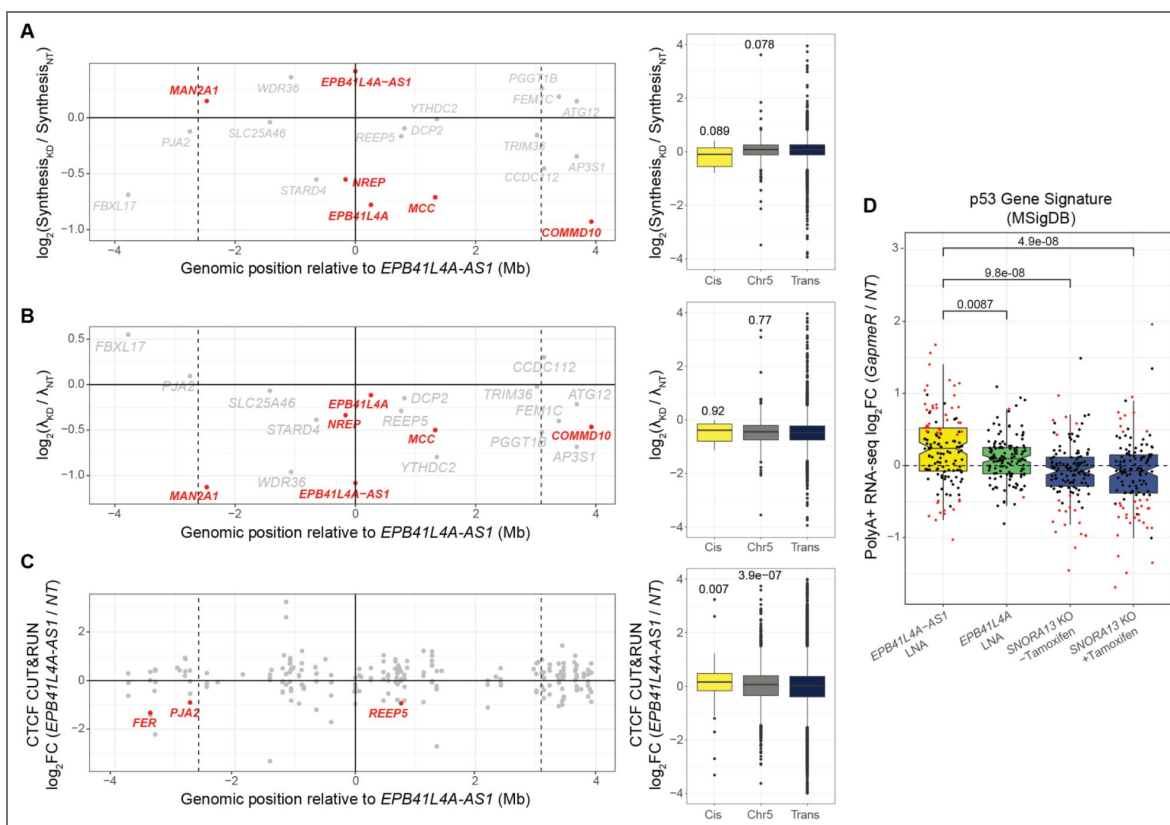
Supplementary Figure 9. GAS5 depletion does not affect snoRNAs expression.

(A) Schematics of the *GAS5* locus with tracks depicting the eCLIP peaks for both SUB1 and NPM1 (source: ENCODE), as well as the location of the two GapmeRs that were used. (B) RT-qPCR upon *GAS5* depletion with GapmeRs to assess KD efficiency. (C) MA plot showing the genome-wide gene expression changes after *GAS5* KD with GapmeRs. (D) Violin/boxplots of gene expression changes ($\log_2 FC$) after *GAS5* depletion for the indicated RNA classes. (E) Boxplot of gene expression changes ($\log_2 FC$) after *GAS5* depletion for the indicated RNA categories. (F) Correspondence between the changes in gene expression of the color-coded snoRNA categories after *EPB41L4A-AS1* and *GAS5* depletion. The trendline and Spearman correlation coefficient are also reported. All experiments were performed in $n = 3$ biological replicates, with the error bars in the barplots representing the standard deviation. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test, and a global ANOVA p-value is also reported. A gene was considered to be differentially expressed if both adjusted $P < 0.05$ and $|\log_2 \text{Fold-change}| > 0.41$ (corresponding to a change of 33%). In all cases, ns - $P > 0.05$ = ns; * - $P < 0.05$ = *; ** - $P < 0.01$ = **; *** - $P < 0.001$ = *** (two-sided Student's t-test).



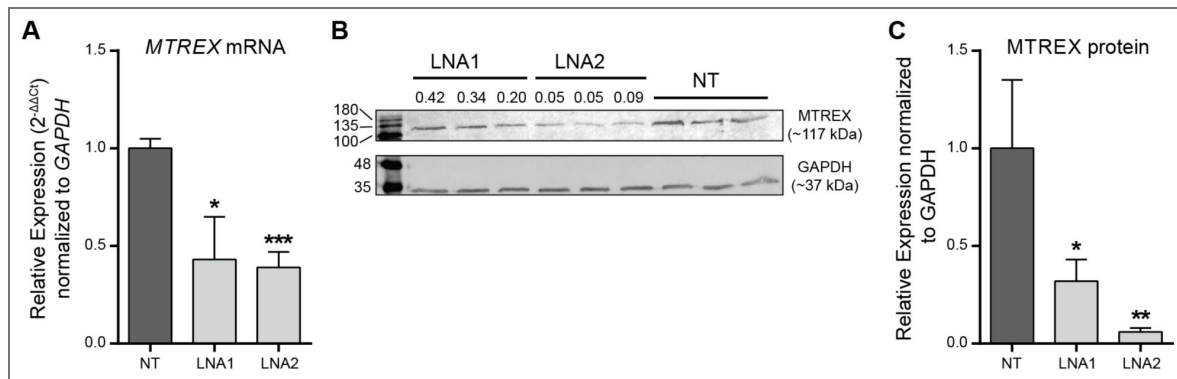
Supplementary Figure 10. Nucleolar stress induced by CX-5461 treatment affects SUB1 and NPM1 nuclear patterns.

(A) Representative immunofluorescence images for NPM1 after treating cells with the indicated CX-5461 concentrations and time (scale bar = 20 μm). (B) Boxplot depicting the changes of the Kurtosis of the nuclear NPM1 signal after treating MCF-7 cells with the indicated CX-5461 concentrations and time. (C) Same as in (A), but for SUB1. (D) Same as in (B), but for SUB1. All experiments were performed in n = 3 biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test.



Supplementary Figure 11. *EPB41L4A-AS1* affects local RNA metabolism.

(A) Changes in synthesis rate in KD cells of the genes in the two flanking TADs of *EPB41L4A-AS1*. The vertical dotted lines represent the TAD boundaries (as assessed by TADmap), the continuous vertical line the lncRNA locus and inter-TAD boundary, and the horizontal continuous line a $\log_2\text{Fold-change}$ equal to 0. The dots represent individual genes, with the significant (adjusted $P < 0.05$ and $|\log_2\text{Fold-change}| > 0.41$) ones highlighted in red. (B) Same as in (A), but for RNA half lives. (C) Same as in (A) and (B), but for CTCF binding as assessed by CUT&RUN. Each dot and name represent a single CTCF peak and the closest gene, respectively. (D) Changes in gene expression of the p53 signature genes (from MSigDB) after KD with GapmeRs targeting either *EPB41L4A-AS1* or *EPB41L4A* (polyA+ RNA-seq data), and *SNORA13* KO cells²⁷. The points in the boxplot represent individual genes, and their color whether they were found to be significantly dysregulated (red) or not (black). All experiments were performed in $n = 3$ biological replicates. In the boxplots, the thick line, edges of the box, and whiskers represent the median, first and third quartiles, and the upper and lower 1.5 interquartile ranges (IQRs), respectively. Outliers (observations outside the 1.5 IQRs) are drawn as single points, the significance of the different comparisons was computed by a Mann-Whitney test.



Supplementary Figure 12. *EPB41L4A-AS1* depletion consistently represses *MTREX*.

(A) RT-qPCR and **(B)** Western blot for *MTREX* mRNA and protein following *EPB41L4A-AS1* KD with LNA1 and LNA2. **(C)** Quantification of *MTREX* protein levels from (B). All experiments were performed in n=3 biological replicates, with the error bars in the barplots representing the standard deviation. ns - $P > 0.05$; * - $P < 0.05$; ** - $P < 0.01$; *** - $P < 0.001$ (two-sided Student's t-test).

Data availability

All the RNA-seq, RIP-seq, UMI-4C, and CUT&RUN datasets were deposited to the GEO database under the accession GSE292272.

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Additional information

Author contributions

A.M. and I.U. conceived the study. A.M. carried out most experiments and analyzed the data. J.P.U. helped with experimental designs, UMI-4C experiments and provided insightful input. T.C. assisted with stress induction experiments. A.M. and I.U. wrote the manuscript with input from the other authors.

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Additional files

[Table S1](#) 

[Table S2](#) 

[Table S3](#) 

[Table S4](#) 

[Table S5](#) 

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Peer reviews

Reviewer #1 (Public review):

Monziani and Ulitsky present a large and exhaustive study on the lncRNA EPB41L4A-AS1 using a variety of genomic methods. They uncover a rather complex picture of a RNA transcript that appears to act via diverse pathways to regulate large numbers of genes' expression, including many snoRNAs. The activity of EPB41L4A-AS1 seems to be intimately linked with the protein SUB1, via both direct physical interactions and direct/indirect of SUB1 mRNA expression.

The study is characterised by thoughtful, innovative, integrative genomic analysis. It is shown that EPB41L4A-AS1 interacts with SUB1 protein and that this may lead to extensive changes in SUB1's other RNA partners. Disruption of EPB41L4A-AS1 leads to widespread changes in non-polyA RNA expression, as well as local cis changes. At the clinical level, it is possible that EPB41L4A-AS1 plays disease relevant roles, although these seem to be somewhat contradictory with evidence supporting both oncogenic and tumour suppressive activities.

A couple of issues could be better addressed here. Firstly, the copy number of EPB41L4A-AS1 is an important missing piece of the puzzle. It is apparently highly expressed from the FISH experiments. To get an understanding of how EPB41L4A-AS1 regulates SUB1, an abundant protein, we need to know the relative stoichiometry of these two factors. Secondly, while many of the experiments use two independent Gapmers for EPB41L4A-AS1 knockdown, the RNA-sequencing experiments apparently use just one, with one negative control (?). Evidence is emerging that Gapmers produce extensive off target gene expression effects in cells,

potentially exceeding the amount of on-target changes arising through the intended target gene. Therefore, it is important to estimate this through use of multiple targeting and non-targeting ASOs, if one is to get a true picture of EPB41L4A-AS1 target genes. In this Reviewer's opinion, this casts some doubt over interpretation of RNA-seq experiments until that work is done. Nonetheless, the Authors have designed thorough experiments, including overexpression rescue overexpression constructs, to quite confidently assess the role of EPB41L4A-AS1 in snoRNA expression.

It is possible that EPB41L4A-AS1 plays roles in cancer, either as oncogene or tumour suppressor. However it will in future be important to extend these observations to a greater variety of cell contexts.

This work is valuable in providing an extensive and thorough analysis of the global mechanisms of an important regulatory lncRNA, and highlights the complexity of such mechanisms via cis and trans regulation and extensive protein interactions.

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Reviewer #2 (Public review):

Summary:

In this manuscript, Monziani et al. identified long noncoding RNAs (lncRNAs) that act in cis and are coregulated with their target genes located in close genomic proximity. The authors mined the GeneHancer database, and this analysis led to the identification of four lncRNA-target pairs. The authors decided to focus on lncRNA EPB41L4A-AS1.

They thoroughly characterised this lncRNA, demonstrating that it is located in the cytoplasm and the nuclei, and that its expression is altered in response to different stimuli. Furthermore, the authors showed that EPB41L4A-AS1 regulates EPB41L4A transcription, leading to a mild reduction in EPB41L4A protein levels. This was not recapitulated with sirna-mediated depletion of EPB41L4AAS1. RNA-seq in EPB41L4A-AS1 depleted cells with single LNA revealed 2364 DEGs linked to pathways including the cell cycle, cell adhesion, and inflammatory response. To understand the mechanism of action of EPB41L4A-AS1, the authors mined the ENCODE eCLIP data and identified SUB1 as an lncRNA interactor. The authors also found that the loss of EPB41L4A-AS1 and SUB1 leads to the accumulation of snoRNAs, and that SUB1 localisation changes upon the loss of EPB41L4A-AS1. Finally, the authors showed that EPB41L4A-AS1 deficiency did not change the steady-state levels of SNORA13 nor RNA modification driven by this RNA. The phenotype associated with the loss of EPB41L4A-AS1 is linked to increased invasion and EMT gene signature.

Overall, this is an interesting and nicely done study on the versatile role of EPB41L4A-AS1 and the multifaceted interplay between SUB1 and this lncRNA, but some conclusions and claims need to be supported with additional experiments before publication. My primary concerns are using a single LNA gapmer for critical experiments, increased invasion and nucleolar distribution of SUB1- in EPB41L4A-AS1-depleted cells.

Strengths:

The authors used complementary tools to dissect the complex role of lncRNA EPB41L4A-AS1 in regulating EPB41L4A, which is highly commendable. There are few papers in the literature on lncRNAs at this standard. They employed LNA gapmers, siRNAs, CRISPRi/a, and exogenous overexpression of EPB41L4A-AS1 to demonstrate that the transcription of EPB41L4A-AS1 acts in cis to promote the expression of EPB41L4A by ensuring spatial proximity between the TAD boundary and the EPB41L4A promoter. At the same time, this lncRNA binds to SUB1 and regulates snoRNA expression and nucleolar biology. Overall, the manuscript is easy to read,

and the figures are well presented. The methods are sound, and the expected standards are met.

Weaknesses:

The authors should clarify how many lncRNA-target pairs were included in the initial computational screen for cis-acting lncRNAs and why MCF7 was chosen as the cell line of choice. Most of the data uses a single LNA gapmer targeting EPB41L4A-AS1 lncRNA (eg, Fig. 2c, 3B and RNA-seq), and the critical experiments should be using at least 2 LNA gapmers. The specificity of SUB1 CUT&RUN is lacking, as well as direct binding of SUB1 to lncRNA EPB41L4A-AS1, which should be confirmed by CLIP qPCR in MCF7 cells. Finally, the role of EPB41L4A-AS1 in SUB1 distribution (Fig. 5) and cell invasion (Fig. 8) needs to be complemented with additional experiments, which should finally demonstrate the role of this lncRNA in nucleolus and cancer-associated pathways. The use of MCF7 as a single cancer cell line is not ideal.

Revised version of the manuscript:

The authors have addressed many of my concerns in their revised manuscript:

The use of single gapmers has been adequately addressed in the revised version of the manuscript, as well as CUT RUN for SUB1.

Future studies will address the role of this lncRNA in invasion and migration using more relevant and appropriate cellular assays. In addition, nucleolar fractionation and analysis of rRNA synthesis are recommended in the follow-up studies for EPB41L4A-AS1.

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Reviewer #3 (Public review):

Summary:

In Monziani et al. paper entitled: "EPB41L4A-AS1 long noncoding RNA acts in both cis- and trans-acting transcriptional regulation and controls nucleolar biology", the authors made some interesting observations that EPB41L4A-AS1 lncRNA can regulate the transcription of both the nearby coding gene and genes on other chromosomes. They started by computationally examining lncRNA-gene pairs by analyzing co-expression, chromatin features of enhancers, TF binding, HiC connectome and eQTLs. They then zoomed in on four pairs of lncRNA-gene pairs and used LNA antisense oligonucleotides to knock down these lncRNAs. This revealed EPB41L4A-AS1 as the only one that can regulate the expression of its cis-gene target EPB41L4A. By RNA-FISH, the authors found this lncRNA to be located in all three parts of a cell: chromatin, nucleoplasm and cytoplasm. RNA-seq after LNA knockdown of EPB41L4A-AS1 showed that this increased >1100 genes and decreased >1250 genes, including both nearby genes and genes on other chromosomes. They later found that EPB41L4A-AS1 may interact with SUB1 protein (an RNA binding protein) to impact the target genes of SUB1. EPB41L4A-AS1 knockdown reduced the mRNA level of SUB1 and altered the nuclear location of SUB1. Later, the authors observed that EPB41L4A-AS1 knockdown caused increase of snRNAs and snoRNAs, likely via disrupted SUB1 function. In the last part of the paper, the authors conducted rescue experiments that suggested that the full-length, intron- and SNORA13-containing EPB41L4A-AS1 is required to partially rescue snoRNA expression. They also conducted SLAM-Seq and showed that the increased abundance of snoRNAs is primarily due to their hosts' increased transcription and stability. They end with data showing that EPB41L4A-AS1 knockdown reduced MCF7 cell proliferation but increased its migration, suggesting a link to breast cancer progression and/or metastasis.

Strengths:

The strength of the paper includes: it is overall well-written; the results are overall presented with good technical rigor and appropriate interpretation. The observation that a complex lncRNA EPB41L4A-AS1 regulates both cis and trans target genes, if fully proven, is interesting and important.

Weaknesses:

The weakness includes: the paper is a bit disjointed as it started from cis and trans gene regulation, but later it switched to a partially relevant topic of snoRNA metabolism via SUB1; the paper was limited in the mechanisms as to how these trans genes (including SUB1 or NPM1 genes themselves) are affected by EPB41L4A-AS1 knockdown; there are discrepancy of results upon EPB41L4A-AS1 knockdown by LNA versus by CRISPR activation, or by plasmid overexpression of this lncRNA.

Overall, the data is supportive of a role of this lncRNA in regulating cis and trans target genes, and thereby impacting cellular phenotypes.

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

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It is possible that EPB41L4A-AS1 plays roles in cancer, either as an oncogene or a tumour suppressor. However, it will in the future be important to extend these observations to a greater variety of cell contexts.

This work is valuable in providing an extensive and thorough analysis of the global mechanisms of an important regulatory lncRNA and highlights the complexity of such mechanisms via cis and trans regulation and extensive protein interactions.

Reviewer #2 (Public review):

Summary:

In this manuscript, Monziani et al. identified long noncoding RNAs (lncRNAs) that act in cis and are coregulated with their target genes located in close genomic proximity. The authors mined the GeneHancer database, and this analysis led to the identification of four lncRNA-target pairs. The authors decided to focus on lncRNA EPB41L4A-AS1.

They thoroughly characterised this lncRNA, demonstrating that it is located in the cytoplasm and the nuclei, and that its expression is altered in response to different stimuli. Furthermore, the authors showed that EPB41L4A-AS1 regulates EPB41L4A transcription, leading to a mild reduction in EPB41L4A protein levels. This was not recapitulated with siRNA-mediated depletion of EPB41L4A-AS1. RNA-seq in EPB41L4A-AS1-depleted cells with single LNA revealed 2364 DEGs linked to pathways including the cell cycle, cell adhesion, and inflammatory response. To understand the mechanism of action of EPB41L4A-AS1, the authors mined the ENCODE eCLIP data and identified SUB1 as an lncRNA interactor. The authors also found that the loss of EPB41L4A-AS1 and SUB1 leads to the accumulation of snoRNAs, and that SUB1 localisation changes upon the loss of EPB41L4A-AS1. Finally, the authors showed that EPB41L4A-AS1 deficiency did not change the steady-state levels of SNORA13 nor RNA modification driven by this RNA. The phenotype associated with the loss of EPB41L4A-AS1 is linked to increased invasion and EMT gene signature.

Overall, this is an interesting and nicely done study on the versatile role of EPB41L4A-AS1 and the multifaceted interplay between SUB1 and this lncRNA, but some conclusions and claims need to be supported with additional experiments. My primary concerns are using a single LNA gapmer for critical experiments, increased invasion, and nucleolar distribution of SUB1- in EPB41L4A-AS1-depleted cells. These experiments need to be validated with orthogonal methods.

Strengths:

The authors used complementary tools to dissect the complex role of lncRNA EPB41L4A-AS1 in regulating EPB41L4A, which is highly commendable. There are few papers in the literature on lncRNAs at this standard. They employed LNA gapmers, siRNAs, CRISPRi/a, and exogenous overexpression of EPB41L4A-AS1 to demonstrate that the transcription of EPB41L4A-AS1 acts in cis to promote the expression of EPB41L4A by ensuring spatial proximity between the TAD boundary and the EPB41L4A promoter. At the same time, this lncRNA binds to SUB1 and regulates snoRNA expression and nucleolar biology. Overall, the manuscript is easy to read, and the figures are well presented. The methods are sound, and the expected standards are met.

Weaknesses:

The authors should clarify how many lncRNA-target pairs were included in the initial computational screen for cis-acting lncRNAs and why MCF7 was chosen as the cell line of choice. Most of the data uses a single LNA gapmer targeting EPB41L4A-AS1 lncRNA (eg,

Fig. 2c, 3B, and RNA-seq), and the critical experiments should be using at least 2 LNA gapmers. The specificity of SUB1 CUT&RUN is lacking, as well as direct binding of SUB1 to lncRNA EPB41L4A-AS1, which should be confirmed by CLIP qPCR in MCF7 cells. Finally, the role of EPB41L4A-AS1 in SUB1 distribution (Figure 5) and cell invasion (Figure 8) needs to be complemented with additional experiments, which should finally demonstrate the role of this lncRNA in nucleolus and cancer-associated pathways. The use of MCF7 as a single cancer cell line is not ideal.

Reviewer #3 (Public review):

Summary:

In this paper, the authors made some interesting observations that EPB41L4A-AS1 lncRNA can regulate the transcription of both the nearby coding gene and genes on other chromosomes. They started by computationally examining lncRNA-gene pairs by analyzing co-expression, chromatin features of enhancers, TF binding, HiC connectome, and eQTLs. They then zoomed in on four pairs of lncRNA-gene pairs and used LNA antisense oligonucleotides to knock down these lncRNAs. This revealed EPB41L4A-AS1 as the only one that can regulate the expression of its cis-gene target EPB41L4A. By RNA-FISH, the authors found this lncRNA to be located in all three parts of a cell: chromatin, nucleoplasm, and cytoplasm. RNA-seq after LNA knockdown of EPB41L4A-AS1 showed that this increased >1100 genes and decreased >1250 genes, including both nearby genes and genes on other chromosomes. They later found that EPB41L4A-AS1 may interact with SUB1 protein (an RNA-binding protein) to impact the target genes of SUB1. EPB41L4A-AS1 knockdown reduced the mRNA level of SUB1 and altered the nuclear location of SUB1. Later, the authors observed that EPB41L4A-AS1 knockdown caused an increase of snRNAs and snoRNAs, likely via disrupted SUB1 function. In the last part of the paper, the authors conducted rescue experiments that suggested that the full-length, intron- and SNORA13-containing EPB41L4A-AS1 is required to partially rescue snoRNA expression. They also conducted SLAM-Seq and showed that the increased abundance of snoRNAs is primarily due to their hosts' increased transcription and stability. They end with data showing that EPB41L4A-AS1 knockdown reduced MCF7 cell proliferation but increased its migration, suggesting a link to breast cancer progression and/or metastasis.

Strengths:

Overall, the paper is well-written, and the results are presented with good technical rigor and appropriate interpretation. The observation that a complex lncRNA EPB41L4A-AS1 regulates both cis and trans target genes, if fully proven, is interesting and important.

Weaknesses:

The paper is a bit disjointed as it started from cis and trans gene regulation, but later it switched to a partially relevant topic of snoRNA metabolism via SUB1. The paper did not follow up on the interesting observation that there are many potential trans target genes affected by EPB41L4A-AS1 knockdown and there was limited study of the mechanisms as to how these trans genes (including SUB1 or NPM1 genes themselves) are affected by EPB41L4A-AS1 knockdown. There are discrepancies in the results upon EPB41L4A-AS1 knockdown by LNA versus by CRISPR activation, or by plasmid overexpression of this lncRNA.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

(1) Copy number:

Perhaps I missed it, but it seems that no attempt is made to estimate the number of copies of EPB41L4A-AS1 transcripts per cell. This should be possible given RNAseq and FISH. At least an order of magnitude estimate. This is important for shedding light on the later observations that EPB41L4A-AS1 may interact with SUB1 protein and regulate the expression of thousands of mRNAs.

We thank the reviewer for the insightful suggestion. We agree that an estimate of EPB41L4A-AS1 copy number might further strengthen the hypotheses presented in the manuscript. Therefore, we analyzed the smFISH images and calculated the copy number per cell of this lncRNA, as well as that of GAPDH as a comparison.

Because segmenting MCF-7 cells proved to be difficult due to the extent of the cell-cell contacts they establish, we imaged multiple ($n = 14$) fields of view, extracted the number of EPB41L4A-AS1/GAPDH molecules in each field and divided them by the number of cells (as assessed by DAPI staining, 589 cells in total). We detected an average of 33.37 ± 3.95 EPB41L4A-AS1 molecules per cell, in contrast to 418.27 ± 61.79 GAPDH molecules. As a comparison, within the same qPCR experiment the average of the Ct values of these two RNAs is about 22.3 and 17.5, the FPKMs in the polyA+ RNA-seq are ~ 2479.4 and 35.6, and the FPKMs in the rRNA-depleted RNA-seq are ~ 3549.9 and 19.3, respectively. Thus, our estimates of the EPB41L4A-AS1 copy number in MCF-7 cells fits well into these observations.

The question whether an average of ~ 35 molecules per cell is sufficient to affect the expression of thousands of genes is somewhat more difficult to ascertain. As discussed below, it is unlikely that all the genes dysregulated following the KD of EPB41L4A-AS1 are all direct targets of this lncRNA, and indeed SUB1 depletion affects an order of magnitude fewer genes. It has been shown that lncRNAs can affect the behavior of interacting RNAs and proteins in a substoichiometric fashion (Unfried & Ulitsky, 2022), but whether this applies to EPB41L4A-AS1 remains to be addressed in future studies. Nonetheless, this copy number appears to be sufficient for a trans-acting functions for this lncRNA, on top of its cis-regulatory role in regulating EPB41L4A. We added this information in the text as follows:

“Using single-molecule fluorescence in-situ hybridization (smFISH) and subcellular fractionation we found that EPB41L4A-AS1 is expressed at an average of 33.37 ± 3.95 molecule per cell, and displays both nuclear and cytoplasmic localization in MCF-7 cells (Fig. 1D), with a minor fraction associated with chromatin as well (Fig. 1E).”

We have updated the methods section as well:

“To visualize the subcellular localization of EPB41L4A-AS1 in vivo, we performed single-molecule fluorescence in situ hybridization (smFISH) using HCR™ amplifiers. Probe sets ($n = 30$ unique probes) targeting EPB41L4A-AS1 and GAPDH (positive control) were designed and ordered from Molecular Instruments. We followed the Multiplexed HCR v3.0 protocol with minor modifications. MCF-7 cells were plated in 8-well chambers (Ibidi) and cultured O/N as described above. The next day, cells were fixed with cold 4% PFA in 1X PBS for 10 minutes at RT and then permeabilized O/N in 70% ethanol at -20°C . Following permeabilization, cells were washed twice with 2X SSC buffer and incubated at 37°C for 30 minutes in hybridization buffer (HB). The HB was then replaced with a probe solution containing 1.2 pmol of EPB41L4A-AS1 probes and 0.6 pmol of GAPDH probes in HB. The slides were incubated O/N at 37°C . To remove excess probes, the slides were washed four times with probe wash buffer at 37°C for 5 minutes each, followed by two washes with 5X SSCT at RT for 5 minutes. The samples were then pre-amplified in amplification buffer for 30 minutes at RT and subsequently incubated O/N in the dark at RT in amplification buffer supplemented with 18 pmol of the appropriate hairpins. Finally, excess hairpins were removed by washing the slides five times in 5X SSCT at RT. The slides were mounted with ProLong™ Glass Antifade Mountant (Invitrogen), cured O/N in the dark at RT, and imaged using a Nikon CSU-W1

spinning disk confocal microscope. In order to estimate the RNA copy number, we imaged multiple distinct fields, extracted the number of EPB41L4A-AS1/GAPDH molecules in each field using the “Find Maxima” tool in ImageJ/Fiji, and divided them by the number of cells (as assessed by DAPI staining).”

(2) Gapmer results:

Again, it is quite unclear how many and which Gapmer is used in the genomics experiments, particularly the RNA-seq. In our recent experiments, we find very extensive off-target mRNA changes arising from Gapmer treatment. For this reason, it is advisable to use both multiple control and multiple targeting Gapmers, so as to identify truly target-dependent expression changes. While I acknowledge and commend the latter rescue experiments, and experiments using multiple Gapmers, I'd like to get clarification about how many and which Gapmers were used for RNAseq, and the authors' opinion on the need for additional work here.

We agree with the Reviewer that GapmeRs are prone to off-target and unwanted effects (Lai et al., 2020; Lee & Mendell, 2020; Maranon & Wilusz, 2020). Early in our experiments, we found out that LNA1 triggers a non-specific CDKN1A/p21 activation (Fig. S5A-C), and thus, we have initially performed some experiments such as RNA-seq with only LNA2.

Nonetheless, other experiments were performed using both GapmeRs, such as multiple RT-qPCRs, UMI-4C, SUB1 and NPM1 imaging, and the in vitro assays, among others, and consistent results were obtained with both LNAs.

To accommodate the request by this and the other reviewers, we have now performed another round of polyA⁺ RNA-seq following EPB41L4A-AS1 knockdown using LNA1 or LNA2, as well as the previously used and an additional control GapmeR. The FPKMs of the control samples are highly-correlated both within replicates and between GapmeRs (Fig. S6A). More importantly, the fold-changes to control are highly correlated between the two on-target GapmeRs LNA1 and LNA2, regardless of the GapmeR used for normalization (Fig. S6B), thus showing that the bulk of the response is shared and likely the direct result of the reduction in the levels of EPB41L4A-AS1. Notably, key targets NPM1 and MTREX (see discussion, Fig. S12A-C and comments to Reviewer 3) were found to be downregulated by both LNAs (Fig. S6C).

However, we acknowledge that some of the dysregulated genes are observed only when using one GapmeR and not the other, likely due to a combination of indirect, secondary and non-specific effects, and as such it is difficult to infer the direct response. Supporting this, LNA2 yielded a total of 1,069 DEGs (617 up and 452 down) and LNA1 2,493 DEGs (1,328 up and 1,287 down), with the latter triggering a stronger response most likely as a result of the previously mentioned CDKN1A/p21 induction. Overall, 45.1% of the upregulated genes following LNA2 transfection were shared with LNA1, in contrast to only the 24.3% of the downregulated ones.

We have now included these results in the Results section (see below) and in Supplementary Figure (Fig. S6).

“Most of the consequences of the depletion of EPB41L4A-AS1 are thus not directly explained by changes in EPB41L4A levels. An additional trans-acting function for EPB41L4A-AS1 would therefore be consistent with its high expression levels compared to most lncRNAs detected in MCF-7 (Fig. S5G). To strengthen these findings, we have transfected MCF-7 cells with LNA1 and a second control GapmeR (NT2), as well as the previous one (NT1) and LNA2, and sequenced the polyadenylated RNA fraction as before. Notably, the expression levels (in FPKMs) of the replicates of both control samples are highly correlated with each other (Fig. S6A), and the global transcriptomic changes triggered by the two EPB41L4A-AS1-targeting LNAs are largely concordant (Fig. S6B and S6C). Because of this concordance and the cleaner

(i.e., no CDKN1A upregulation) readout in LNA2-transfected cells, we focused mainly on these cells for subsequent analyses.”

(3) Figure 1E:

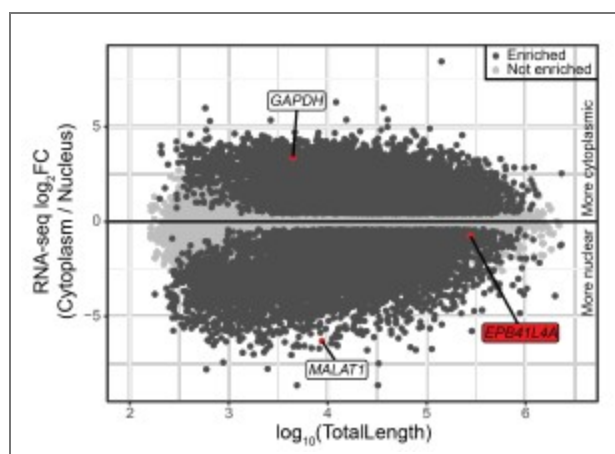
Can the authors comment on the unusual (for a protein-coding mRNA) localisation of EPB41L4A, with a high degree of chromatin enrichment?

We acknowledge that mRNAs from protein-coding genes displaying nuclear and chromatin localizations are quite unusual. The nuclear and chromatin localization of some mRNAs are often due to their low expression, length, time that it takes to be transcribed, repetitive elements and strong secondary structures (Bahar Halpern et al., 2015; Didiot et al., 2018; Lubelsky & Ulitsky, 2018; Ly et al., 2022).

We now briefly mention this in the text:

“In contrast, both EPB41L4A and SNORA13 were mostly found in the chromatin fraction (Fig. 1E), the former possibly due to the length of its pre-mRNA (>250 kb), which would require substantial time to transcribe (Bahar Halpern et al., 2015; Didiot et al., 2018; Lubelsky & Ulitsky, 2018; Ly et al., 2022).”

Supporting our results, analysis of the ENCODE MCF-7 RNA-seq data of the cytoplasmic, nuclear and total cell fractions indeed shows a nuclear enrichment of the EPB41L4A mRNA (Author response image 1), in line with what we observed in Fig. 1E by RT-qPCR.



Author response image 1. The EPB41L4A transcript is nuclear-enriched in the MCF-7 ENCODE subcellular RNA-seq dataset. Scatterplot of gene length versus cytoplasm/nucleus ratio (as computed by DESeq2) in MCF-7 cells. Each dot represents an unique gene, color-coded reflecting if their DESeq2 adjusted p-value < 0.05 and absolute log₂FC > .41 (33% enrichment or depletion). GAPDH and MALAT1 are shown as representative cytoplasmic and nuclear transcripts, respectively. Data from ENCODE.

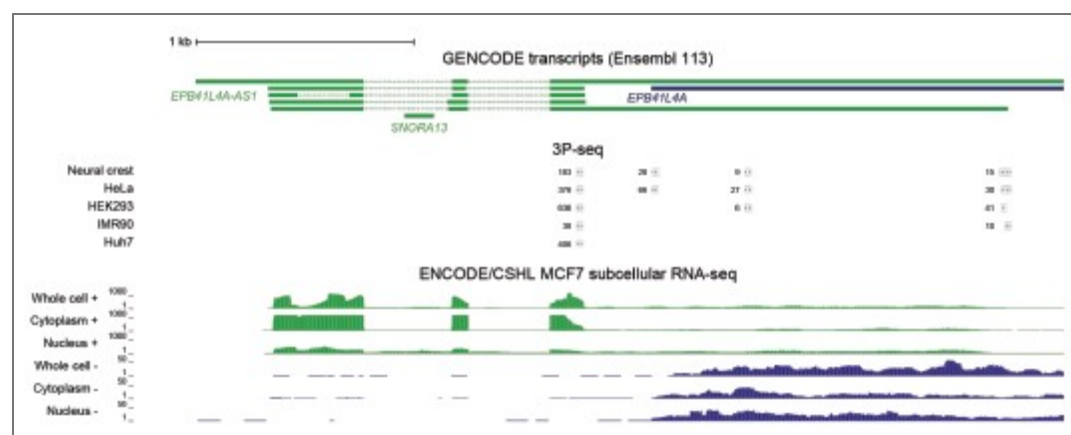
(4) Annotation and termini of EPB41L4A-AS1:

The latest Gencode v47 annotations imply an overlap of the sense and antisense, different from that shown in Figure 1C. The 3' UTR of EPB41L4A is shown to extensively overlap EPB41L4A-AS1. This could shed light on the apparent regulation of the former by the latter that is relevant for this paper. I'd suggest that the authors update their figure of the EPB41L4A-AS1 locus organisation with much more detail, particularly evidence for the true polyA site of both genes. What is more, the authors might consider performing RACE experiments for both RNAs in their cells to definitely establish whether these transcripts contain complementary sequence that could cause their Watson-Crick

hybridisation, or whether their two genes might interfere with each other via some kind of polymerase collision.

We thank the reviewer for pointing this out. Also in previous GENCODE annotations, multiple isoforms were reported with some overlapping the 3' UTR of EPB41L4A. In the EPB41L4A-AS1 locus image (Fig. 1C), we report at the bottom the different transcripts isoforms currently annotated, and a schematics of the one that is clearly the most abundant in MCF-7 cells based on RNA-seq read coverage. This is supported by both the polyA(+) and ribo(-) RNA-seq data, which are strand-specific, as shown in the figure.

We now also examined the ENCODE/CSHL MCF-7 RNA-seq data from whole cell, cytoplasm and nucleus fractions, as well as 3P-seq data (Jan et al., 2011) (unpublished data from human cell lines), reported in Author response image 2. All these data support the predominant use of the proximal polyA site in human cell lines. This shorter isoform does not overlap EPB41L4A.



Author response image 2. Most EPB41L4A-AS1 transcripts end before the 3' end of EPB41L4A. UCSC genome browser view showing tracks from 3P-seq data in different cell lines and neural crest (top, with numbers representing the read counts, i.e. how many times that 3' end has been detected), and stranded ENCODE subcellular RNA-seq (bottom).

Based on these data, the large majority of cellular transcripts of EPB41L4A-AS1 terminate at the earlier polyA site and don't overlap with EPB41L4A. There is a small fraction that appears to be restricted to the nucleus that terminates later at the annotated isoform. 3' RACE experiments are not expected to provide substantially different information beyond what is already available.

(5) Figure 3C:

There is an apparent correlation between log2FC upon EPB41L4A-AS1 knockdown, and the number of clip sites for SUB1. However, I expect that the clip signal correlates strongly with the mRNA expression level, and that log2FC may also correlate with the same. Therefore, the authors would be advised to more exhaustively check that there really is a genuine relationship between log2FC and clip sites, after removing any possible confounders of overall expression level.

As the reviewer suggested, there is a correlation between the baseline expression level and the strength of SUB1 binding in the eCLIP data. To address this issue, we built expression-matched controls for each group of SUB1 interactors and checked the fold-changes following EPB41L4A-AS1 KD, similarly to what we have done in Fig. 3C. The results are presented, and are now part of Supplementary Figure 7 (Fig. S7C).

Based on this analysis, while there is a tendency of increased expression with increased SUB1 binding, when controlling for expression levels the effect of down-regulation of SUB1-bound RNAs upon lncRNA knockdown remains, suggesting that it is not merely a confounding effect. We have updated the text as follows:

“We hypothesized that loss of EPB41L4A-AS1 might affect SUB1, either via the reduction in its expression or by affecting its functions. We stratified SUB1 eCLIP targets into confidence intervals, based on the number, strength and confidence of the reported binding sites. Indeed, eCLIP targets of SUB1 (from HepG2 cells profiled by ENCODE) were significantly downregulated following EPB41L4A-AS1 KD in MCF-7, with more confident targets experiencing stronger downregulation (Fig. 3C). Importantly, this still holds true when controlling for gene expression levels (Fig. S7C), suggesting that this negative trend is not due to differences in their baseline expression.”

(6) The relation to cancer seems somewhat contradictory, maybe I'm missing something. Could the authors more clearly state which evidence is consistent with either an Oncogene or a Tumour Suppressive function, and discuss this briefly in the Discussion? It is not a problem if the data are contradictory, however, it should be discussed more clearly.

We acknowledge this apparent contradiction. Cancer cells are characterized by a multitude of hallmarks depending on the cancer type and stage, including high proliferation rates and enhanced invasive capabilities. The notion that cells with reduced EPB41L4A-AS1 levels exhibit lower proliferation, yet increased invasion is compatible with a function as an oncogene. Cells undergoing EMT may reduce or even completely halt proliferation/cell division, until they revert back to an epithelial state (Brabletz et al., 2018; Dongre & Weinberg, 2019). Notably, downregulated genes following EPB41L4A-AS1 KD are enriched in GO terms related to cell proliferation and cell cycle progression (Fig. 2I), whereas those upregulated are enriched for terms linked to EMT processes. Thus, while we cannot rule out a potential function as tumor suppressor gene, our data fit better the notion that EPB41L4A-AS1 promotes invasion, and thus, primarily functions as an oncogene. We now address this in point in the discussion:

“The notion that cells with reduced EPB41L4A-AS1 levels exhibit lower proliferation (Fig. 8C), yet increased invasion (Fig. 8A and 8B) is compatible with a function as an oncogene by promoting EMT (Fig. 8D and 8E). Cells undergoing this process may reduce or even completely halt proliferation/cell division, until they revert back to an epithelial state (Brabletz et al., 2018; Dongre & Weinberg, 2019). Notably, downregulated genes following EPB41L4A-AS1 KD are enriched in GO terms related to cell proliferation and cell cycle progression (Fig. 2I), whereas those upregulated for terms linked to EMT processes. Thus, while we cannot rule out a potential function as tumor suppressor gene, our data better fits the idea that this lncRNA promotes invasion, and thus, primarily functions as an oncogene.”

Reviewer #2 (Recommendations for the authors):

Below are major and minor points to be addressed. We hope the authors find them useful.

(1) Figure 1:

Where are LNA gapmers located within the EPB41L4A-AS1 gene? Are they targeting exons or introns of the EPB41L4A-AS1? Please clarify or include in the figure.

We now report the location of the two GapmeRs in Fig. 1C. LNA1 targets the intronic region between SNORA13 and exon 2, and LNA2 the terminal part of exon 1.

(2) Figure 2B:

Why is a single LNA gapmer used for EPB41L4A Western? In addition, are the qPCR data in Figure 2B the same as in Figure 1B? Please clarify.

The Western Blot was performed after transfecting the cells with either LNA1 or LNA2. We now have replaced Fig. 2C with the full Western Blot image, in order to show both LNAs. With respect to the qPCRs in Fig. 1B and 2B, they represent the results from two independent experiments.

(3) Figure 2F:

2364 DEGs for a single LNA is a lot of deregulated genes in RNA-seq data. How do the authors explain such a big number in DEGs? Is that because this LNA was intronic? Additional LNA gapmer would minimise the "real" lncRNA target and any potential off-target effect.

We agree with the Reviewer that GapmeRs are prone to off-target and unwanted effects (Lai et al., 2020; Lee & Mendell, 2020; Maranon & Wilusz, 2020). Early in our experiments, we found out that LNA1 triggers a non-specific CDKN1A/p21 activation (Fig. S5A-C), and thus, we have initially performed some experiments such as RNA-seq with only LNA2.

Nonetheless, other experiments were performed using both GapmeRs, such as multiple RT-qPCRs, UMI-4C, SUB1 and NPM1 imaging, and the in vitro assays, among others, and consistent results were obtained with both LNAs.

To accommodate the request by this and the other reviewers, we have now performed another round of polyA+ RNA-seq following EPB41L4A-AS1 knockdown using LNA1 or LNA2, as well as the previously used and an additional control GapmeR. The FPKMs of the control samples are highly-correlated both within replicates and between GapmeRs (Fig. S6A). More importantly, the fold-changes to control are highly correlated between the two on-target GapmeRs LNA1 and LNA2, regardless of the GapmeR used for normalization (Fig. S6B), thus showing that despite significant GapmeR-specific effects, the bulk of the response is shared and likely the direct result of the reduction in the levels of EPB41L4A-AS1. Notably, key targets NPM1 and MTREX (see discussion, Fig. S12A-C and comments to Reviewer 3) were found to be downregulated by both LNAs (Fig. S6C).

However, we acknowledge that some of the dysregulated genes are observed only when using one GapmeR and not the other, likely due to a combination of indirect, secondary and non-specific effects, and as such it is difficult to infer the direct response. Supporting this, LNA2 yielded a total of 1,069 DEGs (617 up and 452 down) and LNA1 2,493 DEGs (1,328 up and 1,287 down), with the latter triggering a stronger response most likely as a result of the previously mentioned CDKN1A/p21 induction. Overall, 45.1% of the upregulated genes following LNA2 transfection were shared with LNA1, in contrast to only the 24.3% of the downregulated ones.

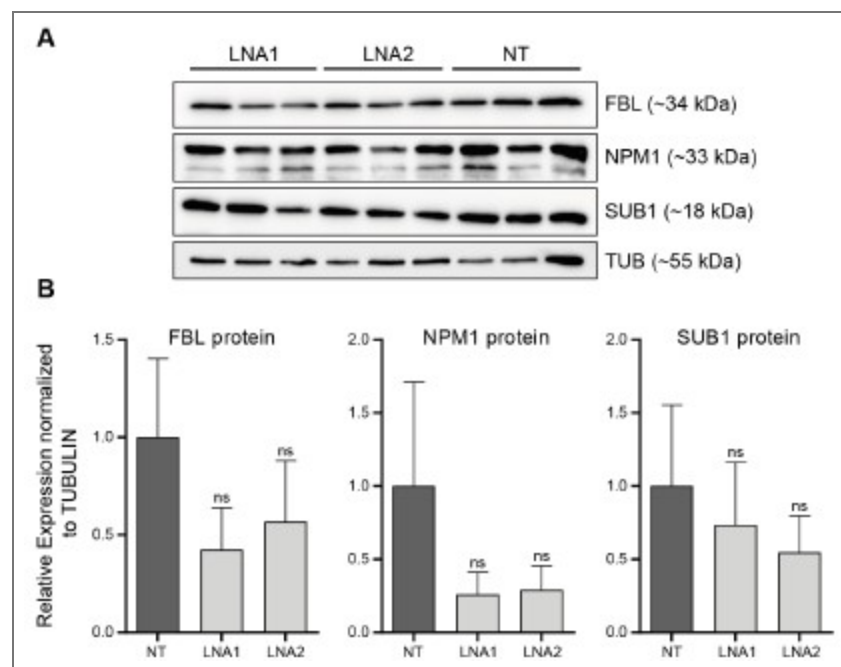
We have now included these results in the Results section (see below) and in Supplementary Figure (Fig. S6).

“Most of the consequences of the depletion of EPB41L4A-AS1 are thus not directly explained by changes in EPB41L4A levels. An additional trans-acting function for EPB41L4A-AS1 would therefore be consistent with its high expression levels compared to most lncRNAs detected in MCF-7 (Fig. S5G). To strengthen these findings, we have transfected MCF-7 cells with LNA1 and a second control GapmeR (NT2), as well as the previous one (NT1) and LNA2, and sequenced the polyadenylated RNA fraction as before. Notably, the expression levels (in FPKMs) of the replicates of both control samples are highly correlated with each other (Fig.

S6A), and the global transcriptomic changes triggered by the two EPB41L4A-AS1-targeting LNAs are largely concordant (Fig. S6B and S6C). Because of this concordance and the cleaner (i.e., no CDKN1A upregulation) readout in LNA2-transfected cells, we focused mainly on these cells for subsequent analyses.”

(4) *Figure 3B: Does downregulation of SUB1 and NPM1 reflect at the protein level with both LNA gapmers? The authors should show a heatmap and metagene profile for SUB1 CUT & RUN. How did the author know that SUB1 binding is specific, since CUT & RUN was not performed in SUB1-depleted cells?*

As requested by both Reviewer #2 and #3, we have performed WB for SUB1, NPM1 and FBL following EPB41L4A-AS1 KD with two targeting (LNA1 and LNA2) and the previous control GapmeRs. Interestingly, we did not detect any significant downregulation of either proteins (Author response image 3), although this might be the result of the high variability observed in the control samples. Moreover, the short timeframe in which the experiments have been conducted—that is, transient transfections for 3 days—might not be sufficient time for the existing proteins to be degraded, and thus, the downregulation is more evident at the RNA (Fig. 3B and Supplementary Figure 6C) rather than protein level.



Author response image 3. EPB41L4A-AS1 KD has only marginal effects on the levels of nucleolar proteins. (A)

Western Blots for the indicated proteins after the transfection for 3 days of the control and targeting GapmeRs. (B) Quantification of the protein levels from (A). All experiments were performed in n=3 biological replicates, with the error bars in the barplots representing the standard deviation. ns - $P > 0.05$; * - $P < 0.05$; ** - $P < 0.01$; *** - $P < 0.001$ (two-sided Student's t-test).

Following the suggestion by the Reviewer, we now show both the SUB1 CUT&RUN metagene profile (previously available as Fig. 3F) and the heatmap (now Fig. 3G) around the TSS of all genes, stratified by their expression level. Both graphs are reported.

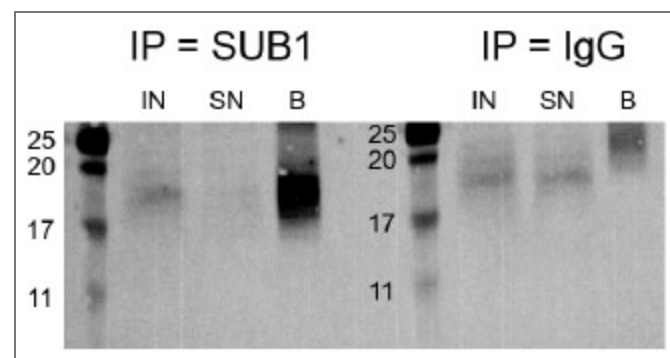
We show that the antibody signal is responsive to SUB1 depletion via siRNAs in both WB (Fig. S8F) and IF (Fig. 5E) experiments. As mentioned below, this and the absence of non-specific signals makes us confident in the CUT&RUN data. Performing CUT&RUN in SUB1 depleted cells would be difficult to interpret as perturbations are typically not complete, and so the remaining protein can still bind the same regions. Since there isn't a clear way to add spike-

ins to CUT&RUN experiments, it is very difficult to show specificity of binding by CUT&RUN in siRNA-knockdown cells.

(5) Figure 3D: The MW for the depicted proteins are lacking. Why is there no SUB1 protein in the input? Please clarify. Since the authors used siRNA to deplete SUB1, it would be good to know if the antibody is specific in their CUT & RUN (see above)

We apologize for the lack of the MW in Fig. 3D. As shown in Fig. S8F, SUB1 is ~18 kDa and the antibody signal is responsive to SUB1 depletion via siRNAs in both WB (Fig. S8F) and IF (Fig. 5E) experiments. Thus, given its 1) established specificity in those two settings and 2) the lack of generalized signal at most open chromatin regions, which is typical of nonspecific CUT&RUN experiments, we are confident in the specificity of the CUT&RUN results.

We now mention the MW of SUB1 in Fig. 3D as well and we provide in Author response image 4 the full SUB1 WB picture, enhancing the contrast to highlight the bands. We agree that the SUB1 band in the input is weak, likely reflecting the low abundance in that fraction and the detection difficulty due to its low MW (see Fig. S8F).



Author response image 4. Western blot for SUB1 following RIP using either a SUB1 or IgG antibody. IN - input, SN - supernatant/unbound, B - bound.

(6) Supplementary Figure 6C:

The validation of lncRNA EPB41L4A-AS1 binding to SUB1 should be confirmed by CLIP qPCR, since native RIP can lead to reassociation of RNA-protein interactions (PMID: 15388877). Additionally, the eclip data presented in Figure 3a were from a different cell line and not MCF7.

We acknowledge that the SUB1 eCLIP data was generated in a different cell line, as we mentioned in the text:

“Indeed, eCLIP targets of SUB1 (from HepG2 cells profiled by ENCODE) were significantly downregulated following EPB41L4A-AS1 KD in MCF-7, with more confident targets experiencing stronger downregulation (Fig. 3C). Importantly, this still holds true when controlling for gene expression levels (Fig. S7C), suggesting that this negative trend is not due to differences in their baseline expression. To obtain SUB1-associated transcripts in MCF-7 cells; we performed a native RNA immunoprecipitation followed by sequencing of polyA+ RNAs (RIP-seq) (Fig. 3D, S7D and S7E).”

Because of this, we resorted to native RIP, in order to get binding information in our experimental system. As we show independent evidence for binding using both eCLIP and RIP, and the substantial challenge in establishing the CLIP method, which has not been successfully used in our group, we respectfully argue that further validations are out of scope of this study. We nonetheless agree that several genes which are nominally significantly

enriched in our RIP data are likely not direct targets of SUB1, especially given that it is difficult to assign the perfect threshold that discriminates between bound and unbound RNAs.

We now additionally mention this at the beginning of the paragraph as well:

“In order to identify potential factors that might be associated with EPB41L4A-AS1, we inspected protein-RNA binding data from the ENCODE eCLIP dataset (Van Nostrand et al., 2020). The exons of the EPB41L4A-AS1 lncRNA were densely and strongly bound by SUB1 (also known as PC4) in both HepG2 and K562 cells (Fig. 3A).”

(7) Figure 3G:

Can the authors distinguish whether loss of EPB41L4A-AS1 affects SUB1 chromatin binding or its activity as RBP? Please discuss.

Distinguishing between altered SUB1 chromatin and RNA binding is challenging, as this protein likely does not interact directly with chromatin and exhibits rather promiscuous RNA binding properties (Ray et al., 2023). In particular, SUB1 (also known as PC4) interacts with and regulates the activity of all three RNA polymerases, and was reported to be involved in transcription initiation and elongation, response to DNA damage, chromatin condensation (Conesa & Acker, 2010; Das et al., 2006; Garavís & Calvo, 2017; Hou et al., 2022) and telomere maintenance (Dubois et al., 2025; Salgado et al., 2024).

Based on our data, genes whose promoters are occupied by SUB1 display marginal, yet highly significant changes in their steady-state expression levels upon lncRNA perturbations. We also show that upon EPB41L4A-AS1 KD, SUB1 acquires a stronger nucleolar localization (Fig. 5A), which likely affects its RNA interactome as well. However, further elucidating these activities would require performing RIP-seq and CUT&RUN in lncRNA-depleted cells, which we argue is out of the scope of the current study. We note that KD of SUB1 with siRNAs have milder effects than that of EPB41L4A-AS1 (Fig. S8G), suggesting that additional players and effects shape the observed changes. Therefore, it is highly likely that the loss of this lncRNA affects both SUB1 chromatin binding profile and RNA binding activity, with the latter likely resulting in the increased snoRNAs abundance.

(8) Figure 4: *Can the authors show that a specific class of snorna is affected upon depletion of SUB1 and EPB41L4A-AS1? Can they further classify the effect of their depletion on H/ACA box snoRNAs, C/D box snoRNAs, and scaRNAs?*

Such potential distinct effect on the different classes of snoRNAs was considered, and the results are available in Fig. S8B and S8H (boxplots, after EPB41L4A-AS1 and SUB1 depletion), as well as Fig. 4F and S9F (scatterplots between EPB41L4A-AS1 and SUB1 depletion, and EPB41L4A-AS1 and GAS5 depletion, respectively). We see no preferential effect on one group of snoRNAs or the other.

(9) Figure 5: *From the representative images, it looks to me that LNA 2 targeting EPB41L4A-AS1 has a bigger effect on nucleolar staining of SUB1. To claim that EPB41L4A-AS1 depletion "shifts SUB1 to a stronger nucleolar distribution", the authors need to perform IF staining for SUB1 and Fibrillarin, a known nucleolar marker. Also, how does this data fit with their qPCR data shown in Figure 3B? It is instrumental for the authors to demonstrate by IF or Western blotting that SUB1 levels decrease in one fraction and increase specifically in the nucleolus. They could perform Western blot for SUB1 and Fibrillarin in EPB41L4A-AS1-depleted cells and isolate cytoplasmic, nuclear, and nucleolar fractions. This experiment will strengthen their finding. The scale bar is missing for all the images in Figure 5. The authors should also show magnified images of a single representative cell at 100x.*

We apologize for the confusion regarding the scale bars. As mentioned here and elsewhere, the scale bars are present in the top-left image of each panel only, in order to avoid overcrowding the panel. All the images are already at 100X, with the exception of Fig. 5E (IF for SUB1 upon siSUB1 transfection) which is 60X in order to better show the lack of signal. We however acknowledge that the images are sometimes confusing, due to the PNG features once imported into the document. In any case, in the submission we have also provided the original images in high-quality PDF and .ai formats. The suggested experiment would require establishing a nucleolar fractionation protocol which we currently don't have available and we argue that it is out of scope of the current study.

(10) Additionally, is rRNA synthesis affected in SUB1- and EPB41L4A-AS1-depleted cells? The authors could quantify newly synthesised rRNA levels in the nucleoli, which would also strengthen their findings about the role of this lncRNA in nucleolar biology.

We acknowledge that there are many aspects of the role of EPB41L4A-AS1 in nucleolar biology that remain to be explored, as well as in nucleolar biology itself, but given the extensive experimental data we already provide in this and other subjects, we respectfully suggest that this experiment is out of scope of the current work. We note that a recent study has shown that SUB1 is required for Pol I-mediated rDNA transcription in the nucleolus (Kaypee et al., 2025). In the presence of nucleolar SUB1, rDNA transcription proceeds as expected, but when SUB1 is depleted or its nucleolar localization is affected—by either sodium butyrate treatment or inhibition of KAT5-mediated phosphorylation at its lysine 35 (K35)—the levels of the 47S pre-rRNA are significantly reduced. In our settings, SUB1 enriches into the nucleolus following EPB41L4A-AS1 KD; thus, we might expect to see a slightly increased rDNA transcription or no effect at all, given that SUB1 localizes in the nucleolus in baseline conditions as well. We now mention this novel role of SUB1 both in the results and discussion.

“SUB1 interacts with all three RNA polymerases and was reported to be involved in transcription initiation and elongation, response to DNA damage, chromatin condensation (Conesa & Acker, 2010; Das et al., 2006; Garavís & Calvo, 2017; Hou et al., 2022), telomere maintenance (Dubois et al., 2025; Salgado et al., 2024) and rDNA transcription (Kaypee et al., 2025). SUB1 normally localizes throughout the nucleus in various cell lines, yet staining experiments show a moderate enrichment for the nucleolus (source: Human Protein Atlas; <https://www.proteinatlas.org/ENSG00000113387-SUB1/subcellular>) (Kaypee et al., 2025).”

“Several features of the response to EPB41L4A-AS1 resemble nucleolar stress, including altered distribution of NPM1 (Potapova et al., 2023; Yang et al., 2016). SUB1 was shown to be involved in many nuclear processes, including transcription (Conesa & Acker, 2010), DNA damage response (Mortusewicz et al., 2008; Yu et al., 2016), telomere maintenance (Dubois et al., 2025), and nucleolar processes including rRNA biogenesis (Kaypee et al., 2025; Tafforeau et al., 2013). Our results suggest a complex and multi-faceted relationship between EPB41L4A-AS1 and SUB1, as SUB1 mRNA levels are reduced by the transient (72 hours) KD of the lncRNA (Fig. 3B), the distribution of the protein in the nucleus is altered (Fig. 5A and 5C), while the protein itself is the most prominent binder of the mature EPB41L4A-AS1 in ENCODE eCLIP data (Fig. 3A). The most striking connection between EPB41L4A-AS1 and SUB1 is the similar phenotype triggered by their loss (Fig. 4). We note that a recent study has shown that SUB1 is required for Pol I-mediated rDNA transcription in the nucleolus (Kaypee et al., 2025). In the presence of nucleolar SUB1, rDNA transcription proceeds as expected, but when SUB1 is depleted or its nucleolar localization is affected—by either sodium butyrate treatment or inhibition of KAT5-mediated phosphorylation at its lysine 35 (K35)—the levels of the 47S pre-rRNA are significantly reduced. In our settings, SUB1 enriches into the nucleolus following EPB41L4A-AS1 KD; thus, we might expect to see a slightly increased rDNA transcription or no effect at all, given that SUB1 localizes in the nucleolus in baseline conditions as well. It is

however difficult to determine which of the connections between these two genes is the most functionally relevant and which may be indirect and/or feedback interactions. For example, it is possible that EPB41L4A-AS1 primarily acts as a transcriptional regulator of SUB1 mRNA, or that its RNA product is required for proper stability and/or localization of the SUB1 protein, or that EPB41L4A-AS1 acts as a scaffold for the formation of protein-protein interactions of SUB1.”

(11) Figure 8: The scratch assay alone cannot be used as a measure of increased invasion, and this phenotype must be confirmed with a transwell invasion or migration assay. Thus, I highly recommend that the authors conduct this experiment using the Boyden chamber. Do the authors see upregulation of N-cadherin, Vimentin, and downregulation of E-cadherin in their RNA-seq?

We agree with the reviewer that those phenotypes are complex and normally require multiple in vitro, as well as in vivo assays to be thoroughly characterized. However, we respectfully consider those as out of scope of the current work, which is more focused on RNA biology and the molecular characterization and functions of EPB41L4A-AS1.

Nevertheless, in Fig. 8D we show that the canonical EMT signature (taken from MSigDB) is upregulated in cells with reduced expression of EPB41L4A-AS1. Notably, EMT has been found to not possess an unique gene expression program, but it rather involves distinct and partially overlapping gene signatures (Youssef et al., 2024). In Fig. 8D, the most upregulated gene is TIMP3, a matrix metalloproteinase inhibitor linked to a particular EMT signature that is less invasive and more profibrotic (EMT-T2, (Youssef et al., 2024)). Interestingly, we observed a strong upregulation of other genes linked to EMT-T2, such as TIMP1, FOSB, SOX9, JUNB, JUN and KLF4, whereas MPP genes (linked to EMT-T1, which is highly proteolytic and invasive) are generally downregulated or not expressed. With regards to N- and E-cadherin, the first does not pass our cutoff to be considered expressed, and the latter is not significantly changing. Vimentin is also not significantly dysregulated. All these examples are reported, which were added as Fig. 8E:

The text has also been updated accordingly:

“These findings suggest that proper EPB41L4A-AS1 expression is required for cellular proliferation, whereas its deficiency results in the onset of more aggressive and migratory behavior, likely linked to the increase of the gene signature of epithelial to mesenchymal transition (EMT) (Fig. 8D). Because EMT is not characterized by a unique gene expression program and rather involves distinct and partially overlapping gene signatures (Youssef et al., 2024), we checked the expression level of marker genes linked to different types of EMTs (Fig. 8E). The most upregulated gene in Fig. 8D is TIMP3, a matrix metalloproteinase inhibitor linked to a particular EMT signature that is less invasive and more profibrotic (EMT-T2) (Youssef et al., 2024). Interestingly, we observed a stark upregulation of other genes linked to EMT-T2, such as TIMP1, FOSB, SOX9, JUNB, JUN and KLF4, whereas MPP genes (linked to EMT-T1, which is highly proteolytic and invasive) are generally downregulated or not expressed. This suggests that the downregulation of EPB41L4A-AS1 is primarily linked to a specific EMT program (EMT-T2), and future studies aimed at uncovering the exact mechanisms and relevance will shed light upon a possible therapeutic potential of this lncRNA.”

(12) Minor points:

(a) What could be the explanation for why only the EPB41L4A-AS1 locus has an effect on the neighbouring gene?

There might be multiple reasons why EPB41L4A-AS1 is able to modulate the expression of the neighboring genes. First, it is expressed from a TAD boundary exhibiting physical contacts with several genes in the two flanking TADs (Fig. 1F and 2A), placing it in the right spot to regulate their expression. Second, it is highly expressed when compared to most of the genes

nearby, with transcription having been linked to the establishment and maintenance of TAD boundaries (Costea et al., 2023). Accordingly, the (partial) depletion of EPB41L4A-AS1 via GapmeRs transfection slightly reduces the contacts between the lncRNA and EPB41L4A loci (Fig. 2E and S4J), although this effect could also be determined by a premature transcription termination triggered by the GapmeRs.

There are a multitude of mechanisms by which lncRNAs with regulatory functions modulate the expression of one or more target genes in cis (Gil & Ulitsky, 2020), and our data do not unequivocally point to one of them. Distinguishing between these possibilities is a major challenge in the field and would be difficult to address in the context of this one study. It could be that the processive RNA polymerases at the EPB41L4A-AS1 locus are recruited to the neighboring loci, facilitated by the close proximity in the 3D space. It could also be possible that chromatin remodeling factors are recruited by the nascent RNA, and then promote and/or sustain the opening of chromatin at the target site. The latter possibility is intriguing, as this mechanism is proposed to be widespread among lncRNAs (Gil & Ulitsky, 2020; Oo et al., 2025) and we observed a significant reduction of H3K27ac levels at the EPB41L4A promoter region (Fig. 2D). Future studies combining chromatin profiling (e.g., CUT&RUN and ATAC-seq) and RNA pulldown experiments will shed light upon the exact mechanisms by which this lncRNA regulates the expression of target genes in cis and its interacting partners.

(b) The scale bar is missing on all the images in the Supplementary Figures as well.

The scale bars are present in the top-left figure of each panel. We acknowledge that due to the export as PNG, some figures (including those with microscopy images) display abnormal font sizes and aspect ratio. All images were created using consistent fonts, sizes and ratio, and are provided as high-quality PDF in the current submission.

(13) Methods:

The authors should double-check if they used sirn and LNA gapmers at 25 and 50um concentrations, as that is a huge dose. Most papers used these reagents in the range of 5-50nM maximum.

We apologize for the typo, the text has been fixed. We performed the experiments at 25 and 50nM, respectively, as suggested by the manufacturer's protocol.

(14) Discussion:

Which cell lines were used in reference 27 (Cheng et al., 2024 Cell) to study the role of SNORA13? It may be useful to include this in the discussion.

We already mentioned the cell system in the discussion, and now we edited to include the specific cell line that was used:

"A recent study found that SNORA13 negatively regulates ribosome biogenesis in TERT-immortalized human fibroblasts (BJ-HRAS^{G12V}), by decreasing the incorporation of RPL23 into the maturing 60S ribosomal subunits, eventually triggering p53-mediated cellular senescence(Cheng et al., 2024)."

Reviewer #3 (Recommendations for the authors):

Major comments on weaknesses:

(1) The paper is quite disjointed:

(a) Figures1/2 studied the cis- and potential trans target genes altered by EPB41L4A-AS1 knockdown. They also showed some data about EPB41L4A-AS1 overlaps a strong chromatin boundary.

(b) Figures 3/4/5 studied the role of SUB1 - as it is altered by EPB41L4A-AS1 knockdown - in affecting genes and snoRNAs, which may partially underlie the gene/snoRNA changes after EPB41L4A-AS1 knockdown.

(c) Figure 6 showed that EPB41L4A-AS1 knockdown did not directly affect SNORA13, the snoRNA located in the intron of EPB41L4A-AS1. Thus, the upregulation of many snoRNAs is not due to SNORA13.

(d) Figure 7 studied whether the changes of cis genes or snoRNAs are due to transcriptional stability.

(e) Figure 8 studied cellular phenotypes after EPB41L4A-AS1 knockdown.

These points are overly spread out and this dilutes the central theme of these results, which this Reviewer considered to be on cis or trans gene regulation by this lncRNA. The title of the paper implies EPB41L4A-AS1 knockdown affected trans target genes, but the paper did not focus on studying cis or trans effects, except briefly mentioning that many genes were changed in Figure 2. The many changes of snoRNAs are suggested to be partially explained by SUB1, but SUB1 itself is affected (>50%, Figure 3B) by EPB41L4A-AS1 knockdown, so it is unclear if these are mostly secondary changes due to SUB1 reduction. Given the current content of the paper, the authors do not have sufficient evidence to support that the changes of trans genes are due to direct effects or indirect effects. And so they are encouraged to revise their title to be more on snoRNA regulation, as this area took the majority of the efforts in this paper.

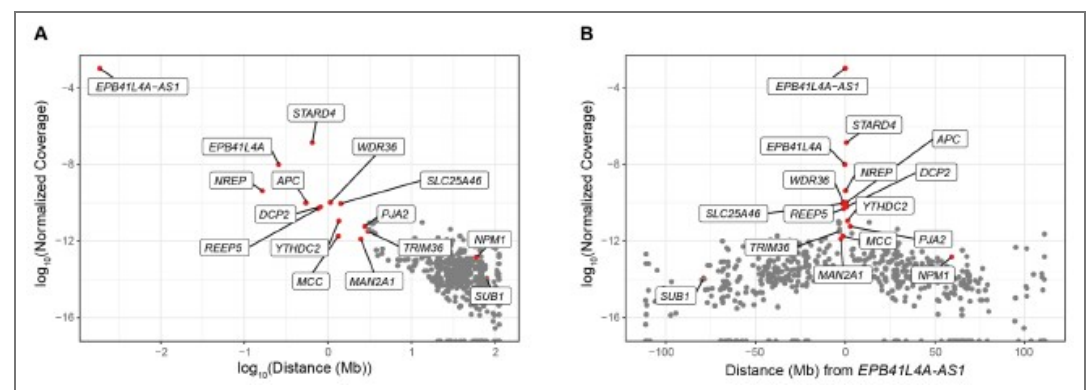
We respectfully disagree with the reviewer. We show that the effect on the proximal genes are cis-acting, as they are not rescued by exogenous expression, whereas the majority of the changes observed in the RNA-seq datasets appear to be indirect, and the snoRNA changes, that indeed might be indirect and not necessarily involve direct interaction partners of the lncRNA, such as SUB1, appear to be trans-regulated, as they can be rescued partially by exogenous expression of the lncRNA. We also show that KD of the main cis-regulated gene, EPB41L4A, results in a much milder transcriptional response, further solidifying the contribution of trans-acting effects. While we agree that the snoRNA effects are interesting, we do not consider them to be the main result, as they are accompanied by many additional changes in gene expression, and changes in the subnuclear distribution of the key nucleolar proteins, so it is difficult for us to claim that EPB41L4A-AS1 is specifically relevant to the snoRNAs rather than to the more broad nucleolar biology. Therefore, we prefer not to mention snoRNAs specifically in the title.

(2) EPB41L4A-AS1 knockdown caused ~2,364 gene changes. This is a very large amount of change on par with some transcriptional factors. It thus needs more scrutiny. First, on Page 9, second paragraph, the authors used $|\log_2\text{Fold-change}| > 0.41$ to select differential genes, which is an unusual cutoff. What is the rationale? Often $|\log_2\text{Fold-change}| > 1$ is more common. How many replicates are used? To examine how many gene changes are likely direct target genes, can the authors show how many of the cis-genes that are changed by EPB41L4A-AS1 knockdown have direct chromatin contacts with EPB41L4A-AS1 in HiC data? Is there any correlation between HiC contact with their fold changes? Without a clear explanation of cis target genes as direct target genes, it is more difficult to establish whether any trans target genes are directly affected by EPB41L4A-AS1 knockdown.

A $|\log_2\text{Fold-change}| > 0.41$ equals a change of 33% or more, which together with an adjusted $P < 0.05$ is a threshold that has been used in the past. All RNA-seq experiments have been performed in triplicates, in line with the standards in the field. While it is possible that the EPB41L4A-AS1 establishes multiple contacts in trans—a process that has been observed in at least another lncRNA, namely Firre but involving its mature RNA product—we do believe this

to be less likely that the alternative, namely that the > 2,000 DEGs are predominantly result from secondary changes rather than genes directly regulated by EPB41L4A-AS1 contacts.

In any case, we have inspected our UMI-4C data to identify other genes exhibiting higher contact frequencies than background levels, and thus, potentially regulated in cis. To this end, we calculated the UMI-4C coverage in a 10kb window centered around the TSS of the genes located on chromosome 5, which we subsequently normalized based on the distance from EPB41L4A-AS1, in order to account for the intrinsic higher DNA recovery the closer to the target DNA sequence. However, in our UMI-4C experiment we have employed baits targeting three different genes—EPB41L4A-AS1, EPB41L4A and STARD4—and therefore such approach assumes that the lncRNA locus has the most regulatory features in this region. As expected, we detected a strong negative correlation between the normalized coverage and the distance from the EPB41L4A-AS1 locus ($\rho = -0.51$, $p\text{-value} < 2.2e-16$), and the genes in the two neighboring TADs exhibited the strongest association with the bait region (Author response image 5). The genes that we see are down-regulated in the adjacent TADs, namely NREP, MCC and MAN2A1 (Fig. 2F) show substantially higher contacts than background with the EPB41L4A-AS1 gene, thus potentially constituting additional cis-regulated targets of this lncRNA. We note that both SUB1 and NPM1 are located on chromosome 5 as well, albeit at distances exceeding 75 and 50 Mb, respectively, and they do not exhibit any striking association with the lncRNA locus.



Author response image 5. UMI-4C coverage over the TSS of the genes located on chromosome 5. (A) Correlation between the normalized UMI-4C coverage over the TSS (± 5 kb) of chromosome 5 genes and the absolute distance (in megabases, Mb) from EPB41L4A-AS1. (B) Same as in (A), but with the x axis showing the relative distance from EPB41L4A-AS1. In both cases, the genes in the two flanking TADs are colored in red and their names are reported.

To increase the confidence in our RNA-seq data, we have now performed another round of polyA+ RNA-seq following EPB41L4A-AS1 knockdown using LNA1 or LNA2, as well as the previously used and an additional control GapmeR. The FPKMs of the control samples are highly-correlated both within replicates and between GapmeRs (Fig. S6A). More importantly, the fold-changes to control are highly correlated between the two on-target GapmeRs LNA1 and LNA2, regardless of the GapmeR used for normalization (Fig. S6B), thus showing that despite significant GapmeR-specific effects, the bulk of the response is shared and likely the direct result of the reduction in the levels of EPB41L4A-AS1. Notably, key targets NPM1 and MTREX (see discussion, Fig. S12A-C and comments to Reviewer 3) were found to be downregulated by both LNAs (Fig. S6C).

However, we acknowledge that some of the dysregulated genes are observed only when using one GapmeR and not the other, likely due to a combination of indirect, secondary and non-specific effects, and as such it is difficult without short time-course experiments (Much et al., 2024) to infer the direct response. Supporting this, LNA2 yielded a total of 1,069 DEGs (617

up and 452 down) and LNA1 2,493 DEGs (1,328 up and 1,287 down), with the latter triggering a stronger response most likely as a result of the previously mentioned CDKN1A/p21 induction. Overall, 45.1% of the upregulated genes following LNA2 transfection were shared with LNA1, in contrast to only the 24.3% of the downregulated ones.

We have now included these results in the Results section (see below) and in Supplementary Figure (Fig. S6).

“Most of the consequences of the depletion of EPB41L4A-AS1 are thus not directly explained by changes in EPB41L4A levels. An additional trans-acting function for EPB41L4A-AS1 would therefore be consistent with its high expression levels compared to most lncRNAs detected in MCF-7 (Fig. S5G). To strengthen these findings, we have transfected MCF-7 cells with LNA1 and a second control GapmeR (NT2), as well as the previous one (NT1) and LNA2, and sequenced the polyadenylated RNA fraction as before. Notably, the expression levels (in FPKMs) of the replicates of both control samples are highly correlated with each other (Fig. S6A), and the global transcriptomic changes triggered by the two EPB41L4A-AS1-targeting LNAs are largely concordant (Fig. S6B and S6C). Because of this concordance and the cleaner (i.e., no CDKN1A upregulation) readout in LNA2-transfected cells, we focused mainly on these cells for subsequent analyses.”

Figure 3B, SUB1 mRNA is reduced >half by EPB41L4A-AS1 KD. How much did SUB1 protein reduce after EPB41L4A-AS1 KD? Similarly, how much is the NPM1 protein reduced? If these two important proteins were affected by EPB41L4A-AS1 KD simultaneously, it is important to exclude how many of the 2,364 genes that changed after EPB41L4A-AS1 KD are due to the protein changes of these two key proteins. For SUB1, Figures S7E,F,G provided some answers. But NPM1 KD is also needed to fully understand such. Related to this, there are many other proteins perhaps changed in addition to SUB1 and NPM1, this renders it concerning how many of the EPB41L4A-AS1 KD-induced changes are directly caused by this RNA. In addition to the suggested study of cist targets, the alternative mechanism needs to be fully discussed in the paper as it remains difficult to fully conclude direct versus indirect effect due to such changes of key proteins or ncRNAs (such as snoRNAs or histone mRNAs).

As requested by both Reviewer #2 and #3, we have performed WB for SUB1, NPM1 and FBL following EPB41L4A-AS1 KD with two targeting (LNA1 and LNA2) and the previous control GapmeRs. Interestingly, we did not detect any significant downregulation of either proteins (Author response image 3), although this might be the result of the high variability observed in the control samples. Moreover, the short timeframe in which the experiments have been conducted—that is, transient transfections for 3 days—might not be sufficient time for the existing proteins to be degraded, and thus, the downregulation is more evident at the RNA (Fig. 3B and Supplementary Figure 6C) rather than protein level.

We acknowledge that many proteins might change simultaneously, and to pinpoint which ones act upstream of the plethora of indirect changes is extremely challenging when considering such large-scale changes in gene expression. In the case of SUB1 and NPM1—which were prioritized for their predicted binding to the lncRNA (Fig. 3A)—we show that the depletion of the former affects the latter in a similar way than that of the lncRNA (Fig. 5F). Moreover, snoRNAs changes are also similarly affected (as the reviewer pointed out, Fig. 4F), suggesting that at least this phenomenon is predominantly mediated by SUB1. Other effects might also be indirect consequences of cellular responses, such as the decrease in histone mRNAs (Fig. 4A) that might reflect the decrease in cellular replication (Fig. 8C) and cell cycle genes (Fig. 2I) (although a link between SUB1 and histone mRNA expression has been described (Brzek et al., 2018)).

Supporting the notion that additional proteins might be involved in driving the observed phenotypes, one of the genes that most consistently was affected by EPB41L4A-AS1 KD with

GapmeRs is MTREX (also known as MTR4), that becomes downregulated at both the RNA and protein levels (now presented in the main text as Supplementary Figure 12). MTREX is part of the NEXT and PAXT complexes (Contreras et al., 2023), that target several short-lived RNAs for degradation, and the depletion of either MTREX or other complex members leads to the upregulation of such RNAs, that include PROMPTs, uaRNAs and eRNAs, among others. Given the lack in our understanding in snoRNA biogenesis from introns in mammalian systems (Monziani & Ulitsky, 2023), it is tempting to hypothesize a role for MTREX-containing complexes in trimming and degrading those introns and release the mature snoRNAs.

We updated the discussion section to include these observations:

“Beyond its site of transcription, EPB41L4A-AS1 associates with SUB1, an abundant protein linked to various functions, and these two players are required for proper distribution of various nuclear proteins. Their dysregulation results in large-scale changes in gene expression, including up-regulation of snoRNA expression, mostly through increased transcription of their hosts, and possibly through a somewhat impaired snoRNA processing and/or stability. To further hinder our efforts in discerning between these two possibilities, the exact molecular pathways involved in snoRNAs biogenesis, maturation and decay are still not completely understood. One of the genes that most consistently was affected by EPB41L4A-AS1 KD with GapmeRs is MTREX (also known as MTR4), that becomes downregulated at both the RNA and protein levels (Fig. S12A-C). Interestingly, MTREX is part of the NEXT and PAXT complexes (Contreras et al., 2023), that target several short-lived RNAs for degradation, and the depletion of either MTREX or other complex members leads to the upregulation of such RNAs, that include PROMPTs, uaRNAs and eRNAs, among others. It is therefore tempting to hypothesize a role for MTREX-containing complexes in trimming and degrading those introns, and releasing the mature snoRNAs. Future studies specifically aimed at uncovering novel players in mammalian snoRNA biology will both conclusively elucidate whether MTREX is indeed involved in these processes.”

With regards to the changes in gene expression between the two LNAs, we provide a more detailed answer above and to the other reviewers as well.

(3) A Strong discrepancy of results by different approaches of knockdown or overexpression:

(a) CRISPRa versus LNA knockdown: Figure S4 - CRISPRa of EPB41L4A-AS1 did not affect EPB41L4A expression (Figure S4B). The authors should discuss how to interpret this result. Did CRISPRa not work to increase the nuclear/chromatin portion of EPB41L4A-AS1? Did CRISPRa of EPB41L4A-AS1 affect the gene in the upstream, the STARD4? Did CRISPRa of EPB41L4A-AS1 also affect chromatin interactions between EPB41L4A-AS1 and the EPB41L4A gene? If so, this may argue that chromatin interaction is not necessary for cis-gene regulation.

There are indeed several possible explanations, the most parsimonious is that since the lncRNA is already very highly transcribed, the relatively modest effect of additional transcription mediated by CRISPRa is not sufficient to elicit a measurable effect. For this reason, we did not check by UMI-4C the contact frequency between the lncRNA and EPB41L4A upon CRISPRa.

CRISPRa augments transcription at target loci, and thus, the nuclear and chromatin retention of EPB41L4A-AS1 are not expected to be affected. We did not check the expression of STARD4, because we focused on EPB41L4A which appears to be the main target locus according to Hi-C (Fig. 2A), UMI-4C (Fig. 2E and S4J) and GeneHancer (Fig. S1).

We already provide extensive evidence of a cis-regulation of EPB41L4A-AS1 over EPB41L4A, and show that EPB41L4A is lowly-expressed and likely has a limited role in our experimental settings. Thus, we respectfully propose that an in-deep exploration of the mechanism of

action of this regulatory axis is out of scope of the current study, that instead focused more on the global effects of EPB41L4A-AS1 perturbation.

(b) Related to this, while CRISPRa alone did not show an effect, upon LNA knockdown of EPB41L4A-AS1, CRISPRa of EPB41L4A-AS1 can increase EPB41L4A expression. It is perplexing as to why, upon LNA treatment, CRISPRa will show an effect (Figure S4H)? Actually, Figures S4H and I are very confusing in the way they are currently presented. They will benefit from being separated into two panels (H into 2 and I into two). And for Ectopic expression, please show controls by empty vector versus EPB41L4A-AS1, and for CRISPRa, please show sgRNA pool versus sgRNA control.

The results are consistent with the parsimonious assumption mentioned above that the high transcription of the lncRNA at baseline is sufficient for maximal positive regulation of EPB41L4A, and that upon KD, the reduced transcription and/or RNA levels are no longer at saturating levels, and so CRISPRa can have an effect. We now mention this interpretation in the text:

“Levels of EPB41L4A were not affected by increased expression of EPB41L4A-AS1 from the endogenous locus by CRISPR activation (CRISPRa), nor by its exogenous expression from a plasmid (Fig. S4B and S4C). The former suggests that endogenous levels of EPB41L4A-AS1—that are far greater than those of EPB41L4A—are sufficient to sustain the maximal expression of this target gene in MCF7 cells.”

We apologize for the confusion regarding the control used in the rescue experiments in Fig. S4H and S4I. The “-” in the Ectopic overexpression and CRISPRa correspond to the Empty Vector and sgControl, respectively, and not the absence of any vector. We changed the text in the figure legends:

“(H) Changes in EPB41L4A-AS1 expression after rescuing EPB41L4A-AS1 with an ectopic plasmid or CRISPRa following its KD with GapmeRs. In both panels (Ectopic OE and CRISPRa) the “-” samples represent those transfected with the Empty Vector or sgControl. Asterisks indicate significance relative to the -- control (transfected with both the control GapmeR and vector). (I) Same as in (H), but for changes in EPB41L4A expression.”

(c) siRNA versus LNA knockdown: Figure S3A showed that siRNA KD of EPB41L4A-AS1 does not affect EPB41L4A expression. How to understand this data versus LNA?

As explained in the text, siRNA-mediated KD presumably affects mostly the cytoplasmic pool of EPB41L4A-AS1 and not the nuclear one, which we assume explains the different effects of the two perturbations, as observed for other lncRNAs (e.g., (Ntini et al., 2018)). However, we acknowledge that we do not know what aspect of the nuclear RNA biology is relevant, let it be the nascent EPB41L4A-AS1 transcription, premature transcriptional termination or even the nuclear pool of this lncRNA, and this can be elucidated further in future studies.

(d) EPB41L4A-AS1 OE versus LNA knockdown: Figure 6F showed that EPB41L4A-AS1 OE caused reduction of EPB41L4A mRNA, particularly at 24hr. How to interpret that both LNA KD and OE of EPB41L4A-AS1 reduce the expression of EPB41L4A mRNA?

We do not believe that the OE of EPB41L4A-AS1, and in particular the one elicited by an ectopic plasmid affects EPB41L4A RNA levels. In the experiment in Fig. 6F, EPB41L4A relative expression at 24h is ~0.65 (please note the log₂ scale in the graph), which is significant as reported. However, throughout this study (and as shown in Fig. S4C for the ectopic and Fig. S4B for the CRISPRa overexpression, respectively), we observed no such behavior, suggesting that the effect reported in Fig. 6F is the result of either that particular setting, and unlikely to reflect a general phenomenon.

(e) Did any of the effects on snoRNAs or trans target genes after EPB41L4A-AS1 knockdown still appear by CRISPRa?

As mentioned above, we did a limited number of experiments after CRISPRa, prompted by the fact that endogenous levels of EPB41L4A-AS1 are already high enough to sustain its functions. Pushing the expression even higher will likely result in no or artifactual effects, which is why we respectfully propose such experiments are not essential in this current work, which instead mostly relies on loss-of-function experiments.

For issue 3, extensive data repetition using all these methods may be unrealistic, but key data discrepancy needs to be fully discussed and interpreted.

Other comments on weakness:

(1) This manuscript will benefit from having line numbers so comments from Reviewers can be made more specifically.

We added line numbers as suggested by the reviewer.

(2) Figure 2G, to distinguish if any effects of EPB41L4A-AS1 come from the cytoplasmic or nuclear portion of EPB41L4A-AS1, an siRNA KD RNA-seq will help to filter out the genes affected by EPB41L4A-AS1 in the cytoplasm, as siRNA likely mainly acts in the cytoplasm.

This experiment would be difficult to interpret as while the siRNAs mostly deplete the cytoplasmic pool of their target, they can have some effects in the nucleus as well (e.g., (Sarshad et al., 2018)) and so siRNAs knockdown will not necessarily report strictly on the cytoplasmic functions.

(3) Figure 2H, LNA knockdown of EPB41L4A should check the protein level reduction, is it similar to the change caused by knockdown of EPB41L4A-AS1?

As suggested by reviewer #2, we have now replaced the EPB41L4A Western Blot that now shows the results with both LNA1 and LNA2. Please note that the previous Fig. 2C was a subset of this, i.e., we have previously cropped the results obtained with LNA1. Unfortunately, we did not have sufficient antibody to check for EPB41L4A protein reduction following LNA KD of EPB41L4A in a timely manner.

(4) There are two LNA Gapmers used by the paper to knock down EPB41L4A-AS1, but some figures used LNA1, some used LNA2, preventing a consistent interpretation of the results. For example, in Figures 2A-D, LNA2 was used. But in Figures 2E-H, LNA1 was used. How consistent are the two in changing histone H3K27ac (like in Figure 2D) versus gene expression in RNA-seq? The changes in chromatin interaction appear to be weaker by LNA2 (Figure S4J) versus LNA1 (Figure 2E).

As explained above and in response to Reviewer #1, we now provide more RNA-seq data for LNA1 and LNA2. We note that besides the unwanted and/or off-target effects, these two GapmeRs might be not equally effective in knocking down EPB41L4A-AS1, which could explain why LNA1 seems to have a stronger effect on chromatin than LNA2. Nonetheless, when we have employed both we have obtained similar and consistent results (e.g., Fig. 5A-D and 8A-C), suggesting that these and the other effects are indeed on target effects due to EPB41L4A-AS1 depletion.

(5) It will be helpful if the authors provide information on how long they conducted EPB41L4A-AS1 knockdown for most experiments to help discern direct or indirect effects.

The length of all perturbations was indicated in the Methods section, and we now mention them also in the Results. Unless specified otherwise, they were carried out for 72 hours. We

agree with the reviewer that having time course experiments can have added value, but due to the extensive effort that these will require, we suggest that they are out of scope of the current study.

(6) In Figures 1C and F, the authors showed results about EPB41L4A-AS1 overlapping a strong chromatin boundary. But these are not mentioned anymore in the later part of the paper. Does this imply any mechanism? Does EPB41L4A-AS1 knockdown or OE, or CRISPRa affect the expression of genes near the other interacting site, STARD4? Do genes located in the two adjacent TADs change more strongly as compared to other genes far away?

We discuss this point in the Discussion section:

“At the site of its own transcription, which overlaps a strong TAD boundary, EPB41L4A-AS1 is required to maintain expression of several adjacent genes, regulated at the level of transcription. Strikingly, the promoter of EPB41L4A-AS1 ranks in the 99.8th percentile of the strongest TAD boundaries in human H1 embryonic stem cells (Open2C et al., 2024; Salnikov et al., 2024). It features several CTCF binding sites (Fig. 2A), and in MCF-7 cells, we demonstrate that it blocks the propagation of the 4C signal between the two flanking TADs (Fig. 1F). Future studies will help elucidate how EPB41L4A-AS1 transcription and/or the RNA product regulate this boundary. So far, we found that EPB41L4A-AS1 did not affect CTCF binding to the boundary, and while some peaks in the vicinity of EPB41L4A-AS1 were significantly affected by its loss, they did not appear to be found near genes that were dysregulated by its KD (Fig. S11C). We also found that KD of EPB41L4A-AS1—which depletes the RNA product, but may also affect the nascent RNA transcription (Lai et al., 2020; Lee & Mendell, 2020)—reduces the spatial contacts between the TAD boundary and the EPB41L4A promoter (Fig. 2E). Further elucidation of the exact functional entity needed for the cis-acting regulation will require detailed genetic perturbations of the locus, that are difficult to carry out in the polyploid MCF-7 cells, without affecting other functional elements of this locus or cell survival as we were unable to generate deletion clones despite several attempts.”

As mentioned in the text (pasted below) and in Fig. 2F, most genes in the two flanking TADs become downregulated following EPB41L4A-AS1 KD. While STARD4 – which was chosen because it had spatial contacts above background with EPB41L4A-AS1 – did not reach statistical significance, others did and are highlighted. Those included NREP, which we also discuss:

“Consistently with the RT-qPCR data, KD of EPB41L4A-AS1 reduced EPB41L4A expression, and also reduced expression of several, but not all other genes in the TADs flanking the lncRNA (Fig. 2F). Based on these data, EPB41L4A-AS1 is a significant cis-acting activator according to TransCistors (Dhaka et al., 2024) ($P=0.005$ using the digital mode). The cis-regulated genes reduced by EPB41L4A-AS1 KD included NREP, a gene important for brain development, whose homolog was downregulated by genetic manipulations of regions homologous to the lncRNA locus in mice (Salnikov et al., 2024). Depletion of EPB41L4A-AS1 thus affects several genes in its vicinity.”

(7) Related to the description of SUB1 regulation of genes are DNA and RNA levels: "Of these genes, transcripts of only 56 genes were also bound by SUB1 at the RNA level, suggesting largely distinct sets of genes targeted by SUB1 at both the DNA and the RNA levels." SUB1 binding to chromatin by Cut&Run only indicates that it is close to DNA/chromatin, and this interaction with chromatin may still likely be mediated by RNAs. The authors used SUB1 binding sites in eCLIP-seq to suggest whether it acts via RNAs, but these binding sites are often from highly expressed gene mRNAs/exons. Standard analysis may not have examined low-abundance RNAs close to the gene promoters, such as promoter antisense RNAs. The authors can examine whether, for the promoters with

cut&run peaks of SUB1, SUB1 eCLIP-seq shows binding to the low-abundance nascent RNAs near these promoters.

In response to a related comment by Reviewer 1, we now show that when considering expression level-matched control genes, knockdown of EPB41L4A-AS1 still significantly affects expression of SUB1 targets over controls. The results are presented in Supplementary Figure 7 (Fig. S7C).

Based on this analysis, while there is a tendency of increased expression with increased SUB1 binding, when controlling for expression levels the effect of down-regulation of SUB1-bound RNAs upon lncRNA knockdown remains, suggesting that it is not merely a confounding effect. We have updated the text as follows:

“We hypothesized that loss of EPB41L4A-AS1 might affect SUB1, either via the reduction in its expression or by affecting its functions. We stratified SUB1 eCLIP targets into confidence intervals, based on the number, strength and confidence of the reported binding sites. Indeed, eCLIP targets of SUB1 (from HepG2 cells profiled by ENCODE) were significantly downregulated following EPB41L4A-AS1 KD in MCF-7, with more confident targets experiencing stronger downregulation (Fig. 3C). Importantly, this still holds true when controlling for gene expression levels (Fig. S7C), suggesting that this negative trend is not due to differences in their baseline expression.”

(8) Figure 8, the cellular phenotype is interesting. As EPB41L4A-AS1 is quite widely expressed, did it affect the phenotypes similarly in other breast cancer cells? MCF7 is not a particularly relevant metastasis model. Can a similar phenotype be seen in commonly used metastatic cell models such as MDA-MB-231?

We agree that further expanding the models in which EPB41L4A-AS1 affects cellular proliferation, migration and any other relevant phenotype is of potential interest before considering targeting this lncRNA as a therapeutic approach. However, given that 1) others have already identified similar phenotypes upon the modulation of EPB41L4A-AS1 in a variety of different systems (see Results and Discussion), and 2) we were most interested in the molecular consequences following the loss of this lncRNA, we respectfully suggest that these experiments are out of scope of the current study.

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